MIL-T-48804(PA)
27 January 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

TEST PROGRAMMER ASSEMBLY, 6A1

OF THE T4138 TEST SET

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of test programmer entitled Test Programmer Assembly, 6A1 of the T4138 Test Set which is used with the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ARMY BALLISTIC MISSILE AGENCY

10970599 - Switch Stepping.

FSC: 1190

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ORDNANCE CORPS

8833327 - Test Programmer Assembly 6A1.
8833314 - Programmer Assembly (Main) 6A1A9.
8833315 - Programmer Assembly (Continuity) 6A1A10.
8833316 - Programmer Assembly (Open Circuit) 6A1A11.
8833317 - Programmer Assembly (T.D.D.) 6A1A12.
8833345 - Panel Assembly 6A1A13.
8833347 - Relay Module #2 Assembly 6A1A8.
8847727 - Voltage Comparator Assembly 6A1A4.
8847741 - Baro Servo Amplifier Assembly 6A1A1.
8847742 - Programmer Step Circuit Module Assembly 6A1A5.
8847761 - Timer Assembly, Unijunction 6A1A6.
8847780 - Reference Voltage Source Assembly 6A1A3.
8847809 - Timer, Precision Leaf Switch Assembly 6A1A2.
8849338 - Relay Module #1 Assembly 6A1A7.
8850424 - Rack Assembly 6A1A14.
EL-8854174 - Equipment List.
8883505 - Programmer Assembly 6A1A9-12.
8883506 - Baro Server Amplifier Assembly 6A1A1.
8883507 - Relay Module Assembly #2 6A1A8.
8883508 - Relay Module Assembly #1 6A1A7.
8883509 - Unijunction Timer Assembly 6A1A6.
8883510 - Programmer Step Option Module Assembly 6A1A5.
8883511 - Voltage Comparator Assembly 6A1A4.
8883512 - Voltage Source Assembly 6A1A3.
8883513 - Timer Assembly, Precision Leaf Switch 6A1A3.
8882431 - Drum and Storage Container XM535 for T4138 (6A1).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The test programmer and modules shall comply with all requirements specified on the drawings (dwgs) listed below, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawings and herein.

No.	Assembly	Drawing
6A1	Test Programmer Assembly	8833327
6A1A1	Baro Servo Amplifier Assembly	8847741
6A1A2	Timer, Precision Leaf Switch Assembly	8847809
6A1A3	Reference Voltage Source Assembly	8847780
6A1A4	Voltage Comparator Assembly	8847727
6A1A5	Programmer Step Circuit Module Assembly	8847742
6A1A6	Timer Assembly, Unijunction	8847761
6A1A7	Relay Module #1 Assembly	8849338
6A1A8	Relay Module #2 Assembly	8833347
6A1A9	Programmer Assembly (Main)	8833314
6A1A10	Programmer Assembly (Continuity)	8833315
6A1A11	Programmer Assembly (Open Circuit)	8833316
6A1A12	Programmer Assembly (T.D.D.)	8833317
6A1A13	Panel Assembly	8833345
6A1A14	Rack Assembly	8850424

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Operational characteristics.

3.5.1 Module assemblies. Each of the assemblies listed in paragraph 3.2 shall be capable of passing the operational characteristics as defined in paragraph 3.5 of the appropriate appendix as listed below.

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No.	Assembly	Appendix
6A1	Test Programmer Assembly	A
6A1A1	Baro Servo Amplifier Assembly	B
6A1A2	Timer, Precision Leaf Switch Assembly	C
6A1A3	Reference Voltage Source Assembly	D
6A1A4	Voltage Comparator Assembly	E
6A1A5	Programmer Step Circuit Assembly	F
6A1A6	Timer Assembly, Unijunction	G
6A1A7	Relay Module #1 Assembly	H
6A1A8	Relay Module #2 Assembly	I
6A1A9	Programmer Assembly (Main)	J
6A1A10	Programmer Assembly (Continuity)	K
6A1A11	Programmer Assembly (Open Circuit)	L
6A1A12	Programmer Assembly (T.D.D.)	M
6A1A13	Panel Assembly	N
6A1A14	Rack Assembly	O

3.5.2 Spare module assemblies. Modules designated as spare parts shall meet the requirements of 3.2 and 3.5.1. In addition, each spare module must be tested for compatibility with and in a T4138 test set. The compatibility test shall be conducted as specified in 4.3.3.2.2.

3.6 Environmental testing. The test programmer and modules shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The test programmer and modules shall also comply with 4.3.3.1 and 4.3.3.2 before and after each environmental test. Testing shall be as specified in 4.4.1 and 4.4.2. Tests shall be conducted in the order specified.

3.6.1 High temperature. Each assembly shall withstand a temperature of 160 plus or minus (±) 5 degrees Fahrenheit (°F.) without damage and shall comply with section 3.5 of the appropriate appendix. After having been subjected to this temperature, it shall then be operated at 130 ± 5°F. The tests shall be conducted as specified in 4.4.3.1.

3.6.2 Low temperature. Each assembly shall withstand a storage temperature of - 65 ± 5°F. without damage and shall comply with section 3.5 of the appropriate appendix. After having been subjected to this temperature, it shall then be operated at 0 ± 5°F. The test shall be conducted as specified in 4.4.3.2.

3.6.3 Vibration. The programmer in its shipping container (Ord. Part No. 8882431) or module assemblies in their shipping containers (procured as spares) shall withstand vibration as specified in Table I without evidence of damage. The vibration shall be applied in the vertical axis as specified in 4.4.3.3.

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Table I

Frequency Hertz (Hz)	Sweep Time (Min.)	Double Amplitude (inches)	Number of Sweeps
5-10-5	10 ± 0.5	0.4 ± 10%	1
11-15-11	10 ± 0.5	0.25 ± 10%	1
16-20-16	10 ± 0.5	0.13 ± 10%	1
21-60-21	15 ± 0.5	0.06 ± 10%	1

3.6.4 Transportation shock. The programmer in its shipping container or module assemblies in their shipping containers (procured as spares) shall withstand three shocks of 30 ± 5 gravity units (g) in both directions in each of three mutually perpendicular axes for a total of 18 shocks. The time duration of each shock shall be 11 ± 2 milliseconds (ms) as measured between points which are 10 percent of peak value. The time to peak shall be 5.5 ± 1 millisecond. The test shall be performed as specified in 4.4.3.4. After the test, the programmer shall show no evidence of damage and shall comply with section 3.5 of the appropriate appendix.

3.6.5 Humidity. The 6A1 test programmer assembly shall withstand exposure to humidity without evidence of deterioration or impairment of operation when tested as specified in 4.4.3.5. Module assemblies procured as spares shall not be subjected to this test.

3.7 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.8.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 8833327) or one of each type module if the contractor is manufacturing spare parts (see dwgs listed in paragraph 3.2) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2. Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests shall be conducted in order specified in Table II.

Table II

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see appendix A through O paragraph 4.3.2.1)		X
2. Operational characteristics (see 3.5)		X
3. High temperature (see 3.6.1)		X
4. Low temperature (see 3.6.2)		X
5. Vibration (see 3.6.3)		X
6. Transportation shock (see 3.6.4)		X
7. Humidity (see 3.6.5)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

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4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as specified in paragraph 4.3.2.1 of the appropriate appendix.

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below.

Test	Test Classification
a. Calibration capability	Major
b. Operational limits	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Calibration capability (see 3.5.1 of the appropriate appendix). Each assembly unless specified otherwise shall be calibrated in accordance with paragraph 4.4.1 of the appropriate appendix. If any assembly is not capable of being so calibrated, it shall be classified as defective and removed from the lot.

4.3.3.2 Operational limits.

4.3.3.2.1 Module assemblies (see 3.5.2 of the appropriate appendix). Each assembly shall be tested in accordance with paragraph 4.4.2 of the appropriate appendix. If any assembly fails to comply with the applicable requirement, it shall be classified as defective and removed from the lot.

4.3.3.2.2 Spare module assemblies. Compatibility testing shall be performed in accordance with the applicable test sequence as outlined in Table V. All compatibility tests shall be conducted with the test module assembled as part of a test set that is otherwise capable of meeting the requirements of MIL-T-48546(PA).

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No. of Module Being Tested	No. of Applicable Test Sequence
6A1A1	2
6A1A2	1
6A1A3	1
6A1A4	5
6A1A5	5
6A1A6	2 & 5
6A1A7	4
6A1A8	4
6A1A9	1
6A1A10	1
6A1A11	1
6A1A12	1
6A1A13	3

Table V

Test-Sequence No.	M91 Simulator Failure Parameters Set	Failures Required (Nixie Numbers)
1	None	None
2	Run 1 - None	None
	Run 2 - Baro Failure	501, 611, 502.
	Run 3 - None	None
3	Run 1 - None	None
	Run 2 - Baro Failure	501, 611, 502.
	Run 3 - None	None
	Run 4 - AK Hazard	AK battery light glows: No run possible as will fail when power switch is thrown on or at any point where failure is thrown on.
4	Run 1 - Fuze Continuity Open Circuit Capacitor AK Failure Identification Baro Contact	801,802,809,810,811,812, 819,820,821,831,832,839, 840,841,842,849,850,851, 607,907,719,515,516,518, 611.
	Run 2 - None	None
	Run 3 - DUD	810,811,820,840,841,850.
	Baro Failure	501,611,502.
	Run 4 - None	None

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Test-Sequence No.	M91 Simulator Failure Parameters Set	Failures Required (Nixie Numbers)
	Run 5 - Delay Element	501,611,502,802,809,810, 811,812,819,820,832,839, 840,841,842,849,850,505, 507,509,511,513,515,518, 519.
	*Run 6 - Premature Fire Contact Failure	805,806,808,811,841,845, 846,848.
	Run 7 - AK Hazard	AK battery light glows: No run possible as will fail when power switch is thrown on or at any point where failure is thrown on.
5	Run 1 - Use Load Box, Electrical T4165 and TB ORD 1026/121 Programmer section e to calibrate. If calibration cannot be made, reject module.	
	Run 2 - Simulator run with no failure parameters inserted. No failure indications should be present.	

*After failure at 811, open premature switch; after failure at 841, close premature switch.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8854174). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column of the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Calibration capability. The procedures for determining calibration capability of each assembly are shown in the appropriate appendix.

4.4.2 Operational limits. The procedures used in testing for operational limits of each assembly are shown in the appropriate appendix.

4.4.3 Environmental testing (see 3.6).

4.4.3.1 High temperature. Each assembly shall be placed in a test chamber. The temperature of the chamber shall be raised to 160 ± 5°F. and held for four (4) hours or until the temperature stabilizes (see 6.4). The temperature shall then be lowered to 130 ± 5°F. and held at this temperature for four (4) hours or until the temperature has stabilized. At the end of the soak period and while still at 130 ± 5°F., the assembly shall comply with 3.5 of the appropriate appendix.

4.4.3.2 Low temperature. Each assembly shall be placed in a test chamber. The temperature of the chamber shall be lowered to - 65 ± 5°F. and held for four (4) hours or until the temperature stabilizes (see 6.4). The temperature shall then be raised to 0 ± 5°F. and held for four (4) hours or until the temperature stabilizes. At the end of the soak period and while still at 0 ± 5°F., the assembly shall comply with 3.5 of the appropriate appendix.

4.4.3.3 Vibration. Place the programmer assembly in its shipping container on a suitable fixture and vibrate in the vertical axis according to the schedule in Table I. The shipping container shall be closed and all latches fastened in a normal manner.

4.4.3.4 Transportation shock. The programmer assembly in its shipping container shall be subjected to three shocks of 30 ± 5 g in each direction of each of three mutually perpendicular axes for a total of 18 shocks. The duration of shock shall be 11 ± 2 ms as measured between points which are 10 percent of the peaks. The time to peak shall be 5.5 ± 1 ms. Following the test, the assembly shall comply with 3.5 of the appropriate appendix. The assembly shall be examined for damage resulting from the shock test.

4.4.3.5 Humidity. The 6A1 programmer with the cover removed shall be placed in a humidity test chamber and subjected to a 72 hour humidity test. The temperature in the chamber is to be maintained at 80 ± 20°F. and the relative humidity at 95% + 4% - 5%. The water used in the control of the relative humidity shall be distilled, having a pH value

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of between 6.5 and 7.5 at 77°F. The pH value shall be adjusted by the use of chemically pure hydrochloric acid or chemically pure sodium hydroxide. After 72 hours, the assembly shall be removed from the chamber. Moisture may be removed by blowing dry filtered compressed air over the unit for 5 minutes. There shall be no evidence of deterioration and the unit shall comply with the requirements of 3.5.2 of Appendix A.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The test programmer and modules shall be packaged in accordance with dwgs listed in Table VI using the specified moisture barrier.

5.1.2 Level C. The test programmer and modules shall be packaged in accordance with dwgs listed in Table VI except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The test programmer and modules shall be packed in accordance with dwgs listed in Table VI.

5.3 Marking. Marking shall be as specified in accordance with dwgs listed in Table VI.

Table VI

No.	Assembly	Packing Dwg.
6A1	Test Programmer Assembly	8883558
6A1A1	Baro Servo Amplifier Assembly	8883506
6A1A2	Timer Assembly, Precision Leaf Switch	8883513
6A1A3	Voltage Source Assembly, Reference	8883512
6A1A4	Voltage Comparator Assembly	8883511
6A1A5	Programmer Step Circuit Assembly	8883510
6A1A6	Unijunction Timer Assembly	8883509
6A1A7	Relay Module Assembly #1	8883508
6A1A8	Relay Module Assembly #2	8883507
6A1A9	Programmer Assembly (Main)	8883505
6A1A10	Programmer Assembly (Continuity)	8883505
6A1A11	Programmer Assembly (Open Circuit)	8883505
6A1A12	Programmer Assembly (T.D.D.)	8883505
6A1A13	Panel Assembly	MIL-STD-1169
6A1A14	Rack Assembly	MIL-STD-1169

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

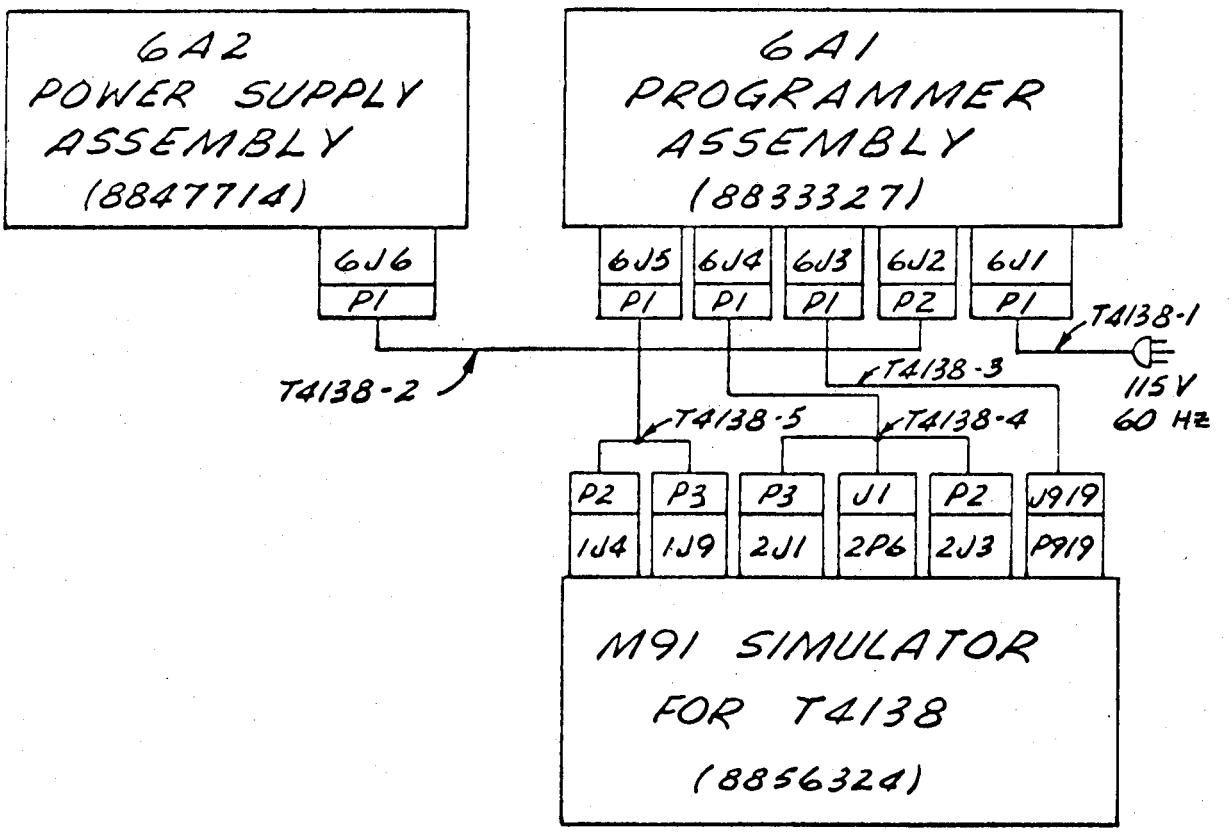
6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2085 dated 11 May 1966.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A081

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APPENDIX A
TEST PROGRAMMER ASSEMBLY
(6A1)

The following sections apply specifically to the test programmer assembly. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.

3.5.1 Calibration capability. The 6A1 Programmer Assembly shall be capable of calibration using Unit A of Load Box, Electrical T4165, with calibration procedures TB ORD 1026/121. Calibration shall be performed as specified in 4.4.1.

3.5.2 Operational limits.

3.5.2.1 Starting procedures. The POWER light and the START light shall light when the POWER switch is operated to the ON position. Depressing the lenses of the AK BATTERIES, AK FAILURE, TESTER FAILURE, TESTING and GO lights shall cause them to turn on momentarily when the assembly is connected as specified in 4.4.2.1.

3.5.2.2 Homing and self test. Depressing the START light button shall turn off the START light and it shall remain off for the duration of testing. The assembly shall then home all stepping switches and the Precision Leaf Switch Timer, if not already in the home position, and thence automatically proceed into the self test mode of operation when tested as specified in 4.4.2.2. Completion of the homing process shall occur within four (4) minutes after depressing the START light button and shall be signified by the lighting of the TESTING light and appearance of TEST NUMBER 001. During self test, the test numbers shall appear rapidly in the order designated in Table I, Column A. Self test is satisfactory if no failure occurs prior to and including the display of TEST NUMBER 021; however, the assembly will automatically proceed into the missile test mode and stop at TEST NUMBER 601, at which point AK FAILURE light shall light within 1.5 seconds. The TESTING light shall be lit during self test and remain lit for all tests.

3.5.2.3 AK Section failure limits. The AK FAILURE light shall turn on within the time specified in Table I, Column C, for each TEST NUMBER specified in Table 1, Column B, unless otherwise specified in Table I, Column H when tested as specified in 4.4.2.3. The OVERRIDE switch shall be ineffective until a failure is indicated. The GO light shall not be lit at the end of the test. The TEST NUMBERS shall appear in the order specified in Table I, Column B.

3.5.2.4 AK Section passing limits. The assembly shall step through the AK section tests when tested as specified in 4.4.2.4. At the end of the test the GO light shall light.

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3.5.2.5 Ledex step signal. The assembly shall provide a signal for stepping Ledex switches. The stepping signal shall be a repetitive On, Off signal of 28 ± 4 volts direct current (vdc) for 50 ± 10 milliseconds (ms) and 0 ± 1 vdc for 50 ± 10 ms and shall have a fall time of less than 5 ms when tested as specified in 4.4.2.5.

3.5.2.6 Precision leaf switch timer. The assembly shall schedule the application of B+ and filament voltages both high and low for each, for operation of electronic equipment within the AK section. When tested as specified in 4.4.2.6, the voltage across load 1 shall be - 2.5 ± 0.5 vdc at t = six (6) seconds. At t = fifteen (15) ± 2 seconds the voltage shall increase to - 6 ± 0.12 vdc and the test programmer assembly shall automatically advance one test number to 804. At TEST NUMBER 804 the voltage across load 3 shall be 330 ± 6.6 vdc prior to an AK FAILURE indication. The - 6 ± 0.12 vdc and 330 ± 6.6 vdc across loads 1 and 3 respectively shall be maintained while the assembly is advanced through TEST NUMBER 812 except that the voltage across load 3 shall be absent when the failure light is lit. At TEST NUMBER 813, the voltage across load 1 shall increase to - 7 ± 0.14 vdc and the voltage across load 3 shall increase to 380 ± 7.4 vdc and these voltages shall be maintained while the assembly is advanced to position 825, except that the voltage across load 3 shall be removed when the K failure light is lit. With the assembly set at position 833, the voltage across load 2 shall be - 2.5 ± 0.5 vdc at t = 126 seconds. At t = 135 ± 2 seconds, the voltage shall increase to - 6 ± 0.12 vdc and the test programmer shall automatically advance to TEST POSITION 834. At TEST NUMBER 834 the voltage across load 4 shall be 330 ± 6.6 vdc prior to AK FAILURE indication. The - 6 ± 0.12 vdc and 330 ± 6.6 vdc voltages shall be maintained while the assembly is advanced through TEST NUMBER 842 except that the voltage across load 4 shall be removed when the failure light is lit. At TEST NUMBER 843 the voltage across load 2 shall increase to - 7 ± 0.14 vdc and the voltage across load 4 shall increase to 380 ± 7.4 vdc and these voltages shall be maintained while the assembly is advanced to position 855, except that the voltage across load 4 shall be removed when failure light is lit. At t = 240 seconds, there shall be no voltages present across any of the loads.

3.5.2.7 Servo drive supplies. The test programmer assembly shall furnish 400 cycle power for the operation of a servo motor. When tested as specified in 4.4.2.7, the voltage across load 1 shall be 115 ± 10 volts, 400 ± 20 cycles per second (Hz). The voltage across load 2 shall be capable of being varied over the range of 30 to 100 volts, 400 ± 20 Hz.

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4.3.2.1 Classification of characteristics.

4.3.2.1.1 Test programmer assembly (6A1) (see dwg 8833327).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location of components	Visual
102.	Incorrect connector orientation and location	Visual
103.	Missing component	Visual
104.	Defective latches	Manual
105.	Objective evidence relative to material improper or missing (see 3.1)	Visual
106.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual
202.	Illegible or improper marking	Visual
203.	Improper finish or color	Visual

4.3.2.1.2 Preparation for delivery (see dwg 8883558).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Packing material and packaging not proper or missing	Visual
102.	Marking not in accordance with dwg	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.

4.4.1 Calibration capability. Prior to operational limits testing as specified in this appendix, the 6A1 programmer shall be calibrated using Unit A of Load Box, Electrical T4165, with calibration procedure TB ORD 1026/121; or acceptable equivalent test equipment and procedures upon obtaining prior Government approval.

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4.4.2 Operational limits.

4.4.2.1 Starting procedure. The test programmer assembly, with AK cables connected to the appropriate jacks, shall be connected to a power supply equivalent to that specified in MIL-P-48725(PA) and 115 volt 60 Hz applied after the following connections have been made:

Jumper Number	From	To
1	T4138-3, J919 pin 3	T4138-3, J919 pin 36
2	T4138-4, P2 pin V	T4138-4, P3 pin B
3	T4138-5, P2 pin $\overline{Y}$	T4138-5, P2 pin Z
4	T4138-5, P3 pin Y	T4138-5, P3 pin Z
5	T4138-5, P2 pin h	T4138-5, P2 pin P

Connect the following resistors to the pins designated:

4000 ± 10% ohm 5 watt - between T4138-4, J919 pins 11 and 17.
680 ± 1% ohm 1/2 watt - between T4138-5, P3 pins Y and h.
680 ± 1% ohm 1/2 watt - between T4138-5, P2 pins $\overline{Y}$ and $\overline{h}$.
220 ± 5% ohm 1/2 watt - between T4138-3, pin 15 to T4138-3 pin 22.

Jumper wires connecting the following pins together on T4138-5: P2V, P2BB, P2p, P3h, P3V, P3p P3BB.
Jumper wires connecting the following pins together on T4138-5: P3EE, P3e, P3v.
Jumper wires connecting the following pins together on T4138-5: P2EE, P2e, P2v.

4.4.2.2 Homing and self test. With the assembly connected as specified in 4.4.2.1, depress the START light button momentarily to initiate the homing process. When the homing process is completed, the START light button will return to its original position and the assembly will automatically proceed into the self test mode of operation. The assembly shall meet the requirements of paragraph 3.5.2.2.

4.4.2.3 AK Section failure limits. Remove power from the test programmer and physically remove the Precision Leaf Switch Timer Assembly (6A1A2). Install a jumper from 6A1A14J6 pin H to 6A1A14J6 pin C. Install a potentiometer as designated in Table I, Column H, for AK SECTION TEST NUMBER 501 and adjust to prevent the test programmer assembly from stepping. With the assembly connected as specified in 4.4.2.1, depress the START light button, and allow the assembly to complete self test and to fail at position 501. Install a no-step parameter meeting the requirements specified in Table I, Column G, for TEST NUMBER 601 between the pins designated in Table 1, Columns D and E. When the AK FAILUTE light lights at TEST NUMBER 501, after a 58 second delay depress the OVERRIDE

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button to advance the TEST PROGRAMMER to TEST NUMBER 601. Proceed through the remaining AK SECTION TEST NUMBERS by sequentially inserting no-step parameters as specified in Table 1, Column G. At each TEST NUMBER be certain that the no-step parameter for the next TEST NUMBER is inserted and the remarks in Table 1 Column H are completed prior to depressing the OVERRIDE button to advance the programmer. Where a resistance value is specified in Table 1, Column G, that resistance value is to be inserted between the pins specified in Table 1, Columns D and E. Where a voltage is specified, that voltage is the voltage applied to the pin specified in Column D with respect to the pin specified in Column E. The internal impedance of the voltage source shall not exceed 200 ohms per volt from DC to 400 Hz. Where upper and lower tolerances are designated, this test shall be repeated to assure compliance to both requirements. The assembly shall comply with the requirements of 3.5.2.3. The OVERRIDE button is ineffective at TEST NUMBERS 803, 813, 825, 833, 843 and 855 so the tester must be advanced through these test numbers by inserting step parameters meeting the requirements of Table 1, Column F. After checking the above referenced test numbers, it is necessary to insert the number of step parameters required to advance to the next test position and depress the START button. The test set will return to TEST NUMBER 501 and must be advanced to the desired TEST NUMBER by either inserting step parameters or by use of the OVERRIDE button. An alternate method may be used to step the programmer to the next respective position if desired. This is accomplished by shorting pin V to pin B on the calibration plug J80 and depressing the OVERRIDE button. The short from V to B on J80 must be removed after overriding in order for the failure lights to be operative.

4.4.2.4 AK Section passing limits. Remove power from the test programmer assembly and physically remove the Unijunction Timer (6A1A6). Repeat the procedure of 4.4.2.3 except use the step parameters as specified in Table 1, Column F. At each TEST NUMBER, the test programmer assembly shall be advanced by adjusting the no-step parameter to meet the step parameter rather than by use of the OVERRIDE button. The test programmer assembly shall meet the requirements of 3.5.2.4.

4.4.2.5 Ledex step. Connect the test programmer assembly as specified in 4.4.4.4 and install components as tabulated:

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Mode No.	Component	Connector	From	To
I	3 K ± 1% ohms resistor	T4138-3J919	50	68
	Ledex stepping switch coil*	T4138-3J919	24	18
II	3 K ± 1% ohms resistor	T4138-3J919	70	68
	Ledex stepping switch coil*	T4138-3J919	31	18

*Ledex stepping switch per dwg 10970599.

With the test programmer assembly at TEST NUMBER 801 and components in Mode I installed, there shall be a pulsing voltage applied to the Ledex coil by the test programmer assembly. With the test programmer assembly at TEST NUMBER 831 and components in Mode II installed, there shall be a pulsing voltage applied to the Ledex coil by the test programmer assembly. The pulsing voltage at T4138-3 connector J919 pins 24 or 31 as applicable with respect to pin 18, when recorded on a Honeywell 906 Visicorder, shall be as specified in 3.5.2.5. Isolate instrumentation from power ground. This test may be conducted in conjunction with 4.4.2.4.
(CAUTION: Pulsing voltage shall not be applied to the Ledex coil for more than 15 seconds maximum. A one minute non-operative internal shall follow each pulsing period.)

4.4.2.6 Precision leaf switch timer assembly. Connect the test programmer assembly as specified in 4.4.2.2 (6A1A2 and 6A1A6 installed in the unit) and install the loads and jumpers tabulated below:

Load No.	Load	From	To
1	0.6 ohms ± 10% 50 W	T4138-5 P3EE	T4138-5 P3V
2	0.6 ohms ± 10% 50 W	T4138-5 P2EE	T4138-5 P2V
3	1.8K ± 10% 80 W	T4138-5 P3AA	T4138-5 P3V
4	1.8 K ± 10% 80 W	T4138-5 P2AA	T4138-5 P2V
5	1.5K ± 1% 1/2 W	T4138-5 P3EE	T4138-5 P3$\overline{y}$
6	1.5K ± 1% 1/2 W	T4138-5 P2EE	T4138-5 P2$\overline{y}$
7	Jumper	T4138-5 P3$\overline{t}$	T4138-5 P3$\overline{h}$
8	Jumper	T4138-5 P2$\overline{t}$	T4138-5 P2$\overline{h}$
9	Jumper	T4138-5 P3$\overline{u}$	T4138-5 P3$\overline{h}$
10	Jumper	T4138-5 P2$\overline{u}$	T4138-5 P2$\overline{h}$
11	Jumper	T4138-5 P3$\overline{x}$	T4138-5 P3$\overline{h}$
12	Jumper	T4138-5 P2$\overline{x}$	T4138-5 P2$\overline{h}$

Depress the START light button and allow the test programmer assembly to complete self test. Advance the assembly to TEST NUMBER 803 by using the OVERRIDE button when necessary.

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Testing time (t) is the time the TESTING light is lit. Testing time (t = 0) begins the instant the test programmer steps to TEST NUMBER 803. Advance the assembly thrugh the 800 series of test numbers by use of the OVERRIDE button where necessary. Voltages across the loads shall meet the requirements of 3.5.2.6. The voltage polarity specified in 3.5.2.6 shall be the polarity of the voltage on pin EE or pin AA with respect to pin V of their respective connectors.

4.4.2.7 Servo drive supplies. Connect the test programmer assembly as specified in 4.4.2.4. Connect an 1100 ± 10% ohm 20 watt resistor, Load 1, from T4138-3, P919 pins 4 and 11. Load 2 is the 4000 ohm resistor installed under 4.4.2.1. Apply power to the test programmer assembly. The voltage across Load 1 shall meet the requirements of 3.5.2.7. Adjust the potentiometer, referenced in Table 1, Column H, for TEST NUMBER 501, to supply a step signal, and insert a no-step parameter for TEST NUMBER 601. Depress the START button and allow the assembly to proceed to 501. Following lighting of the AK FAILURE light at TEST NUMBER 501 the voltage across Load 2 shall meet the requirements of 3.5.2.7 when the potentiometer is varied over its entire range.

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Table I

A	B	C	D From		E To		F	G	H
Self Test Number	AK Section Test Number	Time to Failure (seconds) (nominal)	External Cable & Connector	Pin	External Cable & Connector	Pin	Step Requirement (Ohms or Volts)	No Step Requirement (Ohms or Volts)	Remarks
	(Baro)								
001	501	58							With a calibrated 30 K ± 5% potentiometer connected between T4138-3 J919 pin 28 and T4138-s J919 pin 35 and the wiper connected to T4138-3 J919 pin 41, the assembly shall advance to the next test position when the resistance from T4138-3 J919 pin 41 to T4138-3 J919 pin 28 is slowly varied between 28.5 and 29.2 percent of the total pot resistance. With the calibrated pot set outside the range specified, the tester shall not step to the next test position.
	(Continuity)								
101	601	1.5	4-P2	c	4-P2	b	R≤1.5	R≧2.2	
102	602	1.5	4-P2	Y	4-P2	a	R<1.5	R≧2.2	Jumper 4J1Y to 4J1a
103	603	1.5	4-P2	G	4-P3	n	R<1.5	R≧2.2	
104	604	1.5	4-P2	G	4-P3	k	R<1.5	R≧2.2	
105	605	1.5	4-P2	L	4-P2	BB	R<1.5	R≧2.2	Jumper 4J1L to 4J1BB
106	606	1.5	4-P2	p	4-P2	F	R<1.5	R≧2.2	Jumper 4J1p to 4J1F
107	607	1.5	4-P2	y	4-P2	DD	R<1.5	R≧2.2	
108	608	1.5	4-P2	d	4-P2	E	R<1.5	R≧2.2	Jumper 4J1d to 4J1E
109	609	1.5	4-P2	h	4-P2	H	R<1.5	R≧2.2	Jumper 4J1h to 4J1H
110	610	1.5	4-P2	x	4-P3	K	R<1.5	R≧2.2	

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Table I (continued)

A	B	C	D		E		F	G	H
111	611	1.5	4-P2	CC	4-P2	a	R<1.5	R≧2.2	
112	612	1.5	3-J919	5	4-P2	d	R<1.5	R≧2.2	
113	613	1.5	4-P2	r	4-P2	T	R<1.5	R≧2.2	Jumper 4J1T to 4J1i
114	614	1.5	4-P2	i	4-P2	V	R<1.5	R≧2.2	Jumper 4J1r to 4J1V
115	615	1.5	3-J919	63	4-P2	R	R<1.5	R≧2.2	
116	616	1.5	4-P3	H	4-P2	z	R<1.5	R≧2.2	
117	617	1.5	4-P2	N	4-P2	g	R<1.5	R≧2.2	Jumper 4J1N to 4J1g and 4P2S to 4P2P and 4J1S to 4J1P
118	618	1.5	4-P3	V	4-P2	K	R<1.5	R≧2.2	
119	619	1.5	4-P3	U	4-P2	M	R<1.5	R≧2.2	
120	620	1.5	4-P2	J	4-P2	W	R<1.5	R≧2.2	Jumper 4J1J to 4J1W
121	621	1.5	4-P2	AA	4-P2	EE	R<1.5	R≧2.2	Jumper 4J1AA to 4J1EE
122	622	1.5	4-P2	FF	4-P3	L	R<1.5	R≧2.2	
123	623	1.5	4-P2	FF	4-P3	M	R<1.5	R≧2.2	
124	624	1.5	4-P2	GG	4-P2	w	R<1.5	R≧2.2	Jumper 4J1GG to 4J1w
125	625	1.5	4-P2	m	4-P2	t	R<1.5	R≧2.2	Jumper 4J1m to 4J1t
002	(Baro) 502	58							With the calibrated pot connected as specified in test number 501 the assy shall advance to the next test position when the resistance from T4138-3 J919 pin 35 is slowly varied between 3.8 and 4.4 percent of the total pot resistance. With the calibrated pot set outside the specified limits the test set shall not step to the next test position.

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Table I (continued)

A	B	C	D	E		F	G	H
201	(open circuit) 701	1.5	4-P2	FF 4-P2	r	R>2 Meg	R≦1 Meg	
202	702	1.5	4-P2	K 4-P2	r	R>2 Meg	R≦1 Meg	
203	703	1.5	4-P2	M 4-P2	r	R>2 Meg	R≦1 Meg	
204	704	1.5	4-P2	z 4-P2	r	R>2 Meg	R≦1 Meg	
205	705	1.5	4-P2	x 4-P2	r	R>2 Meg	R≦1 Meg	
206	706	1.5	4-P2	c 4-P2	r	R>2 Meg	R≦1 Meg	
207	707	1.5	4-P2	FF 4-P2	j	R>2 Meg	R≦1 Meg	
208	708	1.5	4-P2	K 4-P2	j	R>2 Meg	R≦1 Meg	
209	709	1.5	4-P2	M 4-P2	j	R>2 Meg	R≦1 Meg	
210	710	1.5	4-P2	z 4-P2	j	R>2 Meg	R≦1 Meg	
211	711	1.5	4-P2	x 4-P2	j	R>2 Meg	R≦1 Meg	
212	712	1.5	4-P2	c 4-P2	j	R>2 Meg	R≦1 Meg	
213	713	1.5	4-P2	FF 4-P2	E	R>2 Meg	R≦1 Meg	Jumper 4J1E to 4J1d
214	714	1.5	4-P2	K 4-P2	E	R>2 Meg	R≦1 Meg	
215	715	1.5	4-P2	M 4-P2	E	R>2 Meg	R≦1 Meg	
216	716	1.5	4-P2	z 4-P2	E	R>2 Meg	R≦1 Meg	
217	717	1.5	4-P2	x 4-P2	E	R>2 Meg	R≦1 Meg	
218	718	1.5	4-P2	c 4-P2	E	R>2 Meg	R≦1 Meg	
219	719	1.5	4-P2	Z 4-P2	H	R>2 Meg	R≦1 Meg	Jumper 4J1H to 4J1h
220	720	1.5						Spare position. Assembly must step to next position.
221	721*	1.5	4-P2	Z 4-P2	G	R>2 Meg	R≦1 Meg	*2 second failure
222	722	1.5	4-P2	CC 4-P2	a	R>2 Meg	R≦1 Meg	
223	723	1.5						Spare position. Assembly must step to next test position.

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Table I (continued)

A	B	C	D		E		F	G	H
224	724	1.5							Spare position. Assembly must step to next test position.
225	725	1.5							Spare position. Assembly must step to next test position.
	(Pneumatic system)								
003	503*	20							*23 second failure. Leak check on pneumatic system. Pneumatic hose on 6A2 must be closed to provide a step signal. There shall be a potential difference of -56 +4 volts on T4138-3 J919 pin 53 with respect to the following pins: T4138-3 J919 pin 1 T4138-3 J919 pin 8 T4138-3 J919 pin 14 T4138-3 J919 pin 21 T4138-3 J919 pin 40 T4138-3 J919 pin 27
	(TDD #1)								
301	801	1.5	3-J919	50	3-J919	69	2.97K<R<3.03K	2.4K>R>5K	
302	802	1.5	3-J919	50	5-P3	N	2.97K<R<3.03K	2.4K>R>5K	
303	803	20	5-P3	Y	5-P3	h	-1.93<E<-1.77	-1.40<E<-2	The tester failure light shall come on when no step parameters are inserted and 6A1A6 is in the circuit.
304	804	58	5-P3	x	5-P3	h	2.45>E>1.60	1.40>E>2.60	
305	805	20	5-P3	t	5-P3	h	0<E<1	-0.70>E>1.55	
306	806	1.5	5-P3	u	5-P3	h	0<E<0.30	-0.70>E>0.70	
307	807	1.5	5-P3	t	5-P3	h	2.45>E>1.60	1.40>E>2.60	
308	808	1.5	5-P3	u	5-P3	h	0<E<0.30	-0.70>E>0.70	

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Table I (continued)

A	B	C	D		E		F	G	H
309	809	1.5	3-J919	50	3-J919	68	2.97K<R<3.03K	2.4K>R>5K	
310	810	1.5	5-P3	u	5-P3	h	2.45>E>1.60	1.40>E>2.60	
311	611	1.5							Provide connections between the following pins in the sequence shown. The assembly shall advance to the next test position when the final connection is made. Connect 4P2Y to 4P2s and 4P2u. Disconnect 4P2Y from 4P2u only and connect to 4P2HH. Disconnect 4P2Y from 4P2HH and 4P2s and connect to 4P2u. Connect 4P2Y to 4P2s.
312	812	1.5	3-J919	50	5-P3	N	2.97K<R<3.03K	2.4K>R>5K	
313	813	1.5	5-P3	y	5-P3	h	-2.18<E<-2.12	-2<E<-2.60	
314	814	1.5	5-P3	x	5-P3	h	2.45>E>1.60	1.40>E>2.60	
315	815	20	5-P3	t	5-P3	h	0<E<1	-0.70>E>1.55	There is an 8-12 second delay before assembly can be stepped off this position.
316	816	1.5	5-P3	u	5-P3	h	0<E<0.30	-0.70>E>0.70	
317	817	1.5	5-P3	t	5-P3	h	2.45>E>1.60	1.40>E>2.60	
318	818	1.5	5-P3	u	5-P3	h	0<E<0.30	-0.70>E>0.70	
319	819	1.5	3-J919	50	3-J919	68	2.97K<R<3.03K	2.4K>R>5K	
320	820	1.5	5-P3	u	5-P3	h	2.45>E>1.60	1.40>E>2.60	
321	821	1.5	3-J919	50	3-J919	69	2.97K<R<3.03K	2.4K>R>5K	
322	822	1.5							Spare position. The assembly shall advance to the next test position.

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Table I (continued)

A	B	C	D		E		F	G	H
323	823	1.5							Spare position. The assembly shall advance to the next test position.
324	824	1.5							Spare position. The assembly shall advance to the next test position.
325	825*	58	5-P3	y	5-P3	h	-0.30<E<0.30	-0.80>E>0.60	*61 second failure
3--	8--								Non-usable position. Assembly shall advance to the next test position.
331	(TDD #21) 831*	1.5	3-J919	70	3-J919	69	2.97K R 3.03K	2.4K>R>5K	*2 second failure
332	832	1.5	3-J919	70	5-P2	N	2.97K R 3.03K	2.4K>R>5K	
333	833	20	5-P2	y	5-P2	h	1.93 E -1.77	-1.40<E<-2	The Tester Failure light shall come on when no step parameters are inserted and 6A1A6 is in the circuit.
334	834	58	5-P2	x	5-P2	h	2.45>E>1.60	1.40>E>2.60	
335	835	20	5-P2	t	5-P2	h	0<E<1	0.70>E>1.55	
336	836	1.5	5-P2	u	5-P2	h	0<E<0.30	-0.70>E>0.70	
337	837	1.5	5-P2	t	5-P2	h	2.45>E>1.60	1.40>E>2.60	
338	838	1.5	5-P2	u	5-P2	h	0<E<0.30	-0.70>E>0.70	
339	839	1.5	3-J919	70	3-J919	68	2.97K<R<3.03K	2.4K>R>5K	
340	840	1.5	5-P2	u	5-P2	h	2.45>E>1.60	1.40>E>2.60	
341	841	1.5							Provides connections between the following pins in the sequence shown. The assembly shall step to the next test position when the final connector is made. Connect

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Table I (continued)

A	B	C	D		E		F	G	H
									4P2Y to 4P2n and 4P2u. Disconnect 4P2Y from 4P2n only and connect to 4P2i. Disconnect 4P2Y from 4P2i and connect to 4P2n.
342	842	1.5	3-J919	70	5-P2	H	2.97K<R<3.03K	2.4K>R>5K	
343	843	1.5	5-P2	y	5-P2	h	-2.18<E<-2.12	-2<E<-2.60	
344	844	1.5	5-P2	x	5-P2	h	2.45>E>1.60	1.40>E>2.60	
345	845	20	5-P2	t	5-P2	h	0<E<1	-0.70>E>1.55	There is an 8-12 second delay before assembly can be stepped off this position.
346	846	1.5	5-P2	u	5-P2	h	0<E<0.30	-0.70>E>0.70	
347	847	1.5	5-P2	t	5-P2	h	2.45>E>1.60	1.40>E>2.60	
348	848	1.5	5-P2	u	5-P2	h	0<E<0.30	-0.70>E>0.70	
349	849	1.5	3-J919	70	3-J919	68	2.97K<R<3.03K	2.4K>R>5K	
350	850	1.5	5-P2	u	5-P2	h	2.45>E>1.60	1.40>E>2.60	
351	851	1.5	3-J919	70	3-J919	69	2.97K<R<3.03K	2.4K>R>5K	
352	852	1.5							Spare position. The assembly shall advance to the next test position.
353	853	1.5							Spare position. The assembly shall advance to the next test position.
354	854	1.5							Spare position. The assembly shall advance to the next test position.
355	855*	58	5-P2	y	5-P2	h	-0.30<E<0.30	-0.80>E>0.60	*61 second failure

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Table I (continued)

A	B	C	D		E		F	G	H
004	(Capacitor) 504	1.5							Supplementary self check to insure that a charging voltage was applied to the capacitor tested in the following test number.
005	505*	1.5	3-J919	1	3-J919	53	37.5<E<58.5	35>E>62	There should be no potential difference between the connections designated in Columns D and E. *2 second failure
006	506	1.5							Supplementary self check to insure that a charging voltage was applied to the capacitor tested in the following test number.
007	507*	1.5	3-J919	8	3-J919	53	37.5<E<58.5	35>E>62.5	There should be no potential difference between the connections designated in Columns D and E. *2 second failure
008	508	1.5							Supplementary self check to insure that a charging voltage was applied to the capacitor tested in the following test number.
009	509*	1.5	3-J919	14	3-J919	53	37.5<E<58.5	35>E>62	There should be no potential difference between the connections designated in Columns D and E. *2 second failure

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Table I (continued)

A	B	C	D		E		F	G	H
010	510	1.5							Supplementary self check to insure that a charging voltage was applied to the capacitor tested in the following test number.
011	511*	1.5	3-J919	21	3-J919	53	37<E<58.5	35>E>62	There should be no potential difference between the connections designated in Columns D and E. *2 second failure
012	512	1.5							Supplementary self check to insure that a charging voltage was applied to the capacitor tested in the following test number.
013	513*	1.5	3-J919	40	3-J919	53	37<E<58.5	35>E>62	There should be no potential difference between the connections designated in Columns D and E. *2 second failure
014	514	1.5							Supplementary self check to insure that a charging voltage was applied to the capacitor tested in the following test number.
015	515	1.5	3-J919	27	3-J919	53	37<E<58.5	35>E>62	There should be no potential difference between the connections designated in Columns D and E.
	(Continuity)								
401	901	1.5	4-P2	c	4-P2	b	R<1.5	R≥2.2	
402	902	1.5	4-P2	Y	4-P2	a	R<1.5	R≥2.2	Jumper 4J1Y to 4J1a
403	903	1.5	4-P2	G	4-P3	n	R<1.5	R≥2.2	
404	904	1.5	4-P2	G	4-P3	k	R<1.5	R≥2.2	

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Table I (continued)

A	B	C	D		E		F	G	H
405	905	1.5	4-P2	L	4-P2	BB	R<1.5	R≧2.2	Jumper 4J1L to 4J1BB
406	906	1.5	4-P2	p	4-P2	F	R<1.5	R≧2.2	Jumper 4J1p to 4J1F
407	907	1.5	4-P2	y	4-P2	DD	R<1.5	R≧2.2	
408	908	1.5	4-P2	d	4-P2	E	R<1.5	R≧2.2	Jumper 4J1d to 4J1E
409	909	1.5	4-P2	h	4-P2	H	R<1.5	R≧2.2	Jumper 4J1h to 4J1H
410	910	1.5	4-P2	x	4-P3	K	R<1.5	R≧2.2	
411	911	1.5							Corresponds to Baro contact test 611 but since the baro has not been reset to high HOB the contacts are open and an automatic override signal is supplied by the assembly. The assembly shall advance to the next test number.
412	912*	1.5	3-J919	5	4-P2	d	R<1.5	R≧2.2	*2 second failure
413	913	1.5	4-P2	r	4-P2	T	R<1.5	R≧2.2	Jumper 4J1j to 4J1T
414	914	1.5	4-P2	J	4-P2	V	R<1.5	R≧2.2	Jumper 4J1r to 4J1V
415	915	1.5	3-J919	63	4-P2	R	R<1.5	R≧2.2	
416	916	1.5	4-P3	H	4-P2	z	R<1.5	R≧2.2	
417	917	1.5	4-P2	N	4-P2	g	R<1.5	R≧2.2	Jumper 4J1N to 4J1g and 4P2S to 4P2P and 4J1S to 4J1P
418	918	1.5	4-P3	V	4-P2	K	R<1.5	R≧2.2	
419	919	1.5	4-P3	U	4-P2	M	R<1.5	R≧2.2	
420	920	1.5	4-P2	J	4-P2	W	R<1.5	R≧2.2	Jumper 4J1J to 4J1W
421	921	1.5	4-P2	AA	4-P2	EE	R<1.5	R≧2.2	Jumper 4J1AA to 4J1EE
422	922	1.5	4-P2	FF	4-P3	L	R<1.5	R≧2.2	
423	923	1.5	4-P2	FF	4-P3	M	R<1.5	R≧2.2	

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Table I (continued)

A	B	C	D		E		R	G	H
424	924	1.5	4-P2	GG	4-P2	w	R<1.5	R≥2.2	Jumper 4J1GG to 4J1w
425	925	1.5	4-P2	m	4-P2	t	R<1.5	R≥2.2	Jumper 4J1m to 4J1t
	(AK & TDD Failure)								
016	516	1.5	3-J919	9	3-J919	3	R<1K	R>43K	
017	517	1.5							Spare position. The assembly shall advance to the next test position.
	(Warhead Identification)								
018	518*	1.5	4-P3	S	3-J919	68	5.01K>R>4.99H	5.15<R<4.85R	*2 second failure
019	519	1.5	4-P3	P	3-J919	68	7.01K>R>6.99K	7.21<R<6.79K	
	(Spares)								
020	520	1.5							Spare position. The assembly shall advance to the next test position.
021	521	1.5							Spare position. The assembly shall advance to the next test position.

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APPENDIX B
BARO SERVO AMPLIFIER ASSEMBLY (6A1A1)

The following sections apply specifically to the baro servo amplifier assembly. In addition to the requirements of this appendix, the assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.

3.5.1 Calibration capability. The baro servo amplifier assembly (6A1A1) shall be calibrated so that the voltage across the motor control output (terminals R and P) shall be 33 ± 3.3 volts RMS with an input voltage of 30 ± 0.5 millivolts dc across terminals U and X. The calibration shall be performed in accordance with 4.4.1.

3.5.2 Operational limits. With the baro servo amplifier assembly (6A1A1) adjusted as per 3.5.1, the null test point voltage (terminals S and P) during the run-down shall be 0.90 ± 0.30 vdc and 0 ± 0.45 vdc at null conditions. The input differential (terminals V and X) shall be 55 millivolts dc or less at null. The system shall null without oscillation. The phase difference, θ p, between motor control output and motor line voltage, shall be 90° ± 20° while running in each direction. At null conditions θ p shall be a minimum of 29° when the test point voltage (S to P) is a maximum of 0.45 vdc.

3.5.2.1 Continuity and resistance. Resistance between the terminals listed below shall be within the specified values when tested as specified in 4.4.2.3.

Terminals		Resistance
From	To	(ohms)
P1CC	P1DD	0.10 max.
P1Z	P1DD	8.66K ± 1%
P1Z	P1AA	20K ± 1%
P1W	P1AA	1.24K ± 1%
*P1S	P1P	1.40K ± 2%

*Positive ohmmeter lead.

3.5.2.2 Insulation resistance. The insulation resistance shall be greater than 20 megohms between each of the following terminals and groups of terminals and all others listed and case when tested as specified in 4.4.2.4.

P1S, *P1BB, P1M and *P1W

*The following groups of terminals shall be connected together and each group tested as one terminal. Connect together for this test only.

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1. P1BB, P1Y, P1T
2. P1W, P1U, P1X

The following unused pins may be connected together and tested as one terminal.

P1A, P1B, P1C, P1D, P1E, P1F, P1H, P1J, P1K, P1L and P1V

3.5.3 Noise level. The output voltage with 0 input (jumper between terminals U and X) shall be less than 1 VRMS across terminals R and P. This test shall be performed in accordance with 4.4.3.

4.3.2.1 Classification of characteristics.
4.3.2.1.1 Baro servo amplifier (6A1A1) (see dwg 8847741).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connector with respect to alignment pins	Gage
102.	Defect solder joints	Visual/Manual
103.	Fastenings loose or missing	Visual/Manual
104.	Component missing	Visual
105.	Cracked or broken connectors or damaged pins	Visual
106.	Board mounting screws loose	Manual
107.	Objective evidence relative to material improper or missing (see 3.1)	Visual
108.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual
202.	Illegible or improper marking	Visual
203.	Improper finish or color	Visual

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4.3.1.1.2 Preparation for delivery (see 8883506).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Packing material and packaging not proper or missing	Visual
102.	Marking not in accordance with dwg	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.

4.4.1 Calibration capability. Connect the baro servo amplifier assembly to the circuit shown in Figure 1 and make the following adjustments.

4.4.1.1 Tach adjustment. Close S2; place S3 in position 1, S4 in position 2; and open S5. Adjust the 400 Hz sine wave generator to 1.6 + 0.06 volts rms from Y to BB. Adjust R40 until 20 + 0.5 volts rms is obtained across pins R and P.

4.4.1.2 Error signal adjustment. Close S1, open S2, place S3 in position 2, close S5 and place S6 in "Hi" position. Disconnect the 30K 10 turn potentiometer (R) from the gear train shaft and adjust it until a differential voltage of 30 ± 0.5 millivolts dc is obtained across pins U to X. Then open S1 and adjust the calibration potentiometer R36 until 33 ± 3.3 vrms is obtained across pins R and P. The assembly shall be calibrated at a temperature of 75° ± 10°F.

4.4.2 Operational limits.

4.4.2.1 Operational limits at null. Disconnect the 30K 10 turn potentiometer from the gear train shaft. Open S1 and S2; place S3 in position 2, S4 in position 1; close S5; and place S6 in "Hi" position. Adjust the potentiometer to obtain the test point voltage (pins S to P) of 0.45 vdc. Check the phase difference (θ p) between motor control output and motor line voltage using a dual beam oscilloscope to compare the wave form zero crossing points. Repeat the foregoing procedure except place S6 in "Lo" position. θ p shall conform to the requirements of 3.5.2.

4.4.2.2 Operational limits while running. Connect the 30K 10 turn potentiometer to the gear train shaft. Open S1 and S2; place S3 in position 2, S4 in position 1; close S5; and place S6 in "Hi" position. Check the phase difference (θ p) between motor control, output and motor line voltage using a dual beam oscilloscope to compare the wave form zero

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crossing points. Repeat the foregoing procedures except place S6 in "Lo" position. θ p shall conform to the requirements of 3.5.2. As the system approaches null condition, a visual check of the shaft shall be made for oscillations; the differential voltage (pins U to X) shall also be observed for evidence of oscillations. The input null (U to X) and the output null test point voltage (S to P) shall be in accordance with 3.5.2 at both "Hi" and "Lo" positions.

4.4.2.3 Continuity and resistance. Resistance between the terminals listed in 3.5.2.1 shall be measured using a digital ohmmeter of sufficient accuracy or its equivalent. Resistance under 1 ohm is to be measured employing a 4 wire system.

4.4.2.4 Insulation resistance. The insulation resistance between the terminals listed in 3.5.2.2 shall be measured using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma for this test.

4.3.3 Noise level. Open S1, close S2, place S3 in position 2, place S4 in position 2 and close S5. The output voltage at R to P shall be in accordance with 3.5.3.

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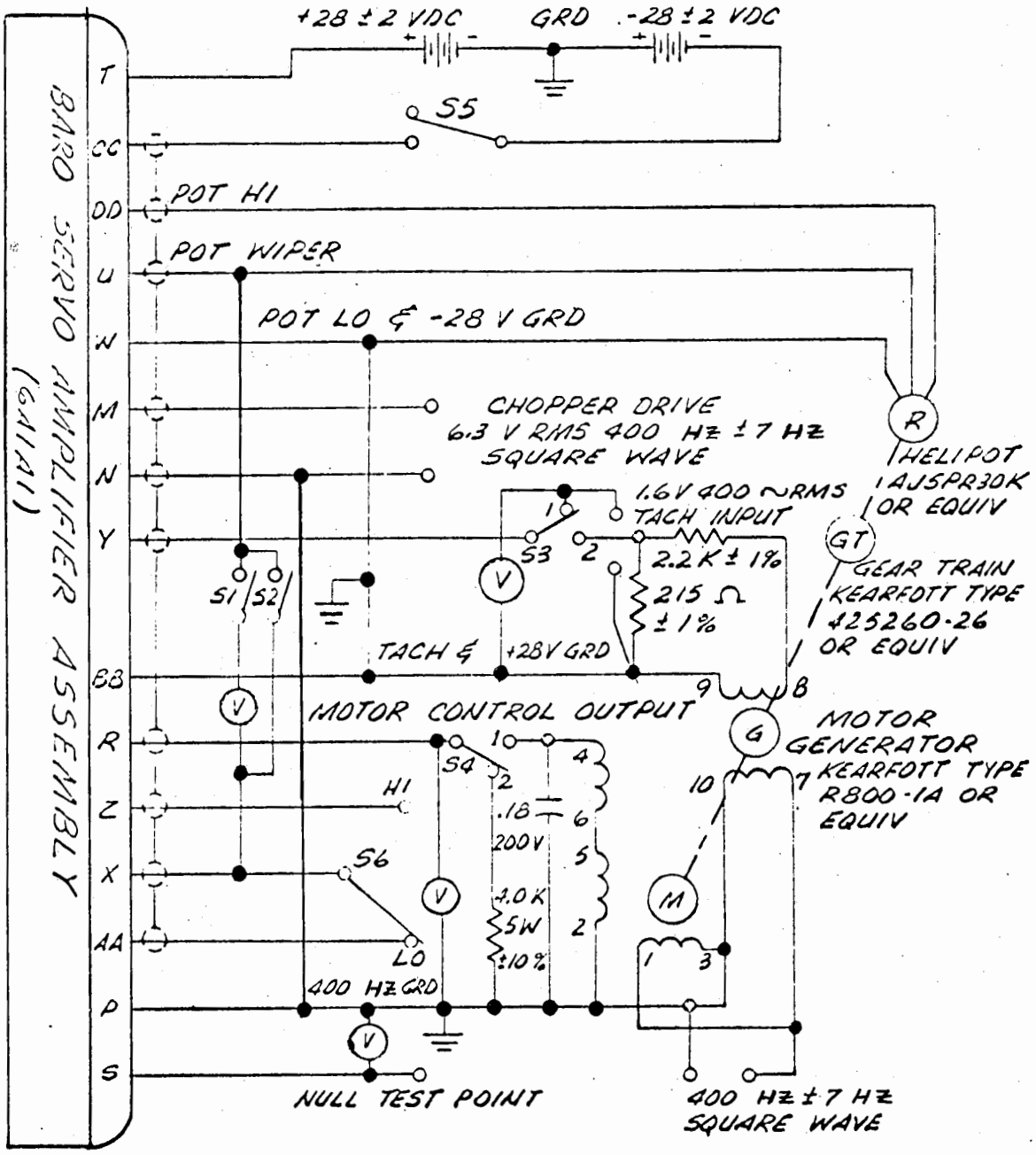


Figure I. Baro Servo Amplifier Assembly
Test Circuit

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APPENDIX C
TIMER, PRECISION LEAF SWITCH ASSEMBLY (6A1A2)

The following sections apply specifically to the precision leaf switch timer assembly. In addition to the requirements of this appendix, this assembly shall meet the requirements of MIL-T-48804(PA).

3.5 Operational characteristics.

3.5.1 Calibration capability. Not applicable.

3.5.2 Operational limits.

3.5.2.1 Motor operation. The time between successive switch closures as seen across terminals A and B shall be 240 ± 1 second when measured as specified in 4.4.2.1.

3.5.2.2 Closed circuit resistance. The resistance between terminals A and B, and between all others and terminal H during the "CLOSED" portion of the operating cycle for each switch shall not exceed 1 ohm maximum when measured as specified in 4.4.2.2.

3.5.2.3 Insulation resistance. During the "OPEN" portion of the cycle, the resistance (1) between terminal A and terminal B, (2) between terminal H and all other terminals, and (3) between all terminals and the case shall not be less than 20 megohms when tested as specified in 4.4.4.2. Unused terminals D, L, M and N shall be connected together and tested as one terminal.

3.5.2.4 Timing sequence. Timing for each of the six switch closures (ON) and openings (OFF) shall be as shown in Table 1. It is required that switch S1-C open at 116 ± 1 second and before switch S1-F closes (Note A) and switch S1-D open at 236 ± 1 second and before switch S1-E opens (Note B). It is also required that the time interval between closure of S1-F at 118 ± 1 second and the opening of S1-F or the closure of S1-E at 135 ± 1 second, whichever occurs last, shall not exceed 17.4 seconds (Note C). Times will be measured as specified in 4.4.2.4.

4.3.2.1 Classification of characteristics.

4.3.2.1.1 Timer, precision leaf switch assembly (6A1A2) (see dwg 8847809).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect connector orientation and location with respect to alignment pins	Gage

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102. Defective solder joints.....Visual/Manual
103. Fastenings loose or missing.....Visual/Manual
104. Component missing.....Visual
105. Cracked or broken connectors or damaged pins.....Visual
106. Board mounting screws loose.....Manual
107. Objective evidence relative to material improper or missing (see 3.1).....Visual
108. Serialization improper or missing (see 3.4).....Visual

Minor:
201. Evidence of poor workmanship (see 3.8).....Visual
202. Illegible or improper marking.....Visual
203. Improper finish or color.....Visual

4.3.2.1.2 Preparation for delivery (see 8883513).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Packing material and packaging not proper or missing	Visual
	102. Marking not in accordance with dwg	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.
4.4.1 Calibration capability. Not applicable.
4.4.2 Operational limits.
4.4.2.1 Motor operation. Closing S1 of the test circuit connects 115 ± 10 V 60 Hz power source across terminals J and K causing the timer motor to operate. Using a suitable test instrument, determine the time between successive switch closures across terminals A and B. The time shall be within limits specified in 3.5.2.1. The test equipment used on terminals A and B shall be limited to 50 ma maximum current at 32 volts dc resistive.

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4.4.2.2 Closed circuit resistance. Using Table I as a guide to switch closures, operate timer motor until the switch being tested is closed. Employing a four wire measuring system and limiting test equipment to a maximum of 50 ma current, the resistance shall be within the values given in 3.5.2.2.

4.4.2.3 Insulation resistance. Using Table I as a guide to switch openings, operate timer motor until zero time occurs (see 4.4.2.4) and all switches are open. Potential used shall be 500 ± 50 vdc with a current of 5 ma. Resistance values shall be as specified in 3.5.2.3.

4.4.2.4 Timing sequence (see Figure 2). Operate the timer motor. Using a suitable test instrument, record the opening (OFF) and closure (ON) times and the sequence of all six switch contacts listed in Table I and illustrated in Figure 1. It will be necessary to run the motor for at least 240 seconds in order to complete a full cycle of operation. Zero time is the time when switch S1-A opens. All other times of switch closures (ON) and openings (OFF) are measured from zero time. The timing sequence of the six switches shall comply with 3.5.2.4.

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Table I

Switch settings

Switch	Module Terminals	Timing ± 1 Second	
S1-A	A & B	Open	0 - 5 second
		Closed	5 - 240 second
S1-B	C & H	Open	0-75,195 - 240 second
		Closed	75 - 195 second
S1-C	P & H	Open	0-65,116 - 240 second
		Closed	65 - 116 second
S1-D	E & H	Open	0-185,236 - 240 second
		Closed	185 - 236 second
S1-E	F & H	Open	0-15,125-135,238 - 240 second
		Closed	15-125,135 - 238 second
S1-F	R & H	Open	0-5,15-118,135 - 240 second
		Closed	5-15,118 - 135 second

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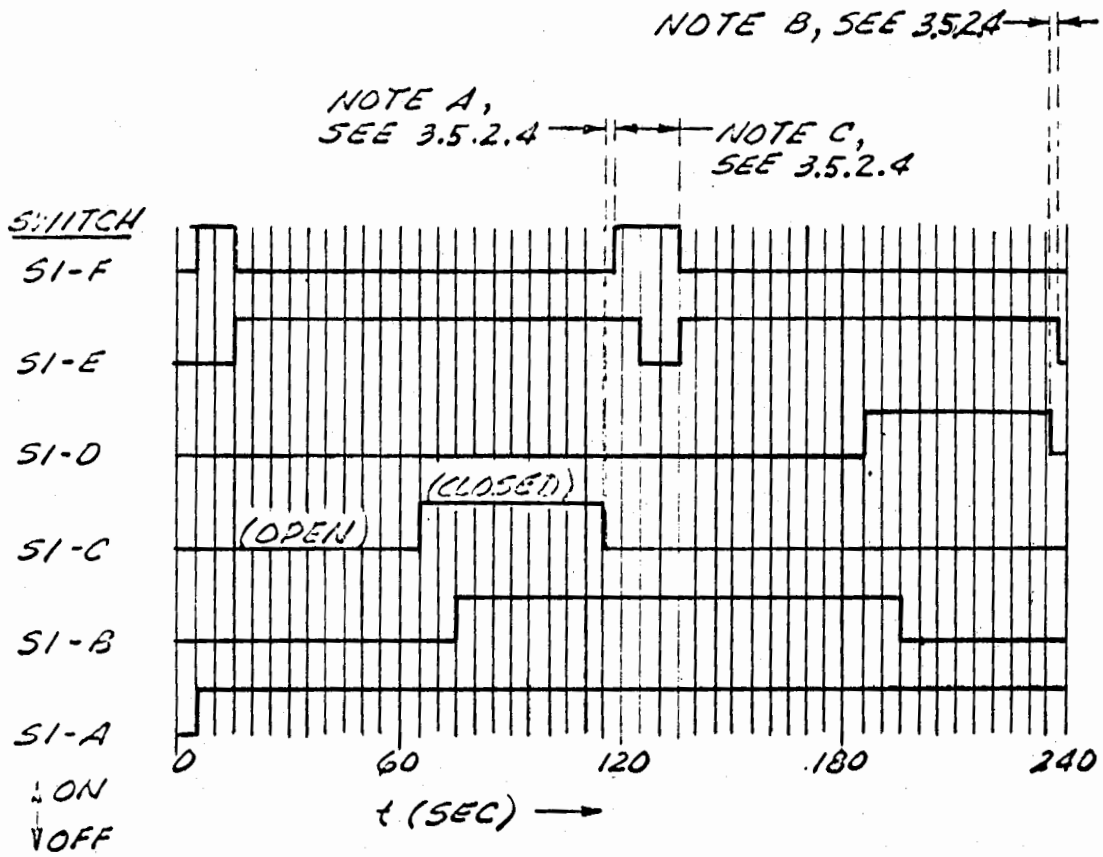


Figure I

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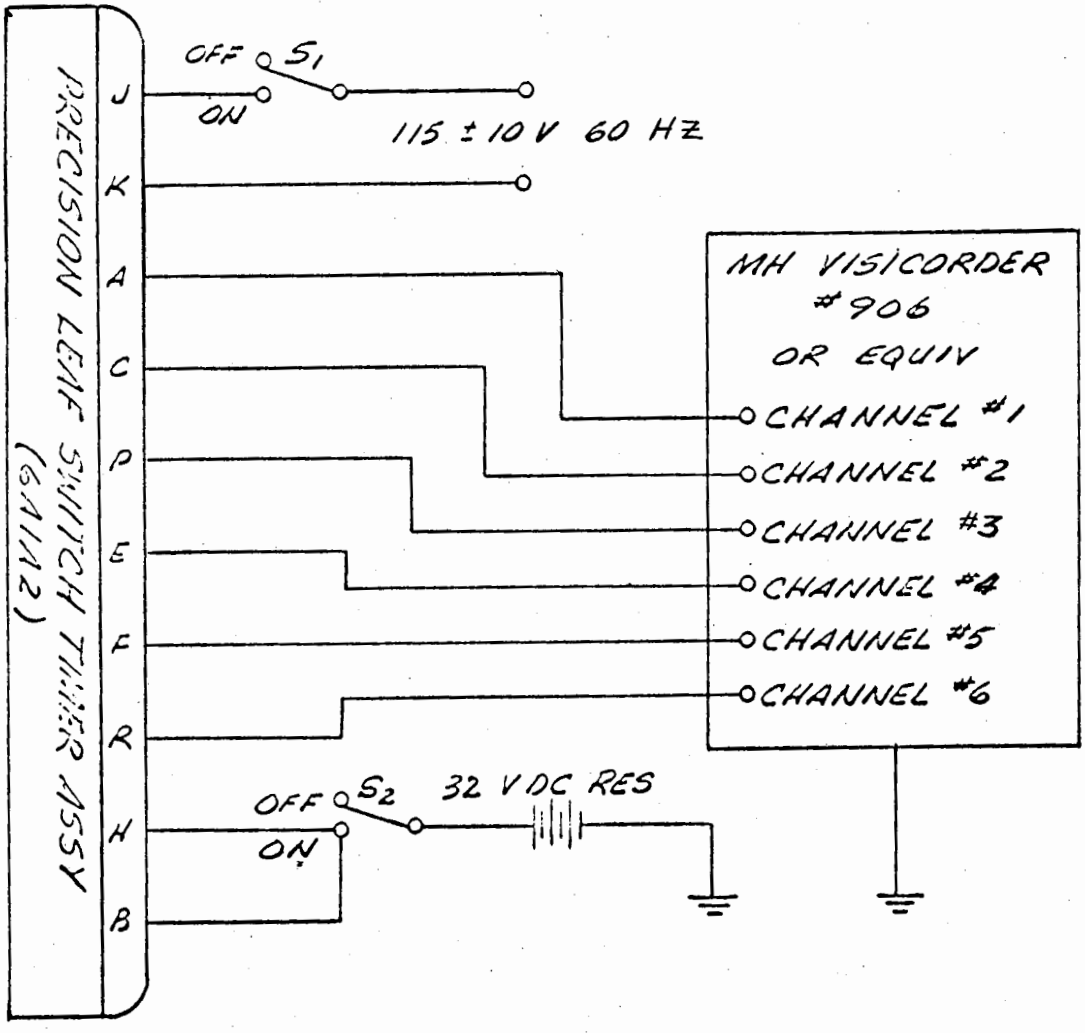


Figure II. Precision Leaf Switch Timer
Timing Sequence Test
Test Circuit

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APPENDIX D
REFERENCE VOLTAGE SOURCE ASSEMBLY (6A1A3)

The following sections apply specifically to the reference voltage source assembly. In addition to the requirements of this appendix, the assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.

3.5.1 Calibration capability. The reference voltage source assembly (6A1A3) shall be capable of calibration such that with input voltages of + 28 vdc on terminal A and - 28 vdc on terminal V (terminal N grounded) adjustment of the positive calibration pot R3 shall give a voltage of + 2.510 vdc on terminal C. Turning R3 in a clockwise direction will increase the indicated voltage. Similarly, adjustment of the negative calibration pot R12 shall give - 2.510 vdc on terminal T. Turning R12 in a clockwise direction will make the indicated voltage more negative. The calibration shall be performed in accordance with 4.4.1.

3.5.2 Operational limits. With the reference voltage source assembly (6A1A3) adjusted to give + 2.510 and - 2.510 vdc on terminals C and T respectively, the voltages at the output terminals shall be as specified below in Table I. The output voltages read at terminals C and T shall not vary by more than ± 0.003 vdc when the input voltages are varied ± 2 vdc. The tests shall be performed in accordance with 4.4.2.

Table I

Terminal	Nominal Voltage	Tolerance
C	+ 2.510 vdc*	0 (adjusted)
E	+ 1.998	± 0.016
J	+ 1.486	± 0.015
K	+ 0.509	± 0.008
L	+ 0.0215	± 0.0005
N	0	Ground
P	- 0.020	± 0.0005
M	- 0.287	± 0.005
R	- 0.719	± 0.012
S	- 1.488	± 0.015
U	- 1.999	± 0.015
T	- 2.510*	0 (adjusted)

*Voltage externally adjusted. Readings verified through 3 decimal places.

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3.5.2.1 Continuity and resistance. Continuity and resistance between the terminals listed below shall be within the specified resistance when tested as specified in 4.4.2.2.

Check

From (Positive)	To (Negative)	Resistance (Ohms)
Terminal P1-C	Terminal P1-E	511 ± 1%
Terminal P1-E	Terminal P1-J	511 ± 1%
Terminal P1-J	Terminal P1-K	976 ± 1%
Terminal P1-K	Terminal P1-L	487 ± 1%
Terminal P1-L	Terminal P1-N	21.5 ± 1%
Terminal P1-N	Terminal P1-P	20 ± 1%
Terminal P1-P	Terminal P1-M	267 ± 1%
Terminal P1-M	Terminal P1-R	432 ± 1%
Terminal P1-R	Terminal P1-S	770 ± 1%
Terminal P1-S	Terminal P1-U	511 ± 1%
Terminal P1-U	Terminal P1-T	511 ± 1%

3.5.2.2 Insulation resistance. The insulation resistance shall be greater than 20 Megohms between P1-A, the unused terminals P1-B, P1-D, P1-F and P1-H tied together, and case. Testing shall be as specified in 4.4.2.3.

4.3.2.1 Classification of characteristics.

4.3.2.1.1 Reference voltage source assembly (6A1A3) (see dwg 8847780).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connector with respect to alignment pins	Gage
102.	Defective solder joints	Visual/Manual
103.	Component missing	Visual
104.	Fastenings loose or missing	Visual
105.	Cracked or broken connectors or damaged pins	Visual
106.	Board mounting screws loose	Manual
107.	Objective evidence relative to material improper or missing (see 3.1)	Visual

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108. Serialization improper or missing (see 3.4).....Visual

Minor:

201. Evidence of poor workmanship (see 3.8).....Visual
202. Illegible or improper marking.....Visual
203. Improper finish or color.....Visual

4.3.2.1.2 Preparation for delivery (see 8883512).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Packing material and packaging not proper or missing	Visual
	102. Marking not in accordance with dwg	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.

4.4.1 Calibration capability. The reference voltage source assembly shall be connected to the suggested test circuit shown in Figure 1. With + 28 vdc connected to terminal A, - 28 vdc on terminal V and terminal N grounded, adjust the positive calibration pot R3 to give + 2.510 vdc at terminal C and the negative calibration pot R12 to give - 2.510 vdc on terminal T. The assembly shall be calibrated at 80 ± 30°F. The calibration setting shall not be changed for the remainder of the tests.

4.4.2 Operational limits. With the assembly calibrated in accordance with 4.4.1, the voltages at terminals E, J, K, L, P, M, R, S and U shall be within the nominal values given in Table I. Readings shall be made at 80 ± 30°F.

4.4.2.1 Regulation. The input voltages shall be adjusted to + 26 vdc and -26 vdc on terminals A and V respectively. The voltages at terminal C shall be + 2.510 ± 0.003 vdc and at terminal T - 2.510 ± 0.003 vdc. The test shall be repeated for + 30 and - 30 vdc on terminals A and V respectively. This test shall be performed at 80 ± 30°F.

4.4.2.2 Continuity and resistance. Resistance between terminals listed in 3.5.2.1 shall be measured with a digital ohmmeter of sufficient accuracy or its equivalent. Observe indicated testing polarity.

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4.4.2.3 Insulation resistance. The insulation resistance shall be measured between the terminals listed in 3.5.2.2 using a test voltage of 500 ± 50 vdc. Current shall be limited to 5 ma for this test.

Differential dc voltmeters or equivalent digital voltmeters shall be used for all dc voltage measurement.

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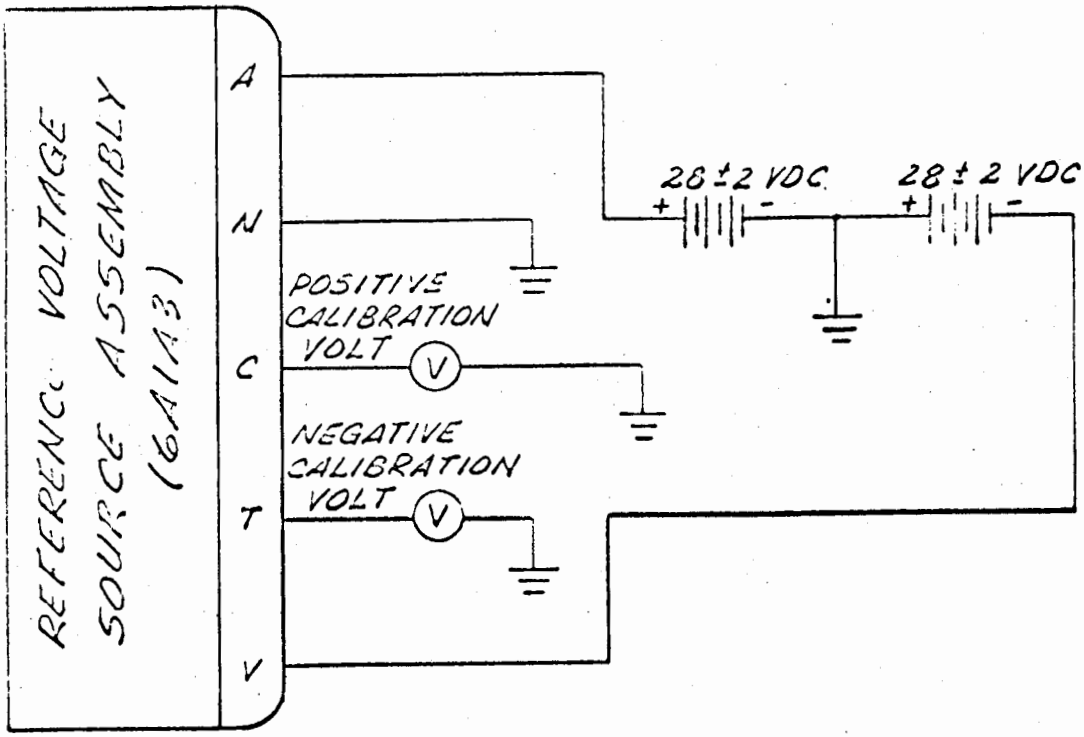


Figure I. Test Circuit For Reference Voltage Source Assembly

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APPENDIX E
VOLTAGE COMPARATOR ASSEMBLY
(6A1A4)

The following sections apply specifically to the voltage comparator assembly. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.

3.5.1 Calibration capability. Not applicable.

3.5.2 Operational limits. The voltage comparator assembly (6A1A4) shall provide the positive and negative reference amplifier outputs specified in Table I when tested with the circuit shown in Figure 1. The test shall be conducted in accordance with 4.4.2.

Table I

Test	S6 Switch Setting	S1 and S2 Switch Setting	V Out
1	10 MV	500 Ohms	450 ± 90 MV
2	10 MV	5,000 Ohms	2650 ± 530 MV
3	10 MV	10,000 Ohms	3500 ± 700 MV
4	1000 MV	500 Ohms	1200 ± 240 MV
5	1000 MV	10,000 Ohms	8210 ± 1640 MV

3.5.2.1 Insulation resistance. The insulation resistance shall be greater than 20 Megohms between terminal P1-K, the terminals P1-A, P1-U, P1-X and P1-Y tied together, the unused terminals P1-B, P1-C, P1-F, P1-H, P1-J, P1-L, P1-M, P1-R, P1-S, P1-T, P1-V, P1-W, P1-Z, P1-AA, P1-BB and P1-DD tied together, and the case. Testing shall be as specified in 4.4.2.1.

4.3.2.1 Classification of characteristics.

4.3.2.1.1 Voltage comparator assembly (6A1A4) (see dwg 8847727).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect location and orientation of connector with respect to alignment pins	Gage
	102. Board mounting screws loose	Manual
	103. Defective solder joints	Visual/Manual

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104. Cracked or broken connectors or damaged pins.....Visual
105. Component missing.....Visual
106. Fastenings loose or missing.....Visual
107. Objective evidence relative to material improper or missing (see 3.1).....Visual
108. Serialization improper or missing (see 3.4).....Visual

Minor:
201. Evidence of poor workmanship (see 3.8).....Visual
202. Illegible or improper marking.....Visual
203. Improper finish or color.....Visual

4.3.2.1.2 Preparation for delivery (see 8883511).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Packing material and packaging not proper or missing.....Visual
102. Marking not in accordance with dwg.....Visual

Minor:
201. Evidence of poor workmanship (see 3.8).....Visual

4.4 Test methods.
4.4.1 Calibration capability. Not applicable.
4.4.2 Operational limits. The voltage comparator assembly shall be connected to the test circuit shown in Figure 1. Close switches S3, S4, S5, and S7. For test #1, set S1 and S2 to the 500 ohm position. Set S6 to the "10 MV in" position. Adjust R1 so that Ammeter A reads 1 ma. Change switch positions of S1, S2 and S6 to set combination of five test positions required in Table I. Measurements of the positive and negative reference amplifier voltage outputs (Vp and Vn) on pins CC and P for each of the settings specified in Table I shall be made. Output voltage readings shall comply with 3.5.2.

Differential dc voltmeters or equivalent digital voltmeters shall be used for all dc voltage measurements.

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A suitable dc milliammeter with an accuracy of 1% or better shall be used for measuring current.

4.4.2.1 Insulation resistance. The insulation resistance shall be measured between the terminals listed in 3.5.2.1 using a test voltage of 500 ± 50 vdc. Current shall be limited to 5 ma for this test.

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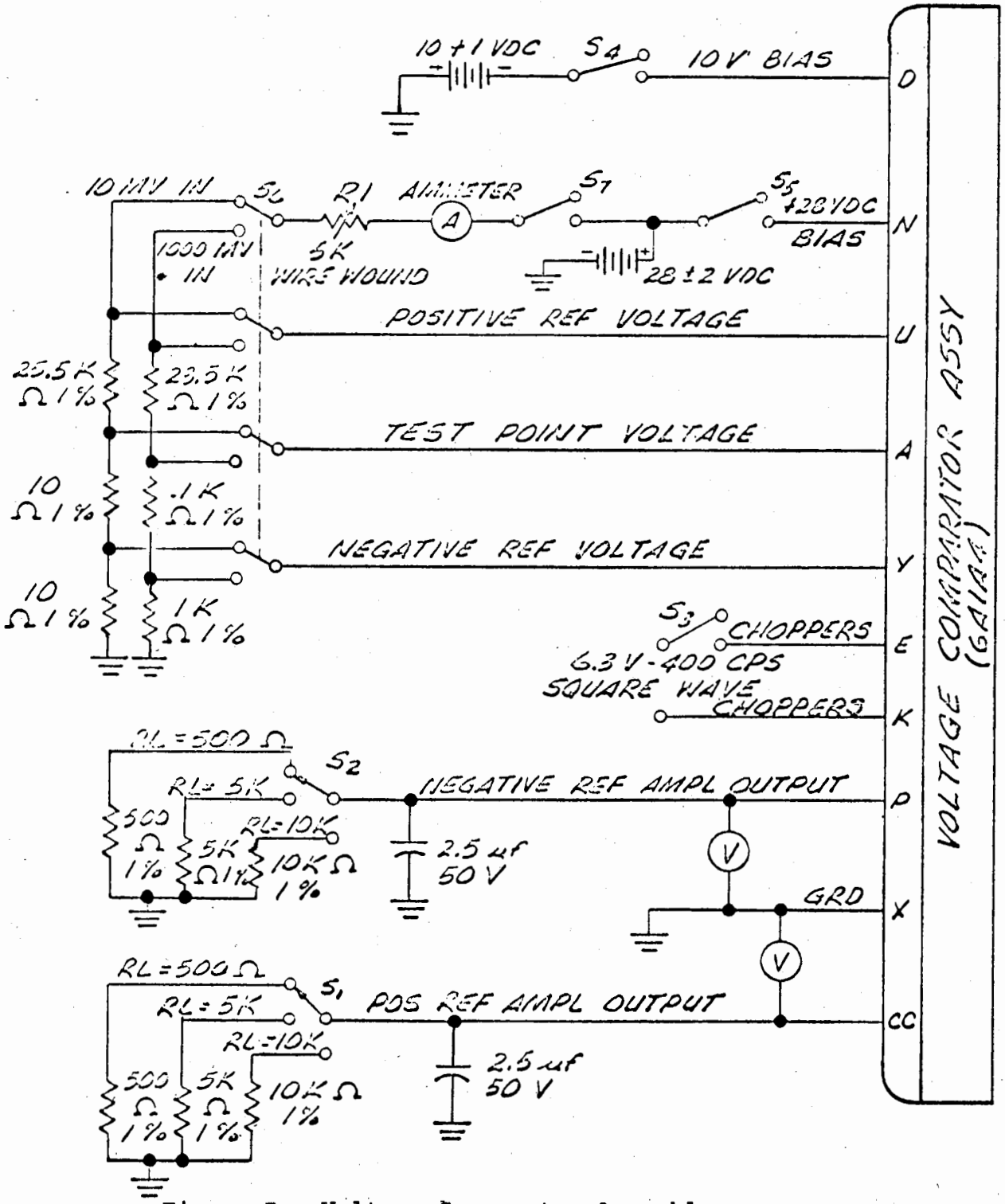


Figure I. Voltage Comparator Assembly Test Circuit (6A1A4)

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APPENDIX F
PROGRAMMER STEP CIRCUIT ASSEMBLY
(6A1A5)

The following sections apply specifically to the programmer step circuit assembly. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.
3.5.1 Calibration capability. Not applicable.
3.5.2 Operational limits. The output pulse delivered to a 20 ohm, 10 watt resistor at terminal K shall have the following characteristics when measured as specified in 4.4.2.

a. Step pulse "On" time: 45 ± 5 ms
b. Step pulse "Off" time: 225 ± 25 ms
c. Pulse amplitude: 19.5 to 23.5 volts

3.5.2.1 Continuity and resistance. Continuity and resistance between the terminals listed below shall be within the specified resistance when tested as specified in 4.4.2.4.

From	To	Resistance (Ohm)
Terminal P1-F	Terminal P1-L	38.7K + 10% - 5%

3.5.2.2 Insulation resistance. The insulation resistance shall be greater than 20 megohms between terminals P1-A, P1-D and P1-L tied together, the unused terminals P1-B, P1-E, P1-J, P1-M and P1-R tied together, and case. Testing shall be as specified in 4.4.2.5. Tie P1-A, P1-D and P1-L together for testing only.

4.3.2.1 Classification of characteristics.
4.3.2.1.1 Programmer step circuit assembly (6A1A5) (see dwg 8847742).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connector with respect to alignment pins	Gage
102.	Fastenings loose or missing	Manual/Visual
103.	Cracked or broken connector or damaged pins	Visual

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104. Defective solder joints.....Visual/Manual
105. Board mounting screws loose.....Manual
106. Component missing.....Visual
107. Objective evidence relative to material improper or missing (see 3.1).....Visual
108. Serialization improper or missing (see 3.4).....Visual

Minor:
201. Evidence of poor workmanship (see 3.8).....Visual
202. Illegible or improper marking.....Visual
203. Improper finish or color.....Visual

4.3.2.1.2 Preparation for delivery (see 8883510).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Packing material and packaging not proper or missing.....Visual
102. Marking not in accordance with dwg.....Visual

Minor:
201. Evidence of poor workmanship (see 3.8).....Visual

4.4 Test methods.
4.4.1 Calibration capability. Not applicable.
4.4.2 Operational limits.
4.4.2.1 Adjustment. Connect the programmer step circuit assembly to the test circuit shown in Figure 1. Close switches S1, S2, S4, S5 and S6. Adjust R1 and R2 in the test circuit, so that 800 microamperes ± 1%, as measured by an ammeter of sufficient accuracy, are supplied to pins H and C respectively. Adjust R1, in the test circuit, and R18, in the assembly, so that 800 microamperes are supplied to pin C. An output pulsing to the 20 ohms resistor (R3) shall be present.
4.4.2.2 Override test. When the programmer step circuit assembly has been adjusted as per 4.4.2.1, and with S1, S2 and S6 in "On" position, close S3 for less than one second. A single pulse should be detected across R3. Record pulse voltage amplitude with a suitable recording instrument. This measurement shall be as specified in 3.5.2.

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4.4.2.3 Stepping test. When the programmer step circuit assembly has been adjusted as per 4.4.2.1 and with S1, S2 and S6 in "On" position, close S7. A series of pulses shall be detected. Record pulse "On" time, "Off" time, and pulse voltage amplitude with a suitable recording instrument. These measurements shall be as specified in 3.5.2.

4.4.2.4 Continuity and resistance. Resistance measurements between the terminals listed in 3.5.2.1 shall be measured with a digital ohmmeter of sufficient accuracy or its equivalent.

4.4.2.5 Insulation resistance. The insulation resistance shall be measured between the terminals listed in 3.5.2.2 using a test voltage of 500 ± 50 vdc. Current shall be limited to 5 ma for this test.

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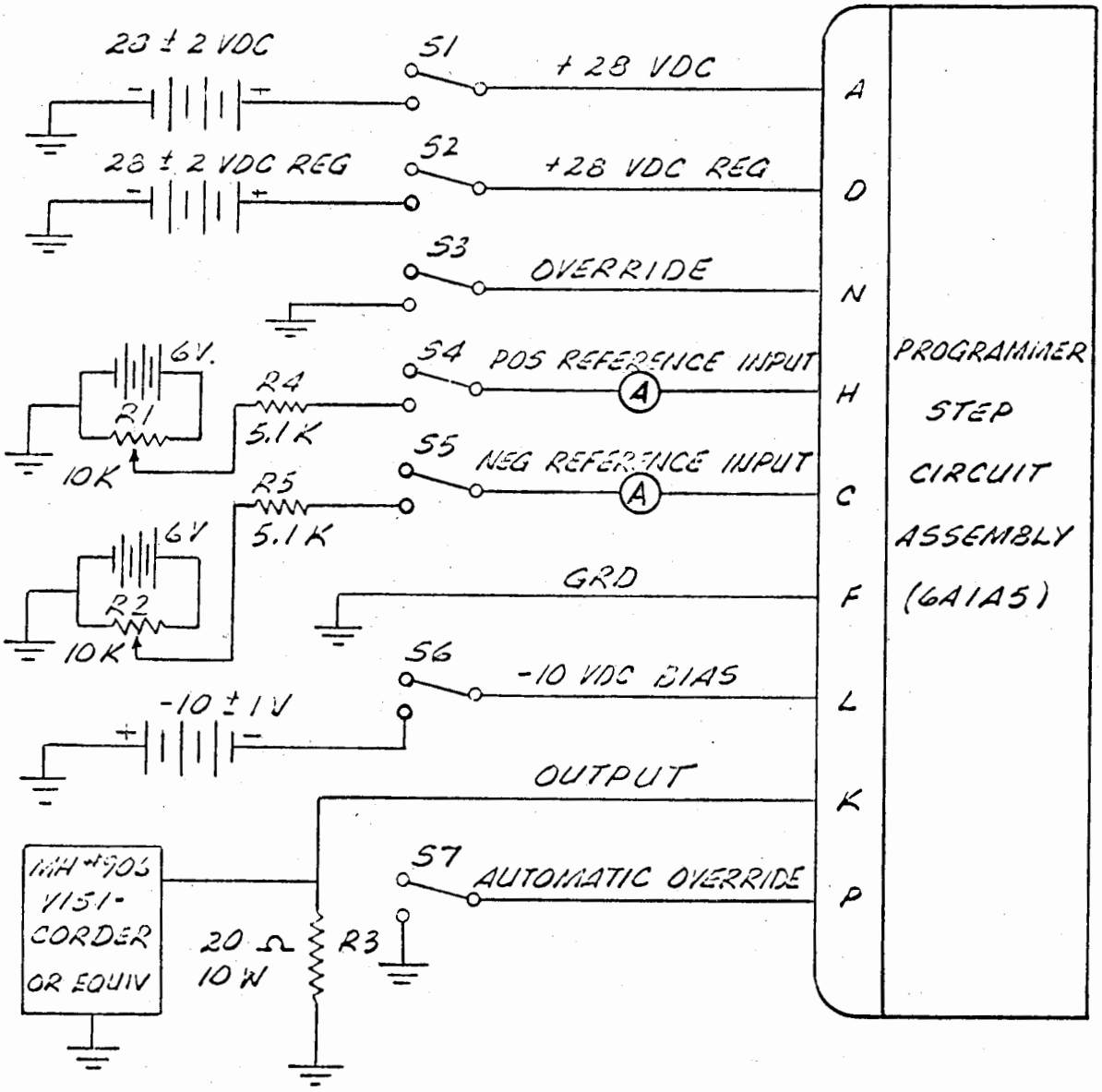


Figure I. Programmer Step Circuit Assembly Test Circuit

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APPENDIX G
TIMER ASSEMBLY, UNIJUNCTION
(6A1A6)

The following sections apply specifically to the unijunction timer assembly. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.

3.5.1 Calibration capability. With the test circuit shown in Figure 1, the unijunction timer assembly shall be capable of being calibrated. This capability is fulfilled when the following conditions are met:

Test Terminal	Timer Trimpot (R4) Voltage Setting	vs.	Time	Max. or Min.
L (55 sec. nominal)	Full CCW		58 sec.	Max.
L (55 sec. nominal)	Full CW		58 sec.	Min.

The calibration capability shall be performed in accordance with 4.4.1.

3.5.1.1 Calibration. With the test circuit shown in Figure 1, the unijunction timer assembly shall meet the following timings after calibration:

Pin L 58 ± 0.25 seconds

The calibration shall be performed in accordance with 4.4.1.1.

3.5.2 Operational limits. With the test circuit shown in Figure 1, the unijunction timer assembly shall provide the following timing limits with the input voltage ranging from 52 to 60 vdc when tested as specified in 4.4.2.

Test Terminal	Input Voltage	Time
L	52 to 60 vdc	58 ± 2 seconds
N	52 to 60 vdc	20 + 1.5, - 1 seconds
P	52 to 60 vdc	1.5 ± 0.1 seconds

3.5.2.1 Continuity and resistance. Resistance between the listed terminals shall be within the specified resistance when tested as specified in 4.4.2.1.

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From	To	Resistance (Ohms)
Terminal P1-B	Terminal P1-C	0.25 max.
Terminal P1-P	Terminal P1-N	348.9K ± 1%
Terminal P1-L	Terminal P1-P	978K ± 1%
Terminal P1-H	Terminal P1-D	1150 ± 30%
Terminal P1-H	Terminal P1-R	1825 ± 30%

3.5.2.2 Insulation resistance. The insulation resistance shall be greater than 20 Megohms between each of the following terminals and all other terminals listed and case. Testing shall be as specified in 4.4.5.2.

P1-A, P1-B, P1-D and P1-K

Unused terminals P1-E and P1-M shall be connected together and tested as one terminal.

4.3.2.1 Classification of characteristics.
4.3.2.1.1 Timer assembly, unijunction (6A1A6) (see dwg 8847761).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connector with respect to alignment pins	Gage
102.	Fastenings loose or missing	Manual/Visual
103.	Cracked or broken connectors or damaged pins	Visual
104.	Board mounting screws loose	Manual
105.	Defective solder joints	Visual/Manual
106.	Component missing	Visual
107.	Objective evidence relative to material improper or missing (see 3.1)	Visual
108.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual
202.	Illegible or improper marking	Visual
203.	Improper finish or color	Visual

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4.3.2.1.2 Preparation for delivery (see 8883509).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Packing material and packaging not proper or missing	Visual
102.	Marking not in accordance with dwg	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.

4.4.1 Calibration capability. Using the test circuit shown in Figure 1, perform the following adjustments:

a. Switch S3 to "OFF" and set the selector switch S4 to "58". Switch S1 and S2 to "OFF". Adjust the supply voltage V2 to 56 vdc. Make the necessary connections to the assembly. Close switch S2 and adjust the trimpot R4 on the assembly until the voltage V3 from the S4 pole or terminal K, to ground is 45 vdc. Turning the trimpot in a clockwise direction will decrease the indicated voltage and increase timing cycle. Allow 5 minutes for the assembly to stabilize.

b. Close switch S1 and set the electronic counter to C. The assembly is now ready to operate. Switch S3 to "ON;" the assembly should begin its timing cycle when 28 volts are removed from terminal R. The timing cycle will be completed when the assembly switches 28 vdc to terminal C. Switch S3 to "OFF." The electronic counter reading shall conform with 3.5.1.

c. Adjust the trimpot R4 on the assembly until the voltage V3, terminal K to ground, is 30 vdc. Set the electronic counter to 0. Switch S3 to "ON," the assembly should begin its timing cycle. After the assembly has completed its timing cycle or has attained a reading of more than 70.4 seconds, switch S3 to "OFF." The electronic counter reading shall conform with 3.5.1.

4.4.1.1 Calibration. The test circuit in Figure 1 is for calibrating the unijunction timer assembly. The assembly shall be calibrated at a temperature of 80 $\pm$ 30°F.

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a. Switch S3 to "OFF" and set the selector switch to "58." Switch S1 and S2 to "OFF". Adjust the supply voltage V2 to 56 vdc. Make the necessary connections to the timer assembly. Close switch S2 and adjust the trimpot R4 on the timer assembly until the voltage V3 from the S4 pole or terminal K to ground is 38.9 vdc. Turning the trimpot in a CW direction will decrease the indicated voltage and increase timing cycle. Allow 5 minutes for the assembly to stabilize.
b. Close switch S1 and set electronic counter to 0. The assembly is now ready to operate. Switch S3 to "ON," the assembly should begin its timing cycle when 28 volts is removed from terminal R. The timing cycle will be completed when the assembly switches 28 volts to terminal C. Switch S3 to "OFF."
c. If the initial time, T_1, is not 58 $\pm$ 0.25 seconds, use the following equation to determine a new value for the trimpot R4 voltage setting:

$$V3_2 = V3_1 - 0.318\ (58.0 - t_1)$$

where

$V3_2$ is unknown

$V3_1$ is 38.9

t_1 is initial time

By substituting the newly obtained T_2 and $V3_2$ for t_1 and V3 respectively in subsequent trial calculations, a time of 58 + 0.25 seconds will be attained. Using the final R4 setting ($V3_x$), take three consecutive time readings. These readings shall be as specified in 3.5.1.1.

All voltage readings shall be made with a differential voltmeter or equivalent digital voltmeter. The assembly shall be calibrated at 80 $\pm$ 30°F.

4.4.2 Operational limits. Using the test circuit in Figure 1, measure the time durations with a variation in input voltage. The timing shall be measured with input voltages of 52 and 60 vdc at V2. The assembly readings shall conform with 3.5.2.

4.4.2.1 Continuity and resistance. Resistance measurements indicating continuity between the terminals listed in 3.5.2.1 shall be measured with a four wire resistance

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measuring instrument except for resistance greater than 1000 ohms which will be measured with a digital ohmmeter of sufficient accuracy or equivalent.

4.4.2.2 Insulation resistance. The insulation resistance shall be measured between the terminals listed in 3.5.2.2 using a test voltage of 500 ± 50 vdc. Current shall be limited to 5 ma for this test.

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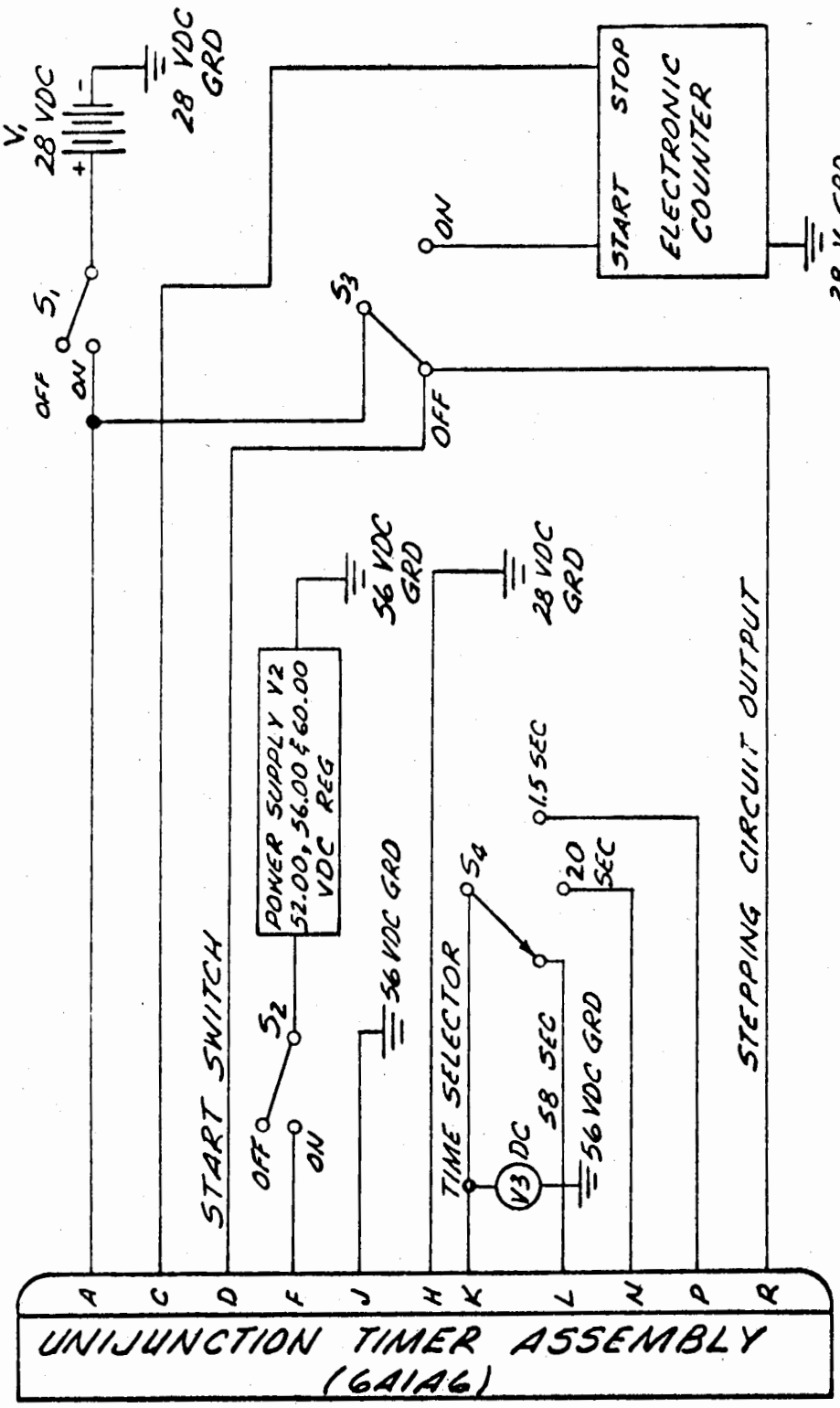


Figure I. Unijunction Timer Assembly Test Circuit

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APPENDIX H
RELAY MODULE #1 ASSEMBLY
(6A1A7)

The following sections apply specifically to the relay module #1 assembly. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.
3.5.1 Calibration capability. Not applicable.
3.5.2 Operational limits.
3.5.2.1 Continuity and resistance. Continuity and resistance between the "Primary Terminal" and each terminal listed in Table I, II, and III under "Secondary Terminals" of the assembly shall be within the specified resistance when tested as specified in 4.4.2.1.

Table I

Mode 1 - Relays de-energized

Primary Terminal	Secondary Terminals	Resistance (Ohms)
1 - P1A	P1J,P1N,P2E,P2c,P2Mm	0.10 maximum
2 - P1B	P1K,P1P,P1X,P2cc, P1LL,P1WW,P1YY,P1r, P1bb,P1uu,P1Rr,P2AA, P2Rr,P2hh,P2jj,P2kk, P2mm,P2nn,P2uu,P2vv, P2Ee,P2Ff,P2Hh	0.10 maximum
3 - P1B	P1Y,P2YY,P2x,P2Cc	0.25 maximum
4 - P1B	P1FF	680 + 8%, - 5%
*5 - P1B	P1e	3.9K + 8%, - 5%
6 - P1Y	P1Dd	220 + 8%, - 5%
7 - P1C	P2A, P2d,P2Nn	0.10 maximum
8 - P1D	P1EE,P1ff,P2Pp	0.10 maximum
9 - P1E	P1R,P2f	0.10 maximum
10 - P1F	P2C,P2e	0.10 maximum
11 - P1H	P2B,P2ZZ	0.10 maximum
12 - P1L	P1Nn,P1Pp,P2J,P2rr, P2ss	0.10 maximum
13 - P1M	P1V,P1Jj,P2S	0.10 maximum
14 - P1U	P1pp,P2Ss	0.25 maximum
15 - P1Z	P1CC	0.25 maximum

*Ohmmeter test voltage applied across these terminals shall be less than 5 VDC. The (+) side of the test voltage shall be connected to P1e.

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Primary Terminal	Secondary Terminals	Resistance (Ohms)
16 - P1b	P1SS,P2T	0.25 maximum
17 - P1f	P1j	750K ± 2%
18 - P1k	P1ee,P2EE	0.25 maximum
19 - P1JJ	P2p	0.10 maximum
20 - P1KK	P2n	0.10 maximum
21 - P1NN	P2r	0.10 maximum
22 - P1PP	P2ww	0.10 maximum
23 - P1nn	P1ss,P2h	0.25 maximum
24 - P1vv	P1Kk	56K + 10%, - 5%
25 - P1ww	P1zz	0.25 maximum
26 - P1xx	P2FF	0.10 maximum
27 - P1yy	P2DD	0.10 maximum
28 - P1Bb	P2ee	0.10 maximum
29 - P1Ee	P2dd	200 + 8%, - 5%
30 - P1d	P2u	0.10 maximum
31 - P1Kk	P2a	0.10 maximum
32 - P1Tt	P2s	0.10 maximum
33 - P2L	P2bb	0.10 maximum
34 - P2U	P2j	27K + 10%, - 5%
35 - P2Y	P2WW,P2Aa	0.25 maximum
36 - P2b	P2Bb	620 + 8%, - 5%
37 - P2t	P2h	2.7K + 8%, - 5%
38 - P2SS	P2Dd	0.10 maximum
39 - P2XX	P1FF	68K + 8%, - 5%
40 - P2XX	P1jj,P1kk,P1mm	100K + 10%, - 5%
41 - P2xx	P1Cc	560 + 8%, - 5%
42 - P2YY	P2Cc	0.10 maximum
43 - P1S	P2Uu	0.25 maximum
44 - P1DD	P2Tt	0.10 maximum
45 - P2b	P2Kk	0.10 maximum
46 - P1W	P1Ee	0.10 maximum
47 - P1Mm	P2yy	0.10 maximum
48 - P1Ss	P2tt	0.10 maximum
49 - P2XX	P1c	68K + 10%, - 5%
50 - P2j	P1B	2K + 8%, - 5%
51 - P2K	P1Y	5.7K + 12%, - 6%

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Table II

*Mode 2 - All relays energized

Primary Terminal	Secondary Terminals	Resistance (Ohms)
1 - P1D	P1FF,P1c	0.25 maximum
2 - P1M	P1Aa	0.25 maximum
3 - P1Z	P1AA	0.25 maximum
4 - P1HH	P1u,P1v,P1x,P1y,P1z	0.25 maximum
5 - P1SS	P1TT,P1ZZ,P2T	0.25 maximum
6 - P1UU	P2CC,P2w	0.25 maximum
7 - P1XX	P1aa,P2VV	0.25 maximum
8 - P1xx	P2JJ	0.25 maximum
9 - P2m	P1h	0.25 maximum
10 - P1k	P1m,P1cc	0.25 maximum
**11 - P2Y	P2PP,P2TT	0.25 maximum

*All relays must be energized simultaneously for these tests. To energize all relays, apply 28 ± 2 vdc across terminals designated in Table IV.

**Remove + 28V from P2PP when measuring P2PP to P2Y and remove + 28 V from P2TT when measuring P2TT to P2Y.

Table III

Mode 3 - Special resistance checks

Primary Terminal	Secondary Terminal	Apply 28 ± 2 vdc From Terminal (Positive)	To Terminal (Negative)	Resistance (Ohms)
*1 - P1B	P1W	P2X	P1B	4.83K ± 0.1%
*2 - P1B	P1W	P2W	P1B	6.75K ± 0.1%
3 - P2Jj	P1Y	P2P,P2X	P1B	5K ± 0.1%
4 - P1e	P1B	P2L	P1B	0.25 maximum
**5 - P1W	P2K	P2W	P1B	6K ± 0.1%
**6 - P2YY	P2K	P2W,P2X	P1B	6K ± 0.1%
7 - P1S	P1aa	P1cc	P1B	0.25 maximum
8 - P1t	P1Y	P2W	P1B	0.25 maximum
**9 - P1nn	P2V	P2V	P1B	0.25 maximum

Note: To perform the above tests, first apply 28 vdc to the indicated positive and negative terminals. Upon completion of each test, remove the 28 vdc.

*Connect the (+) lead of ohmmeter to P1B and allow 20 seconds from the time of voltage application before reading resistance.
**Connect the (+) lead of ohmmeter to the primary terminal.

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Table IV

To energize all relays simultaneously, apply 28 + 2 vdc across all terminals listed.

Column A (Positive Terminals)	Column B (Negative Terminals)
P1U	P1B
*P1V	P1Y
*P1TT	P1VV
*P1ZZ	P1p
*P1m	P1hh
*P1cc	P2Z
*P1nn	P2RR
P1rr	P2SS
*P1Bb	P2x
*P1Ff	P2y
P2F	P2vy
P2L	
P2N	
P2P	
P2R	
P2V	
P2W	
P2X	
P2Y	
P2LL	
P2MM	
P2PP	
P2TT	
P2z	
P2aa	
P2xx	

Notes: A. Apply + 56 vdc to P2k with negative terminal connected to any terminal listed in Column B.
B. Apply + 28 vdc to P1w through a 2.7K ± 5% 1/2 watt resistor.

*Apply + 28 vdc through 1N645 diodes (one diode per terminal).

3.5.2.2 Relay operation. The resistance between terminals for each relay as indicated in Table V shall be less than 0.25 ohm when tested as specified in 4.4.2.2.

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Table V

Relay operation - (Energize all relays simultaneously unless otherwise specified.)

Check Continuity

Relay	From Terminal	To Terminal	Notes
K1	P1Dd	P2b	
K2	P1Bb	P1nn, P1zz, P2ee, P2ff	Remove + 28 V from P1nn when measuring to P1nn. Also checks K3.
K3	See K2		
K4	P1M	P1Ff	Remove + 28 V from P1V.
K5	P1Z	P1BB	Remove - 28 V from P2SS.
K6	P1Z	P1AA	
K7	P1XX	P1aa	Also checks K9.
K8	P1FF	P1c	Also checks K11 and K12.
K9	See K7		
K10	P1f	P1j	
K11	See K8		
K12	See K8		
K13	P1aa	P2VV	
K14	P1UU	P1n	Also checks K15.
K15	See K14		
K16	P2dd	P1Ee	Check for open circuit reading of 50K min.
K17	P2CC	P2w	
K18	P1S	P1rr	Check for open circuit reading of 50K min. The (+) terminal of ohmmeter to pin P1S.
K19	P1U	P1rr	Remove 28 V from P1U when measuring to P1U.
K20	P1T	P1Hh	
K21	P2ff	P1Bb	
K22	P1MM	P1RR	
K23	P2M	P2v	
K24	P1Y	P2Jj	Also checks K26. Resistance = 7000 + 7 ohms.
K25	P1Y	P1s	Remove + 28 V from P2P and P2W.
K26	See K24		
K27	P1ff	P2U	Remove + 28 vdc from P1Ff and P2xx.
K28	P2JJ	P2BB	Remove + 28 V from P2F. Also checks K29 and K30.
K29	See K28		
K30	See K28		

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Check Continuity

Relay	From Terminal	To Terminal	Notes
K31	P2H	P2a	Remove + 28 V from P2z and P2aa.
K32	P1y	P1x	Also checks K35 and K36.
K33	P2y	P2AA	
K34	P2H	P2a	Remove + 28 V from P2F and P2aa.
K35	See K32		
K36	See K32		
K37	P1u	P1y	

3.5.2.3 Capacitor characteristics. The voltage from terminals P1xx and P2K to terminal P1B shall be not less than 9.6 V 20 seconds after a 28 + 2 vdc charging voltage is removed from these terminals when tested as specified in 4.4.2.3.

3.5.2.4 Insulation resistance. The insulation resistance shall be greater than 20 megohms between each of the following terminals and all other terminals listed in Tables VI and VII and case. Testing shall be as specified in 4.4.2.4.

Table VI

Mode 1 - Relays de-energized

P1A, P1C, P1D, P1E, P1F, P1H, P1L, P1T, P1Z, P1AA, P1BB, P1JJ, P1KK, P1MM, P1NN, P1PP, P1RR, P1UU, P1XX, P1b, P1f, P1k, P1n, P1s, P1t, P1aa, P1nn, P1tt, P1Aa, P1Ee, P1Kk, P1Tt, P2H, P2M, P2BB, P2CC, P2LL, P2SS, P2VV, P2b, P2w, P2ff, P1zz, P1rr, P1HH, P1Ss, P1Mm, P1yy, P2m, P1z, P2Jj, and P1B.

Unused Terminals
P1Vv, P1Ww, P2pp, P1Uu, P2D, P2zz, and P2Ww, - shall be connected together and tested as one terminal.

Mode 1A - Special insulation resistance check; relays de-energized

The insulation resistance shall be greater than 20 megohms between terminal P1zz and all other terminals given in Mode 1 above, used and unused, and case. Testing shall be as specified in 4.4.2.4.1.

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Table VII

Mode 2 - *All relays energized

P1A, P1C, P1E, P1F, P1H, P1L, P1S, P1T, P1Z, P1BB, P1CC, P1JJ, P1KK, P1MM, P1NN, P1PP, P1UU, P1b, P1f, P1s, P1t, P1ee, P1pp, P1ss, P1tt, P1Dd, P1Ee, P1Kk, P1d, P1Tt, P2H, P2BB, P2WW, P2dd, and P1ww.

Unused Terminals
P1Vv, P1Ww, P1Uu, P2pp, P2D, P2zz, and P2Ww - shall be connected together and tested as one terminal.

*All relays must be energized simultaneously for these tests. To energize all relays, apply 28 ± 2 vdc across terminals designated in Table IV.

3.5.2.5 Gated diodes. When tested as specified in 4.4.2.5, a voltage of 2 + 0.5 vdc shall be available as specified in Mode 1 and Mode 2 below.

Mode 1 - With 30 ± 2 vdc applied positive to terminals P2Y and P2TT and negative to terminal P1B, the indicated voltages shall be present between "primary" terminal and "secondary" terminal listed in Table VIII when the corresponding listed connections are made. Voltage outputs are dependent upon previous connections and must be made in the listed order. Connections made for a previous voltage indication must be removed before a new one is made.

Table VIII

Primary Terminal (Positive)	Secondary Terminal (Negative)	Connections	Voltage (Volts)
1 - P2KK	P1B	None	0
2 - P2KK	P1B	P1HH to P1z and P1y	4 ± 1
3 - P2KK	P1B	P1HH to P1x	4 ± 1
4 - P2KK	P1B	P1HH to P1x and P1y	2 ± 0.5
5 - P2NN	P1B	None	0
6 - P2NN	P1B	P1HH to P1u and P1y	4 ± 1
7 - P2NN	P1B	P1HH to P1v and P1y	2 ± 0.5

Mode 2 - 2 ± 0.5 vdc shall be present between "primary" and "secondary" terminal listed in Table IX when the 30 + 2 vdc applied in Mode 1 is removed and reapplied as specified in Table IX.

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Table IX

Primary Terminal (Positive)	Secondary Terminal (Negative)	Apply 30 ± 2 vdc From Terminal (Positive)	To Terminal (Negative)	Voltage (Volts)
1 - P2KK	P1B	P2Y, P2LL, P2TT, P2aa	P1B	2 ± 0.5
2 - P2NN	P1B	*P2Y, P2MM, P2TT, P2aa	P1B	2 ± 0.5

*Remove voltage applied in item 1 and reapply.

3.5.2.6 Diode characteristics. The resistance between the terminals listed in Tables X and XI shall be within the specified limits when tested as specified in 4.4.2.6.

Table X

Diode	Terminals From (Primary)	To (Secondary)	Resistance (Ohms)
CR1	P2U	P1DD	10 maximum
	P1DD	P2U	1 meg. min.
CR2	P1S	P1DD	10 maximum.
	P1DD	P1S	1 meg. min.
CR3	P1B	P1e	10 maximum
CR4	P1S	P2JJ	30 maximum
	P2JJ	P1S	200K minimum
CR5	P1a	P1xx	10 maximum
	P1xx	P1a	1 meg. min.
CR6	P1dd	P1xx	10 maximum
	P1xx	P1dd	1 meg. min.
CR8	P2UU	P1xx	10 maximum
	P1xx	P2UU	1 meg. min.
CR14	P1Rr	P2v	10 maximum
	P2v	P1Rr	1 meg. min.
CR13	P2DD	P2HH	30 maximum
	P2HH	P2DD	200K minimum

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Table XI

Diode	Primary Terminal	Secondary Terminal	Resistance (Ohms)	Apply 28 ± 2 vdc From Terminal (Positive)	Apply 28 ± 2 vdc To Terminal (Negative)
CR16	P2H	P2a	0.50 max.	P2z	P2x
	P2H	P2a	open	P2x	P2z
CR17	P1Hh	P1T	0.50 max.	P1U	P1B
	P1Hh	P1T	open	P1B	P1U
CR18	P1Hh	P1T	0.50 max.	P1h	P1B
	P1Hh	P1T	open	P1B	P1h
CR19	P1U	P1PP	0.50 max.	P2Vv	P1rr
	P1U	P1PP	open	P1rr	P1Vv
CR15	P1tt	P2U	0.50 max.	Note: Apply + 28 V to P2N and P2V and connect the return to P1B, then remove + 28 V from P2N.	
	P1tt	P2U	open	Note: Apply + 28 V to P1B and connect the return to P2N and P2V, then remove the return from P2N.	

4.3.2.1 Classification of characteristics.

4.3.2.1.1 Relay module #1 assembly (6A1A7) (see dwg 8849338).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper location and orientation of connector with respect to alignment pins	Gage
102.	Defective solder joints	Visual/Manual
103.	Component missing	Visual
104.	Cracked or broken connectors or damaged pins	Visual

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105. Fastenings loose or missing.....Visual/Manual
106. Board mounting screws loose.....Manual
107. Objective evidence relative to material improper or missing (see 3.1).....Visual
108. Serialization improper or missing (see 3.4).....Visual

Minor:
201. Evidence of poor workmanship (see 3.8).....Visual
202. Illegible or improper marking.....Visual
203. Improper finish or color.....Visual

4.3.2.1.2 Preparation for delivery (see 8883508).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Packing material and packaging not proper or missing	Visual
102.	Marking not in accordance with dwg	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.
4.4.1 Calibration capability. Not applicable.
4.4.2 Operational limits.
4.4.2.1 Continuity and resistance. Resistance values between the terminals listed in 3.5.2.1 shall be measured with a digital ohmmeter of sufficient accuracy or its equivalent. Resistance values of less than one ohm shall be measured utilizing the four-wire system. When necessary, relays shall be energized by application of 28 ± 2 vdc between terminals listed in Table IV. All instrumentation shall be isolated from power line ground.
4.4.2.2 Relay operation. The resistance between terminals for each relay, as indicated in Table V, shall be measured with a four-wire resistance measuring system except for resistances greater than one ohm, which will be measured with a digital ohmmeter of sufficient accuracy and shall conform with 3.5.2.2. Energize all relays simultaneously (see Table IV) unless otherwise specified.

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4.4.2.3 Capacitor characteristics. Place a vtvm having an internal resistance of 11 megohms and a 28 ± 2 vdc source across terminals P1xx (positive) and P1B (negative) to charge the capacitor. Twenty seconds after the voltage source is removed from the terminals, the voltage across the terminals shall read not less than the value specified in 3.5.2.3. Repeat above procedure across terminals P1B (positive) and P2K, P2W, P2X (negative). Remove P2K from - 28 V to check capacitor.

4.4.2.4 Insulation resistance. The insulation resistance shall be measured between the terminals listed in Table VI and Table VII of Section 3.5.2.4 using a test voltage of 500 ± 50 vdc. Current shall be limited to 5 ma for this test.

4.4.2.4.1 Special insulation resistance check. Apply + 28 vdc through a 2.7K ± 5% 1/2 watt resistor to P1w with the return connected to P1B, thereby energizing relay K2. The insulation resistance shall then be measured between terminal P1zz and all other used and unused terminals given to Mode 1, Section 3.5.2.4, Table VI, using a test voltage of 500 ± 50 vdc. Current shall be limited to 5 ma for this test.

4.4.2.5 Gated diodes. The voltages between the terminals listed in Tables VIII and IX of Section 3.5.2.5 shall be as specified using test voltages of 28 and 32 vdc.

4.4.2.6 Diode characteristics. The resistance between the terminals listed in Tables IX and XI of 3.5.2.6 shall be measured using a Triplett Model 630A multimeter or its equivalent. Observe ohmmeter polarity as indicated in Table X. The test lead to the "Primary" terminal will be the COM test lead on the above Triplett multimeter. The COM test lead on this meter is the + side of the internal battery. Apply 28 ± 2 vdc as required in Table XI.

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APPENDIX I
RELAY MODULE #2 ASSEMBLY
(6A1A8)

The following sections apply specifically to the relay Module #2 assembly. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.

3.5.1 Calibration capability. Not applicable.

3.5.2 Operational limits.

3.5.2.1 Continuity and resistance. Continuity and resistance between the "Primary Terminal" and each terminal listed in Table I, Table II, and Table IV under "Secondary Terminals" of the assembly shall be within the specified resistance when tested as specified in 4.4.2.1.

Table I

Mode 1 - Relays in unlatched* or unenergized positions

Primary Terminal	Secondary Terminals	Max. Resistance (Ohms)
1 - P1B	P1U, P1nn	0.10
2 - P1H	P1mm	0.10
3 - P1L	P1kk, P1PP, P1DD	0.10
4 - P1M	P1ss	0.25
5 - P1N	P1a	0.25
6 - P1P	P1VV	0.10
7 - P1T	P1ff	0.10
8 - P1V	P1vv	0.25
9 - P1AA	P1pp	0.10
10 - P1EE	P1FF, P1HH, P1ww, P1Aa, P1Bb, P1tt, P2B, P2J, P2Y, P2y, P2x, P2aa, P2kk, P1Dd	0.10
11 - P1EE	P2Uu	0.25
12 - P1JJ	P1KK, P1LL	0.10
13 - P1JJ	P1NN	0.25
14 - P1MM	P1t	0.10
15 - P1e	P1RR, P1j, P1Cc	0.25
16 - P1e	P1uu, P2yy, P2Aa	0.25
17 - P1h	P1p, P1cc	0.10
18 - P1m	P1s, P1dd	0.10
19 - P1Ee	P1Ff	0.25
20 - P2A	P2hh	0.25
21 - P2C	P2D	0.25
22 - P2F	P2jj	0.10

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Primary Terminal	Secondary Terminals	Max. Resistance (Ohms)
23 - P2H	P2z	0.25
24 - P2K	P2w, P2Vy, P2Ww	0.10
25 - P2JJ	P2Kk	0.10
26 - P2JJ	P2Jj	0.25
27 - P2YY	P2Nn	0.25
28 - P2b	P2Pp	0.25
29 - P2c	P2Rr, P2j	0.25
30 - P2f	P2Ss	0.10
31 - P2n	P2P, P2r, P2uu, P2xx, P2Zz, P2Bb, P2Cc	0.10
32 - P1TT	P1UU, P1WW, P1w, P1xx	0.10
33 - P2ZZ	P2a	0.25
34 - P2t	P2TT	0.10
35 - P2k	P1yy	0.10
36 - P2d	P2E	0.10

*To ensure latching relays are unlatched, momentarily apply + 28 VDC to terminals P2L, P2M, P2SS, P2UU, P2VV, P2WW, P2XX, P1L, P1J, P1T, and P2Tt with the 28 V return applied to terminals P1EE and P1hh.

Table II

Continuity and Resistance

Mode 2 - All relays in latched and energized positions*

Primary Terminal	Secondary Terminals	Max. Resistance (Ohms)
1 - P1K	P1aa, P1N, P1a	0.25
2 - P1BB	P1ZZ, P1b, P1n, P1r, P1z, P1bb	0.25
3 - P1EE	P1k, P1Hh	0.25
4 - P1JJ	P1NN	0.25
5 - P1ee	P2Jj	0.25
6 - P1rr	P2yy, P2Aa	0.25
7 - P2C	P2d	0.25
8 - P2YY	P2ZZ	0.25
9 - P2a	P2Pp	0.25
10 - P2c	P2h	0.25
11 - P1Ee	P1Jj	0.25

*To energize all relays, + 28 vdc must be applied to all terminals listed in Column A of Table III and the 28 V return must be connected to all terminals listed in Column B of the table.

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Table III

To energize all standard relays and to latch all latching type relays simultaneously, apply 28 ± 2 vdc across all terminals listed.

Column A (Positive Terminals)	Column B (Negative Terminals)
P1A	P1Z
P1B	P1CC
P1C	P1EE
P1D	P1hh
P1E	P1jj
P1F	
P1H	
P1M	
P1P	
P1S	
P1V	
P1W	
P1X	
P1Y	
P1ee	
P2N	
P2P	
P2R	
P2KK	
P2LL	
P2MM	
P2NN	
P2PP	
P2RR	
P2TT	
P2Ee	

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Table IV

Continuity and Resistance

Mode 3 - Special resistance checks

At the start of this test, all standard relays shall be energized and all latching type relays latched.* During the test remove voltage from terminals designated under Note. After completion of individual test, the voltage is to be reapplied to the terminal from which it was removed.

Primary Terminal	Secondary Terminal	Note	Resistance (Ohms)
1 - P1N	P1aa	Remove + 28 V from P1H.	0.25 max.
2 - P1JJ	P1h	Remove + 28 V from P1D.	0.25 max.
3 - P1NN	P1v	Remove + 28 V from P1D.	0.25 max.
4 - P1u	P1NN	Remove + 28 V from P1M and P1D and apply to P1J.	0.25 max.
5 - P1KK	P1NN	Remove + 28 V from P1B.	0.25 max.
6 - P1rr	P1uu	Remove + 28 V return from P1CC.	0.25 max.
7 - P1rr	P1e	Remove + 28 V return from P1Z.	0.25 max.
8 - P1m	P1KK	Remove +28 V from P1E and P1M and apply to P1J.	0.63 + 50% - 5%
9 - P1Mm	P1Nn	Apply 28 V to P2K with return applied to P1Kk.	0.25 max.

*To energize relays, apply + 28 VDC to terminals listed in Column A of Table III and the 28 V return to the terminals listed in Column B of the same table.

3.5.2.1.1 Circuit resistance. The circuit resistance between listed terminals shall be as specified in Table V, Table VI, and Table V11 when tested as specified in 4.4.2.1.

Table V

Circuit Resistance

Mode 1 - Relays in unlatched* and de-energized positions

From Terminal	To Terminal	Resistance (Ohms)
1 - P1K	P1EE	1.88K + 8%, - 5%
2 - P1BB	P1SS	75K ± 1%
3 - P1BB	P1EE	432 ± 1%

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From Terminal	To Terminal	Resistance (Ohms)
4 - P1JJ	P1MM	149.9 ± 1%
5 - P1MM	P1f	8.25K ± 1%
6 - P1MM	P1K	8.25K ± 1%
7 - P1e	P1EE	110 + 8%, - 5%
8 - P1rr	P1EE	220 + 8%, - 5%
9 - P2vv	P1EE	15K + 8%, - 5%
10 - P2S	P2k	357 ± 1%
11 - P2S	P2cc	8.25K ± 1%
12 - P2T	P2k	30K + 10%, - 5%
13 - P2U	P2k	357 ± 1%
14 - P2U	P2mm	8.25 ± 1%
15 - P2V	P2k	30K + 10%, - 5%
16 - P2W	P2k	357 ± 1%
17 - P2W	P2ff	8.25K ± 1%
18 - P2X	P2rr	75K + 10%, - 5%
19 - P2X	P2k	3.32K ± 1%
20 - P2Z	P2s	820K + 10%, - 5%
21 - P2Z	P2k	30K + 10%, - 5%
22 - P2AA	P2k	30K + 10%, - 5%
23 - P2BB	P2k	357 ± 1%
24 - P2BB	P2dd	8.25K ± 1%
25 - P2CC	P2k	30K ± 10%, - 5%
26 - P2DD	P2k	357 ± 1%
27 - P2DD	P2ee	8.25K ± 1%
28 - P2EE	P2k	30K + 10%, - 5%
29 - P2FF	P2k	357 ± 1%
30 - P2FF	P2bb	8.25K ± 1%
31 - P2HH	P2k	30K ± 10%, - 5%
32 - P2F	P2vv	15K ± 10%, - 5%
33 - P2ss	P2B	4.3K ± 3%
34 - P1X	P1K	5.86K ± 10%

*To ensure latching relays are unlatched, momentarily apply + 28 vdc to terminals P2L, P2M, P2SS, P2UU, P2VV, P2WW, P2XX, P1L, P1J, P1T and P2Tt with the 28 V return applied to terminals P1EE and P1hh.

Table VI

Circuit Resistance

Mode 2 - All relays in latched or energized positions*

From Terminal	To Terminal	Resistance (Ohms)	Notes
1 - P1K	P1EE	1.88K + 8%, - 5%	All relays de-energized and unlatched.

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From Terminal	To Terminal	Resistance (Ohms)	Notes
			Apply + 28 V to P1P and P1C with the return connected to P1EE, P1hh and P1jj.
2 - P1JJ	P1k	100 ± 1%	
3 - P1MM	P1k	49.3 ± 1%	
4 - P2H	P28	4.3K ± 3%	

*To energize the relays +28 VDC must be applied to all terminals listed in Column A of Table III and the 28 V return must be connected to all terminals listed in Column B of the Table.

Table VII

Circuit Resistance

Mode 3 - For resistance checks, at the start of this test all standard relays will be energized and all latching relays latched.* During the test remove voltage from terminals designated. After completion of individual test the voltage is to be reapplied to the terminal from which it was removed.

From Terminal	To Terminal	Note	Resistance (Ohms)
1 - P2T	P2Hh	Remove + 28 vdc from P2N	820K + 10%, - 5%
2 - P2V	P2Hh	Remove + 28 vdc from P2P	820K + 10%, - 5%
3 - P2AA	P2Hh	Remove + 28 vdc from P2LL	820K + 10%, - 5%
4 - P2CC	P2Hh	Remove + 28 from P2MM	820K + 10%, - 5%
5 - P2EE	P2Hh	Remove + 28 from P2NN	820K + 10%, - 5%
6 - P2HH	P2Hh	Remove + 28 vdc from P2PP	820K + 10%, - 5%

*To energize the relays, + 28 vdc must be applied to all terminals listed in Column A of Table III and the 28 V return must be connected to all terminals listed in Column B of the table.

3.5.2.2 Relay operation. The resistance between terminals for each relay as indicated in Table VIII shall be less than 0.25 ohms unless otherwise specified, when tested as specified in 4.4.2.2.

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Table VIII

Relay Operation 1. (Energize all standard relays and latch all latching relays unless otherwise specified.) After each operation, return relays to original condition.

Check Continuity

Relay	From Terminal	To Terminal	Notes
K1	P1N	P1a	
K2	P1N	P1$\overline{aa}$	Also checks K3 latch.
K3	See K2		
K4	P1r	P1n	
K5	P1$\overline{bb}$	P1$\overline{b}$	
K6	P1$\overline{z}$	P1$\overline{ZZ}$	
K7	P1$\overline{k}$	P1f	
K8	P1$\overline{EE}$	P1$\overline{f}$	
K9	P1SS	P1$\overline{XX}$	Also checks K13 and K19 operation and K10 unlatching. Remove 28 V from P1V and apply to P1T.
K10	P1SS	P1XX	Also checks K19 operation and K9 unlatching. Remove 28 V from P1M and apply to P1J.
K11	P1m	P1JJ	Remove 28 V from P1M and P1D and apply 28 V to P1J.
K12	P1h	P1JJ	Remove 28 V from P1D.
K13	See K9 - latch		
K14	P1JJ	P1NN	Also checks K15 operation.
K15	See K14		
K16	P1M	P1e	Also checks K17 operation. Remove 28 V from P1V, P1M, P2TT, P1ee, and remove P1CC from the negative terminals. Resistance = 3K + 8%, - 5%.
K17	See K16		
K18	P1V	P1uu	Remove 28 V from P1V, P1M, P2TT, P1ee, and remove P1Z from the negative terminals. Resistance = 3K + 8%, - 5%.
K19	See K10		Apply (+) ohmmeter lead to P1V.

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Relay	From Terminal	To Terminal	Notes
K20	P1c	P1SS	Remove 28 V from P1A, P1M and apply 28 V to P1J.
K21	P2k	P2nn	
K22	P2f	P2pp	
K23	P1Mm	P1Nn	Apply 28 V to P2w. Operation of K24, K25, K26, K27, K28, and K29 relays is satisfactory if conditions of Section 3.5.2.3 are fulfilled.
K24			
K25			
K26			
K27			
K28			
K29			
K30	P2Ff	P2Dd	Remove + 28 V from P1S and P1P.
K31	P2YY	P2ZZ	
K32	P2e	P1EE	
K33	P2b	P2Mm	
K34	P2Hh	P2BB	Also checks K35 unlatching. Remove 28 V from P2RR. Apply + 28 V to P2UU. Resistance = 8.25K ± 1% ohm.
K35	P2Hh	P2dd	Remove 28 V from P2LL.
K36	P2Hh	P2DD	Also checks K37 unlatching. Remove 28 V from P2RR. Apply + 28 V to P2VV. Resistance = 8.25K ± 1% ohm.
K37	P2Hh	P2ee	Remove 28 V from P2MM.
K38	P2Hh	P2FF	Also checks K39 unlatching. Remove 28 V from P2RR. Apply + 28 V to P2WW. Resistance = 8.25K ± 1% ohm.
K39	P2Hh	P2bb	Remove 28 V from P2NN.
K40	P2Hh	P2S	Also checks K41 unlatching. Remove 28 V from P2RR. Apply + 28 V to P2XX. Resistance = 8.25K ± 1%.
K41	P2Hh	P2cc	Remove 28 V from P2PP.
K42	P2Hh	P2U	Also checks K43 unlatching. Remove 28 V from P2RR. Apply + 28 V to P2L. Resistance = 8.25K ± 1% ohm.
K43	P2Hh	P2mm	Remove 28 V from P2N.
K44	P2Hh	P2W	Also checks K45 unlatching. Remove 28 V from P2RR. Apply + 28 V to P2M. Resistance = 8.25K ± 1% ohm.

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Relay	From Terminal	To Terminal	Notes
K45	P2Hh	P2ff	Remove 28 V from P2P.
K46	P2Y	P2H	Remove + 28 V from all terminals except P2KK. Resistance = 4300 ± 3% ohm.
K47	P3C	P2E	
K48	P2Ff	P2Dd	Remove 28 V from P1B.

Table IX

Unlatching Relay Operation II. (Remove 28 V from all relays). Operate each relay individually.

Relay	To energize, apply 28 V to: Terminal (Pos.)	& Terminal (Neg.)	Check continuity From Terminal	To Terminal	Notes
K3 - Unlatch	P1L	P1hh	P1N	P1a	
K9 - Unlatch	See K10	Table VIII			
K10 - Unlatch	See K9	Table VIII			
K21 - Unlatch	P2Tt	P2B	P2k	P2nn	Check for open circuit rather than continuity.
K35 - Unlatch	See K34 in Table VIII				
K37 - Unlatch	See K36 in Table VIII				
K39 - Unlatch	See K38 in Table VIII				
K41 - Unlatch	See K40 in Table VIII				
K43 - Unlatch	See K42 in Table VIII				
K45 - Unlatch	See K44 in Table VIII				
K47 - Unlatch	P2SS	P2B	P2D	P2C	

3.5.2.3 Ledex pulse circuits #1 and #2. Output pulses shall be 50 ± 10 milliseconds wide when tested as specified in 4.4.2.3. Fall time of the output pulses shall be less than 5 milliseconds.

3.5.2.4 TDD Premature test delay circuit. The resistance between terminals P1N and P1a shall drop to no more than 0.25 ohms when tested as specified in 4.4.2.4.

3.5.2.5 Insulation resistance. The insulation resistance shall be greater than 20 megohms between each of the following terminals and all other terminals listed in Table X and Table XI and case. Testing shall be as specified in 4.4.2.5.

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Table X

Mode 1 - Standard relays in unenergized position and latching relays in unlatched position.*

P1B, P1N, P1JJ, P1TT**, P1XX, P1ZZ, P1b, P1c, P1h, P1m, P1n, P1r, P1u, P1v, P1z, P1aa, P1bb, P2A, P2C, P2E, P2H, P2JJ, P2YY, P2a, P2b, P2c, P2e, P2f, P2h, P2n, P2nn, P2pp, P2Ff, P2Mm, P1Nn, P1yy, P1Ee, P1F, P1Jj, P2Dd, P2w, P2m, P1tt, P2Hh, and P1Mm.

Unused Terminals
P1zz, P1Pp, P1Rr, P1Ss, P1Tt, P1Uu, P1Vv, and P1Ww shall be connected together and tested as one terminal.

*To ensure latching relays are unlatched, momentarily apply + 28 VDC to terminals P2L, P2M, P2SS, P2UU, P2VV, P2WW, P2X, P1L, P1J, P1T and P2Tt with the 28 V return applied to terminals P1EE and P1hh.
**Terminals P1TT, P1UU, P1WW, P1YY, P1d, P1w, P1x, P1y, and P1xx shall be connected together and tested as one terminal.

Mode 1A - Special circuit, insulation resistance test. Apply + 28 + 2 VDC to terminals P1D, P1M, and P1E with the return connected to terminals P1EE and P1Hh. Delete terminals P1B and P1JJ from the terminals listed under Mode 1 for this special test. Separately test between P1JJ and P1SS and all other terminals for Mode 1 as modified, tied together, and case.

Table XI

Mode 2 - All standard relays in the energized position and latching relays in latched position.*

P1c, P1h, P1m, P1u, P1v, P2C, P2D, P2JJ, P1XX, P2YY, P2a, P2b, P2c, P2f, P2n, P2z, P2bb, P2cc, P2dd, P2ee, P2ff, P2hh, P2mm, P2Hh, P2Nn, P2Rr, P2Uu, P1tt, and P1Kk.

Unused Terminals
P1zz, P1Pp, P1Rr, P1Ss, P1Tt, P1Uu, and P1Vv shall be connected together and tested as one terminal.
Terminals P1TT, P1UU, P1WW, P1YY, P1d, P1w, P1x, P1y and P1xx shall be connected together and tested as one terminal

*All relays must be energized for these tests. To energize all relays, apply 28 + 2 VDC across terminals designated in Table III.

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3.5.2.6 Diode characteristics. The resistance between the terminals listed in Table XII shall be within the specified limits when tested as specified in 4.4.2.6.

Table XII

Diode	From Terminal	To Terminal	Resistance (Ohms)	Apply 28 ± 2 vdc From Terminal (Positive)	To Terminal (Negative)
CR1	P1Ee	P1Ff	Open	P1VV	P1EE
	P1Ee	P1Ff	0.50 max.	P1EE	P1VV
CR2	P1Ee	P1Ff	Open	P1DD	P1EE
	P1Ee	P1Ff	0.50 max.	P1EE	P1DD
CR3	*P1BB	P1AA	10 max	-	-
	*P1AA	P1BB	1 meg min.	-	-
CR4	*P1R	P1X	10 max.	-	-
	*P1X	P1R	1 meg min.	-	-
CR5	P1Ee	P1Ff	Open	P1S	P1EE
	P1Ee	P1Ff	0.50 max.	P1EE	P1S
CR6	*P2k	P2y	30 max.	-	-
	&P2y	P2k	200K min.	-	-
CR8	*P2k	P2u	30 max.	-	-
	*P2u	P2k	200K min.	-	-
CR10	*P2ww	P2m	10 max.	-	-
	*P2m	P2ww	1 meg min.	-	-
CR11	*P2tt	P2m	10 max.	-	-
	*P2m	P2tt	1 meg. min.	-	-
CR14	*P1V	P2t	1 meg min.	-	-

*Indicates COM meter test lead.

4.3.2.1 Classification of characteristics.

4.3.2.1.1 Relay module #2 assembly (6A1A8) (see dwg 8833347).

Categories	Characteristics	Method of Inspection

Critical: None defined.

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Major:

101.	Improper location and orientation of connector with respect to alignment pins	Gage
102.	Defective solder joints	Visual/Manual
103.	Fastenings loose or missing	Visual/Manual
104.	Component missing	Visual
105.	Guide pins incorrectly located	Gage
106.	Cracked or broken connectors or damaged pins	Visual
107.	Objective evidence relative to material improper or missing (see 3.1)	Visual
108.	Serialization improper or missing (see 3.4)	Visual

Minor:

201.	Evidence of poor workmanship (see 3.8)	Visual
202.	Illegible or improper marking	Visual
203.	Improper finish or color	Visual

4.3.2.1.2 Preparation for delivery (see 8883507).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101.	Packing material and packaging not proper or missing	Visual
102.	Marking not in accordance with dwg	Visual

Minor:

201.	Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.

4.4.1 Calibration capability. Not applicable.

4.4.2 Operational limits.

4.4.2.1 Continuity and resistance. Resistance values between the terminals listed in 3.5.2.1 shall be measured with a digital ohmmeter of sufficient accuracy or its equivalent. Resistance values of less than one ohm shall be measured utilizing the four wire system. When necessary, relays

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shall be energized by application of 28 ± 2 vdc between terminals listed in Table III. All instrumentation shall be isolated from power line ground.

4.4.2.2 Relay operation. The resistance between terminals for each relay, as indicated in Table VIII and Table IX, shall be measured with a four wire resistance measuring instrument except for resistances greater than one ohm which will be measured with a digital ohmmeter of sufficient accuracy or a Wheatsone Bridge, and shall conform with 3.5.2.2. Energize all relays (see Table III) unless otherwise specified. Isolate instrumentation from power ground.

4.4.2.3 Ledex pulse circuits.

4.4.2.3.1 Ledex circuit #1. Apply 28 ± 2 vdc (positive) to terminals P1vv, P1B, P1ee, P1J, and P2K. Also apply 28 ± 2 vdc through a 3K ± 1% ohm resistor in series with terminal P1e. The 28 ± 2 vdc return (negative) shall be applied to P1EE and P2k. Connect the coil of a Ledex stepping switch, dwg 10970599, between terminals P2y and P2v to monitor output pulse of the module. The pulse shape shall be recorded using a Minneapolis-Honeywell No. 906 Visicorder or its equivalent. Pulse shape shall conform to parameters as specified in 3.5.2.3. Isolate instrumentation from power ground.

4.4.2.3.2 Ledex Circuit #2. Apply 28 ± 2 vdc (positive) to terminals P1B, P1T, P1ee, P1M, P2K, and P1ss. Also apply 28 ± 2 vdc through a 3K ± 1% ohm resistor in series with terminal P1e. The 28 ± 2 vdc return (negative) shall be applied to P1EE and P2k. Connect the coil of a Ledex stepping switch, dwg 10970599, between terminals P2u and P2v to monitor output pulse of the module. The pulse shape shall be recorded using Minneapolis-Honeywell No. 906 Visicorder or its equivalent. Pulse shape shall conform to parameters as specified in 3.5.2.3. Isolate instrumentation from power ground.

4.4.2.4 TDD Premature test delay circuit. Momentarily apply + 28 vdc between P1 (positive) and P1hh (negative). At time 10 ± 4 seconds from first application of + 28 vdc, a four wire resistance measurement shall be taken between the pins specified in 3.5.2.4.

4.4.2.5 Insulation resistance. The insulation resistance shall be measured between the terminals listed in Table X and Table XI of section 3.5.2.5 using a test voltage of 500 ± 50 vdc. Current shall be limited to 5 ma for this test.

4.4.2.6 Diode characteristics. The resistance between the terminals listed in Table XII of 3.5.2.6 shall be measured using a Triplett Model 630A multimeter or its equivalent. Apply 28 ± 2 vdc between the indicated terminals as required. Observe polarity as indicated. The COM meter test lead on the above Triplett meter is the positive side of the internal battery.

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APPENDIX J
PROGRAMMER ASSEMBLY (MAIN)
(6A1A9)

The following sections apply specifically to the programmer assembly (main). In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.

3.5.1 Calibration capability. Not applicable.

3.5.2 Operational limits.

3.5.2.1 Stepping operation. The assembly shall advance one position each time a voltage pulse as specified below is applied between terminals P1-P2 and P1-63. Testing shall be performed as specified in 4.4.2.1.

a. Step pulse "On" time: 45 ± 5 MS.
b. Step pulse "Off" time: 225 ± 25 MS.
c. Pulse amplitude: 19 to 23 VDC.

3.5.2.2 Homing operation. The assembly shall return to position 1 ("home" position) within 1 second when 28 ± 2 VDC is applied between terminals P1-62 and P1-65. Testing shall be performed as specified in 4.4.2.2.

3.5.2.3 Continuity and resistance. At each assembly position indicated in Table I, the resistance between the corresponding "Primary" and "Secondary" terminals shall be 0.1 ohm max. when tested as specified in 4.4.2.3 unless otherwise noted.

Table I

Assembly Position	Primary Terminal	Secondary Terminal	Assembly Position	Primary Terminal	Secondary Terminal
1 (home)	P2A	P1-76	1	P1-37	P1-46
2	P2A	P1-1	2	P1-37	P1-38
4	P2A	P1-1	3	P1-37	P1-39
6	P2A	P1-2	4	P1-37	P1-40
8	P2A	P1-3	5	P1-37	P1-66
9	P2A	P1-4	6	P1-37	P1-41
10	P2A	P1-5	7	P1-37	P1-50
11	P2A	P1-7	9	P1-37	P1-42
12	P2A	P1-8	11	P1-37	P1-43
13	P2A	P1-10	13	P1-37	P1-44
14	P2A	P1-11	15	P1-37	P1-45
15	P2A	P1-12	17	P1-37	P1-52
16	P2A	P1-13	19	P1-37	P1-51
17	P2A	P1-14	20	P1-37	P1-39
18	P2A	P1-15	21	P1-37	P1-30
19	P2A	P1-16	23	P1-37	P1-49

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Assembly Position	Primary Terminal	Secondary Terminal	Assembly Position	Primary Terminal	Secondary Terminal
21	P2A	P1-17	24	P1-37	P1-48
23	P2A	P1-18	26	P1-37	P1-47
24	P2A	P1-18	2	P1-73	P1-54
25	P2A	P1-20	3	P1-73	P1-55
26	P2A	P1-20	4	P1-73	P1-54
1	P1-21	P1-29	5	P1-73	P1-55
2	P1-21	P1-22	6	P1-73	P2-EE
3	P1-21	P1-36	20	P1-73	P1-55
4	P1-21	P1-22	1	P2-B	P2-D
5	P1-21	P1-23	2	P2-B	P1-56
6	P1-21	P1-24	3	P2-B	P2-C
7	P1-21	P1-33	4	P2-B	P1-56
9	P1-21	P1-25	5	P2-B	P1-72
11	P1-21	P1-26	6	P2-B	P2-D
13	P1-21	P1-27	8	P2-B	P2-D
15	P1-21	P1-28	9	P2-B	P2-D
17	P1-21	P1-35	10	P2-B	P2-D
19	P1-21	P1-34	11	P2-B	P2-D
20	P1-21	P1-36	12	P2-B	P2-D
21	P1-21	P1-32	13	P2-B	P2-D
23	P1-21	P1-31	14	P2-B	P2-D
24	P1-21	P1-31	15	P2-B	P2-D
16	P2-B	P2-D	11	P1-53	P2-Z
17	P2-B	P2-D	12	P1-53	P2-Z
18	P2-B	P2-D	13	P1-53	P2-Z
19	P2-B	P2-D	14	P1-53	P2-Z
20	P2-B	P2-C	15	P1-53	P2-Z
21	P2-B	P2-D	16	P1-53	P2-Z
23	P2-B	P2-E	17	P1-53	P2-Z
24	P2-B	P2-E	18	P1-53	P2-Z
25	P2-B	P2-F	19	P1-53	P2-Z
26	P2-B	P2-F	20	P1-53	P2-d
1	P2-N	P2-P	21	P1-53	P2-$\overline{Z}$
2	P2-N	P1-57	22	P1-53	P2-Z
3	P2-N	P1-58	23	P1-53	P2-Z
4	P2-N	P1-57	24	P1-53	P2-Z
5	P2-N	P1-59	25	P1-53	P2-Z
6	P2-N	P2-P	26	P1-53	P2-Z
8	P2-N	P2-P	1	P1-74	P2-DD
9	P2-N	P2-P	2	P1-74	P2-e
10	P2-N	P2-P	3	P1-74	P2-$\overline{f}$
11	P2-N	P2-P	4	P1-74	P2-$\overline{e}$
12	P2-N	P2-P	5	P1-74	P2-$\overline{h}$
13	P2-N	P2-P	6	P1-74	P2-$\overline{e}$
14	P2-N	P2-P	7	P1-74	P2-$\overline{J}$

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Assembly Position	Primary Terminal	Secondary Terminal	Assembly Position	Primary Terminal	Secondary Terminal
15	P2-N	P2-P	8	P1-74	P2-e
16	P2-N	P2-P	9	P1-74	P2-e
17	P2-N	P2-P	10	P1-74	P2-e
18	P2-N	P2-P	11	P1-74	P2-e
19	P2-N	P2-P	12	P1-74	P2-e
20	P2-N	P1-58	13	P1-74	P2-e
21	P2-N	P2-P	14	P1-74	P2-e
23	P2-N	P2-S	15	P1-74	P2-e
24	P2-N	P2-S	16	P1-74	P2-e
25	P2-N	P2-T	17	P1-74	P2-e
26	P2-N	P2-T	18	P1-74	P2-e
2	P1-53	P2-Z	19	P1-74	P2-e
3	P1-53	P2-a	20	P1-74	P2-k
4	P1-53	P2-Z	21	P1-74	P2-e
5	P1-53	P2-b	22	P1-74	P2-e
6	P1-53	P2-Z	23	P1-74	P2-e
7	P1-53	P2-c	24	P1-74	P2-e
8	P1-53	P2-Z	25	P1-74	P2-e
9	P1-53	P2-Z	26	P1-74	P2-e
10	P1-53	P2-Z	2	P1-73	P2-m
4	P1-73	P2-m	24	P1-73	P2-z
6	P1-73	P2-m	22	P2-N	P2-T
8	P1-73	P2-m	25	P1-73	P2-AA
9	P1-73	P2-m	26	P1-73	P2-r
10	P1-73	P2-m	*1	P1-70	P1-71
11	P1-73	P2-m	2	P1-75	P2-C
12	P1-73	P2-m	2	P1-58	P2-W
13	P1-73	P2-m	2	P2-R	P2-W
14	P1-73	P2-n	2	P2-F	P2-L
15	P1-73	P2-n	2	P2-J	P2-B
16	P1-73	P2-n	2	P2-D	P2-K
17	P1-73	P2-n	2	P2-N	P2-U
18	P1-73	P2-n	2	P2-P	P2-V
19	P1-73	P2-n	2	P2-T	P2-Y
21	P1-73	P2-n	2	P2-S	P2-X
22	P1-73	P2-n	*2	P1-67	P1-70
23	P1-73	P2-n	23	P1-73	P1-77
24	P1-73	P2-n	24	P1-73	P1-77
25	P1-73	P2-p	*24	P1-63	P1-65
26	P1-73	P2-p	*1	P2-BB	P1-65
2	P1-73	P2-r	*1	P1-63	P1-64
4	P1-73	P2-s			
6	P1-73	P2-t	*1.6 ohms max.		
8	P1-73	P2-u			
9	P1-73	P2-v			
10	P1-73	P2-w			

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Assembly Position	Primary Terminal	Secondary Terminal	Assembly Position	Primary Terminal	Secondary Terminal
11	P1-73	P2-x			
12	P1-73	P2-y			
13	P1-73	P2-z			
14	P1-73	P2-AA			
15	P1-73	P2-r			
16	P1-73	P2-s			
17	P1-73	P2-t			
18	P1-73	P2-u			
19	P1-73	P2-v			
20	P1-73	P2-CC			
21	P1-73	P2-w			
22	P1-73	P2-x			
23	P1-73	P2-y			
22	P2-A	P1-20			
22	P2-B	P2-F			

3.5.2.4 Insulation resistance. Insulation resistance shall be 20 megohms minimum in Mode 1 and Mode 2 below when tested as specified in 4.4.2.4.

Mode 1 - At each assembly position, the insulation resistance between each terminal listed below and all other terminals listed and case shall be as specified except for the terminals removed at the indicated assembly positions. Unused terminals may be connected together and tested as one terminal.

P1 - 1, 2, 3, 4, 5, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 54, 55, 56, 57, 58, 59, 65, 66, 70, 72, 76, 77, 64, 67, 71.

P2 - C, D, E, F, P, S, T, Z, a, b, c, d, e, f, h, j, k, m, n, p, r, s, t, u, v, w, x, y, z, AA, BB, CC, DD, EE.

Unused terminals: P1 - 60, 78, 79, 80, and 82.
P2 - H, M, FF, and HH.

Assembly Position	Remove From List
1 (home)	P1-71, P2-BB
2	P2-m, P1-54, 65, 70
3	P1-65, 70
4	P2-m, P1-54, 65, 70
5	P1-65, 70
6	P2-m, EE, P1-65, 70

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Assembly Position	Remove From List
7	P1-65, 70
8	P2-m, P1-65, 70
9	P2-m, P1-65, 70
10	P2-m, P1-65, 70
11	P2-m, P1-65, 70
12	P2-m, P1-65, 70
13	P2-m, P1-65, 70
14	P2-n, P1-65, 70
15	P2-n, P1-65, 70
16	P2-n, P1-65, 70
17	P2-n, P1-65, 70
18	P2-n, P1-65, 70
19	P2-n, P1-65, 70
20	P2-n, P1-55, 65, 70
21	P2-n, P1-65, 70
22	P2-n, P1-65, 70
23	P2-n, P1-65, 70, 77
24	P2-n, P1-65, 70, 77
25	P2-p, P1-65, 70
26	P2-p, P1-65, 70

Mode 2 - At each assembly position, the insulation resistance shall be as specified between each of the terminals listed below and all other terminals listed and case.

P1 - 21, 37, 53, 73, and 74.
P2 - A, B, and N.

4.3.2.1 Classification of characteristics.
4.3.2.1.1 Programmer assembly (main) (6A1A9) (see dwg 8833314).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connector with respect to alignment pins	Gage
102.	Fastenings loose or missing	Manual/Visual
103.	Cracked or broken connector or damaged pins	Visual
104.	Defective solder joints	Visual/Manual
105.	Switch assembly mounting screws loose	Manual

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106. Component missing.....Visual
107. Objective evidence relative to material improper or missing (see 3.1).....Visual
108. Serialization improper or missing (see 3.4).....Visual

Minor:
201. Evidence of poor workmanship (see 3.8).....Visual
202. Illegible or improper marking.....Visual
203. Improper finish or color.....Visual

4.3.1.1.2 Preparation for delivery (see 8883505).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Packing material and packaging not proper or missing	Visual
102.	Marking not in accordance with dwg	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.
4.4.1 Calibration capability. Not applicable.
4.4.2 Operational limits.
4.4.2.1 Stepping test. With the assembly in its home position, apply a pulsing voltage as specified in 3.5.2.1 and the assembly shall advance through one complete cycle and return to its home position after 26 pulses are applied. Pulses may be counted using a Hewlett-Packard Model 522B counter or other suitable means. The above test is to be performed two times in succession.
4.4.2.2 Homing test. With the assembly stepped one position off from home position, apply 28 + 2 VDC as specified in 3.5.2.2. The time interval between application of the 28 VDC and the closing of the circuit between terminals P1-29 and P1-21 (home position) shall be measured using a Hewlett-Packard Model 522B counter or its equivalent. The interval measured shall meet the requirements of 3.5.2.2. The above test is to be performed two times in succession.

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4.4.2.3 Continuity and resistance. Momentarily apply 28 ± 2 vdc to terminals P1-62 and P1-63 as necessary to advance the assembly to the positions indicated in 3.5.2.3, Table I. Resistance measurements, indicating continuity between the terminals listed in Table I, shall be measured using a Keithley milliohmmeter, Model 502, or equivalent. Resistance shall be as specified in 3.5.2.3.

4.4.2.4 Insulation resistance. Momentarily apply 28 ± 2 vdc to terminals P1-62 and P1-63 as necessary to advance the assembly to all positions as specified in 3.5.2.4. Insulation resistance at each assembly position shall be measured between the terminals listed in 3.5.2.4 using a test voltage of 500 ± 50 vdc. Current shall be limited to 5 ma for this test.

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APPENDIX K
PROGRAMMER ASSEMBLY (CONTINUITY)
(6A1A10)

The following sections apply specifically to the programmer assembly (continuity). In addition to the requirements found in this appendix, this assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.

3.5.1 Calibration capability. Not applicable.

3.5.2 Operational limits.

3.5.2.1 Stepping operation. The assembly shall advance one position each time a voltage pulse as specified below is applied between terminals P2x and P2y. Testing shall be performed as specified in 4.4.2.1.

Step pulse "on" time: 45 ± 5 MS
Step pulse "off" time: 225 ± 25 MS
Pulse amplitude: 19 to 23 volts

3.5.2.2 Homing operation. The assembly shall return to position 1 ("home" position) within 1 second when 28 ± 2 VDC is applied between terminals P2BB and P2y. Testing shall be performed as specified in 4.4.2.2.

3.5.2.3 Continuity and resistance. At each assembly position indicated in Table I, the resistance between the corresponding "Primary" and "Secondary" terminals shall be 0.1 ohm max. when tested as specified in 4.4.2.3 unless otherwise noted.

Table I

Assembly Position	Primary Terminal	Secondary Terminal	Assembly Position	Primary Terminal	Secondary Terminal
2	P2-X	P2-Y	10	P2-X	P2-h
3	P2-X	P2-Z	11	P2-X	P2-$\overline{J}$
4	P2-X	P2-a	12	P2-X	P2-$\overline{Y}$
5	P2-X	P2-$\overline{b}$	13	P2-X	P2-Z
6	P2-X	P2-$\overline{c}$	14	P2-X	P2-a
7	P2-X	P2-$\overline{d}$	15	P2-X	P2-$\overline{b}$
8	P2-X	P2-$\overline{c}$	16	P2-X	P2-$\overline{c}$
9	P2-X	P2-$\overline{f}$	17	P2-X	P2-$\overline{d}$
18	P2-X	P2-$\overline{e}$	19	P2-X	P2-$\overline{R}$
19	P2-X	P2-$\overline{f}$	20	P2-X	P2-R
20	P2-X	P2-$\overline{h}$	21	P2-X	P2-S
21	P2-X	P2-$\overline{J}$	22	P2-X	P2-S
22	P2-X	P2-$\overline{Y}$	23	P2-X	P2-S

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Assembly Position	Primary Terminal	Secondary Terminal	Assembly Position	Primary Terminal	Secondary Terminal
23	P2-X	P2-Z	24	P2-X	P2-S
24	P2-X	P2-a	25	P2-X	P2-S
25	P2-X	P2-b	26	P2-X	P2-S
26	P2-X	P2-c	26	P2-L	P2-M
1	P2-w	P2-j	26	P2-K	P1-Bb
1	P2-v	P2-h	26	P2-F	P1-aa
1	P2-u	P2-f	26	P2-E	P1-ZZ
1	P2-t	P2-e	26	P2-A	P1-Z
1	P2-s	P2-d	2	P2-A	P1-A
1	P2-r	P2-c	2	P2-E	P1-AA
1	P2-p	P2-b	2	P2-F	P1-a
1	P2-n	P2-a	2	P2-K	P1-bb
1	P2-m	P2-Z	3	P2-K	P1-cc
1	P2-k	P2-Y	3	P2-F	P1-b
1	P2-B	P2-B	3	P2-E	P1-BB
1	P2-P	P2-U	3	P2-A	P1-B
1	P2-R	P2-V	4	P2-A	P1-C
1	P2-S	P2-W	4	P2-E	P1-CC
1	P2-L	P2-N	4	P2-F	P1-c
1	P2-X	P2-T	4	P2-K	P1-dd
2	P2-X	P2-P	5	P2-K	P1-dd
3	P2-X	P2-P	5	P2-F	P1-d
4	P2-X	P2-P	5	P2-E	P1-CC
5	P2-X	P2-P	5	P2-A	P1-D
6	P2-X	P2-P	6	P2-A	P1-E
7	P2-X	P2-P	6	P2-E	P1-DD
8	P2-X	P2-P	6	P2-F	P1-e
9	P2-X	P2-P	6	P2-K	P1-ee
10	P2-X	P2-P	7	P2-K	P1-ff
11	P2-X	P2-R	7	P2-F	P1-f
12	P2-X	P2-R	7	P2-E	P1-EE
13	P2-X	P2-R	7	P2-A	P1-F
14	P2-X	P2-R	8	P2-A	P1-H
15	P2-X	P2-R	8	P2-E	P1-FF
16	P2-X	P2-R	8	P2-F	P1-h
17	P2-X	P2-R	8	P2-K	P1-hh
18	P2-X	P2-R	9	P2-K	P1-jj
9	P2-F	P1-j	19	P2-A	P1-T
9	P2-E	P1-HH	20	P2-A	P1-U
9	P2-A	P1-J	20	P2-E	P1-TT
10	P2-A	P1-K	20	P2-F	P1-v
10	P2-E	P1-JJ	20	P2-K	P1-vv
10	P2-F	P1-k	21	P2-K	P1-ww
10	P2-K	P1-kk	21	P2-F	P1-w
11	P2-K	P1-mm	21	P2-E	P1-UU

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Assembly Position	Primary Terminal	Secondary Terminal	Assembly Position	Primary Terminal	Secondary Terminal
11	P2-F	P1-m	21	P2-A	P1-v
11	P2-E	P1-KK	22	P2-A	P1-W
11	P2-A	P1-L	22	P2-E	P1-VV
12	P2-A	P1-M	22	P2-F	P1-x
12	P2-E	P1-BB	22	P2-K	P1-xx
12	P2-F	P1-n	23	P2-K	P1-yy
12	P2-K	P1-cc	23	P2-F	P1-y
13	P2-K	P1-j	23	P2-E	P1-WW
13	P2-F	P2-H	23	P2-A	P1-X
13	P2-E	P1-J	24	P2-A	P1-X
13	P2-A	P1-C	24	P2-E	P1-XX
14	P2-A	P1-N	24	P2-F	P1-y
14	P2-E	P1-LL	24	P2-K	P1-zz
14	P2-F	P1-p	25	P2-K	P1-Aa
14	P2-K	P1-nn	25	P2-F	P1-z
15	P2-K	P1-pp	25	P2-E	P1-YY
15	P2-F	P1-r	25	P2-A	P1-Y
15	P2-E	P1-MM	*1	P2-x	P2-DD
15	P2-A	P1-P	*2	P2-x	P2-BB
16	P2-A	P2-D	*1	P2-CC	P2-BB
16	P2-E	P1-NN	*2	P2-AA	P2-BB
16	P2-F	P2-J	**	P2-L	P2-z
16	P2-K	P1-rr			
17	P2-K	P1-ss			
17	P2-F	P1-s			
17	P2-E	P1-PP			
17	P2-A	P1-R			
18	P2-A	P1-S			
18	P2-E	P1-RR			
18	P2-F	P1-t			
18	P2-K	P1-tt			
19	P2-K	P1-uu			
19	P2-F	P1-u			
19	P2-E	P1-SS			

*1.6 ohms max.

**Apply 28 VDC from P2y to P2x for this test. 1.6 ohms max.

Do not apply 28 V for longer than 1/2 minute.

3.5.2.4 Insulation resistance. Insulation resistance shall be 20 megohms minimum in Mode 1 and Mode 2 below when tested as specified in 4.4.2.4.

Mode 1 - At each assembly position, the insulation resistance between each terminal listed below and all other terminals listed and case shall be as specified except for the terminals removed at the indicated assembly positions. Unused terminals may be connected together and tested as one terminal.

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P1 - A, B, C, D, E, F, H, J, K, L, M, N, P, R, S, T, U, V, W, X, Y, Z, a, f, b, c, d, e, h, j, k, m, n. p, r, s, t, u, v, w, x, y, z, AA, BB, CC, DD, EE, FF, HH, JJ, KK, LL, MM, NN, PP, RR, SS, VV, TT, UU, WW, XX, YY, ZZ, aa, bb, cc, dd, ee, ff, hh, jj, kk, mm, nn pp, rr, ss, tt, xx, uu, vv, ww, yy, zz, Aa, and Bb.

P2 - C, D, H, J, L, M, N, P, R, S, T, Y, Z, a, b, c, d, e, f, h, j, z, x, AA, BB, CC, and DD.

Unused Terminals: P1 - Cc, Dd, Ee, Ff, Hh, Jj, Kk, Mm, Nn, Pp, Rr, Ss, Tt, Uu, Vv, Ww, P2EE, FF, and HH.

Assembly Position	Remove From List
1	P2-CC, DD, L
2 thru 10	P2-P, AA, x
11 thru 20	P2-R, AA, x
21 thru 26	P2-S, AA, x

Mode 2 - At each assembly position, the insulation resistance shall be as specified between each of the following terminals and the others listed and case.

P2 - A, E, F, L, K, and X.

4.3.2.1 Classification of characteristics.

4.3.2.1.1 Programmer assembly (continuity) (6A1A10) (see dwg 8833315).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connector with respect to alignment pins	Gage
102.	Switch assembly mounting screws loose	Manual
103.	Defective solder joints	Visual/Manual
104.	Cracked or broken connectors or damaged pins	Visual
105.	Component missing	Visual
106.	Fastenings loose or missing	Visual/Manual
107.	Objective evidence relative to material improper or missing (see 3.1)	Visual

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108. Serialization improper or missing (see 3.4).....Visual

Minor:

201. Evidence of poor workmanship (see 3.8).....Visual
202. Illegible or improper marking.....Visual
203. Improper finish or color.....Visual

4.3.1.1.2 Preparation for delivery (see 8883505).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Packing material and packaging not proper or missing.....Visual
102. Marking not in accordance with dwg.....Visual

Minor:

201. Evidence of poor workmanship (see 3.8).....Visual

4.4 Test methods.

4.4.1 Calibration capability. Not applicable.

4.4.2 Operational limits.

4.4.2.1 Stepping test. With the assembly in its home position, apply a pulsing voltage as specified in 3.5.2.1 and the assembly shall advance through one complete cycle and return to its home position after 26 pulses are applied. Pulses may be counted using a Hewlett-Packard Model 522B counter or other suitable means. Perform the above test two times in succession.

4.4.2.2 Homing test. With the assembly stepped one position off from home position, apply 28 + 2 VDC as specified in 3.5.2.2. The time interval between application of the 28 VDC and the closing of the circuit between terminals P2-BB and P2-CC (home position) shall be measured using a Hewlett-Packard Model 522B counter or its equivalent. The interval measured shall meet the requirements of 3.5.2.2. Perform the above test two times in succession.

4.4.2.3 Continuity and resistance. Momentarily apply 28 + 2 VDC to terminals P2-x and P2-y as necessary to advance the assembly to the position indicated in 3.5.2.3, Table I. Resistance measurements, indicating continuity between the terminals listed in Table I, shall be measured using a Keithley milliohmmeter, Model 502, or equivalent. Resistance shall be as specified in 3.5.2.3.

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4.4.2.4 Insulation resistance. Momentarily apply 28 ± 2 vdc to terminals P2-x and P2-y as necessary to advance the assembly to all positions as specified in 3.5.2.4. Insulation resistance at each position shall be measured between the terminals listed in 3.5.2.4 using a test voltage of 500 ± 50 vdc. Current shall be limited to 5 ma for this test.

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APPENDIX L
PROGRAMMER ASSEMBLY (OPEN CIRCUIT)
(6A1A11)

The following sections apply specifically to the programmer assembly (open circuit). In addition to the requirements found in this appendix, this assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.

3.5.1 Calibration capability. Not applicable.

3.5.2 Operational limits.

3.5.2.1 Stepping operation. The assembly shall advance one position each time a voltage pulse as specified below is applied between terminals P2n and P2m. Testing shall be performed as specified in 4.4.2.1.

Step pulse "On" time: 45 ± 5 milliseconds.
Step pulse "Off" time: 225 ± 25 milliseconds.
Pluse amplitude: 19 to 23 volts.

3.5.2.2 Homing operation. The assembly shall return to position 1(home position) within 1 second when 28 ± 2 vdc is applied between terminals P2n and P2t. Testing shall be performed as specified in 4.4.2.2.

3.5.2.3 Continuity and resistance. At each assembly position indicated in Table I, the resistance between the corresponding "Primary" and "Secondary" terminals, shall be 0.1 ohm maximum when tested as specified in 4.4.2.3 unless otherwise noted.

Table I

Assembly Position	Primary Terminal	Secondary Terminal	Assembly Position	Primary Terminal	Secondary Terminal
1	P2-H	P2-K	13	P1-27	P1-22
1	P2-P	P2-N	14	P2-C	P1-23
2	P2-P	P2-$\overline{v}$	15	P2-C	P1-23
3	P2-P	P2-$\overline{v}$	16	P2-C	P1-23
4	P2-P	P2-$\overline{v}$	17	P2-C	P1-23
5	P2-P	P2-$\overline{v}$	18	P2-C	P1-23
6	P2-P	P2-$\overline{v}$	19	P2-C	P1-23
7	P2-P	P2-$\overline{v}$	19	P1-28	P1-23
8	P2-P	P2-$\overline{v}$	20	P2-C	P1-24
9	P2-P	P2-$\overline{v}$	20	P1-29	P1-24
10	P2-P	P2-$\overline{v}$	21	P2-C	P2-F
11	P2-P	P2-$\overline{L}$	22	P2-C	P1-25
12	P2-P	P2-L	22	P1-30	P1-25
13	P2-P	P2-L	23	P2-C	P1-10
14	P2-P	P2-L	23	P1-18	P1-10
15	P2-P	P2-L	24	P2-C	P2-F

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16	P2-P	P2-L	25	P2-C	P2-F
17	P2-P	P2-L	26	P2-C	P2-F
18	P2-P	P2-L	2	P2-A	P1-1
19	P2-P	P2-L	3	P2-A	P1-2
20	P2-P	P2-L	4	P2-A	P1-3
21	P2-P	P2-M	5	P2-A	P1-4
22	P2-P	P2-M	6	P2-A	P1-5
23	P2-P	P2-M	7	P2-A	P1-7
24	P2-P	P2-M	8	P2-A	P1-1
25	P2-P	P2-M	9	P2-A	P1-2
26	P2-P	P2-M	10	P2-A	P1-3
2	P2-C	P1-21	11	P2-A	P1-4
3	P2-C	P1-21	12	P2-A	P1-5
4	P2-C	P1-21	13	P2-A	P1-7
5	P2-C	P1-21	14	P2-A	P1-1
6	P2-C	P1-21	15	P2-A	P1-2
7	P2-C	P1-21	16	P2-A	P1-3
8	P1-26	P1-21	17	P2-A	P1-4
8	P2-C	P1-22	18	P2-A	P1-5
9	P2-C	P1-22	19	P2-A	P1-7
10	P2-C	P1-22	20	P2-A	P1-8
11	P2-C	P1-22	21	P2-A	P2-D
12	P2-C	P1-22	22	P2-A	P1-8
13	P2-C	P1-22	23	P2-A	P1-11
24	P2-A	P2-D	4	P2-F	P1-47
25	P2-A	P2-D	5	P2-F	P1-47
26	P2-A	P2-D	6	P2-F	P1-47
26	P1-11	P1-20	7	P2-F	P1-47
26	P1-17	P1-7	7	P1-53	P1-47
26	P1-1	P1-16	8	P2-F	P1-48
26	P1-2	P1-15	9	P2-F	P1-48
26	P1-3	P1-14	10	P2-F	P1-48
26	P1-4	P1-13	11	P2-F	P1-48
26	P1-5	P1-12	12	P2-F	P1-48
26	P1-64	P2-$\overline{d}$	13	P2-F	P1-48
26	P1-65	P2-$\overline{c}$	13	P1-54	P1-48
2	P2-D	P1-$\overline{31}$	14	P2-F	P1-49
3	P2-D	P1-32	15	P2-F	P1-49
4	P2-D	P1-33	16	P2-F	P1-49
5	P2-D	P1-34	17	P2-F	P1-49
6	P2-D	P1-35	18	P2-F	P1-49
7	P2-D	P1-36	19	P2-F	P1-49
8	P2-D	P1-31	19	P1-55	P1-49
9	P2-D	P1-32	2	P2-P	P2-R
10	P2-D	P1-33	3	P2-P	P2-S
11	P2-D	P1-34	4	P2-P	P2-T
12	P2-D	P1-35	5	P2-P	P2-U
13	P2-D	P1-36	6	P2-P	P2-V
14	P2-D	P1-31	7	P2-P	P2-W
15	P2-D	P1-32	8	P2-P	P2-X
16	P2-D	P1-33	9	P2-P	P2-Y

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17	P2-D	P1-34	10	P2-P	P2-Z
18	P2-D	P1-35	11	P2-P	P2-a
19	P2-D	P1-36	12	P2-P	P2-R
19	P1-40	P1-31	13	P2-P	P2-S
19	P1-41	P1-32	14	P2-P	P2-T
19	P1-42	P1-33	15	P2-P	P2-U
19	P1-43	P1-34	16	P2-P	P2-V
19	P1-44	P1-35	17	P2-P	P2-W
19	P1-45	P1-36	18	P2-P	P2-X
20	P2-D	P1-38	19	P2-P	P2-Y
22	P2-D	P1-38	20	P2-P	P2-Z
2	P2-F	P1-47	21	P2-P	P2-a
3	P2-F	P1-47	22	P2-P	P2-R
23	P2-P	P2-S	23	P2-F	P1-52
24	P2-P	P2-T	23	P1-37	P1-52
25	P2-P	P2-U	23	P1-39	P1-46
26	P2-P	P2-V	23	P1-39	P2-E
1	P1-67	P2-w	23	P1-39	P2-D
1	P1-58	P2-k	25	P2-H	P2-J
1	P1-59	P2-j	25	P2-b	P1-66
1	P1-60	P2-h	1	P1-56	P1-50
1	P1-62	P2-f	*1	P2-m	P2-u
1	P1-63	P2-e	*1	P2-r	P2-t
20	P2-F	P1-50	*2	P2-s	P2-t
22	P2-F	P1-51	*2	P2-m	P2-t
22	P1-57	P1-51	**	P2-H	P2-p

* - 1.6 ohms max
** - Test with 28 volts dc applied between pins P2n & m. 1.6 ohms max.
Do not apply 28 v longer than 1/2 minute.

3.5.2.4 Insulation resistance. Insulation resistance shall be 20 megohms minimum in Mode 1 and Mode 2 below when specified in 4.4.2.4.

Mode 1 - At each assembly position the insulation resistance between each terminal listed below and all other terminals listed and case shall be as specified except for the terminals removed at the indicated assembly positions. Unused terminals may be connected together and tested as one terminal.

P1- 1,2,3,4,5,7,8,10,11,21,22,23,24,25,31,32,33,34,35, 36,37,38,39,47,48,49,50,51,58,59,60,62,63,64,65, 66 and 67.

P2- H,J,K,L,M,N,R,S,T,U,V,W,X,Y,Z,a,p,v,m,r,s,t and u.

Unused terminals: P1- 70,71,72,73,74,75,76,77,78,79,80 and 82.
P2- B,x,y,z,AA,BB,CC,DD,EE,FF,and HH.

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6	P2-v,A,C,s,t
7	P2-v,A,C,s,t
8	P2-v,A,C,s,t
9	P2-v,A,C,s,t
10	P2-v,A,C,s,t
11	P2-L,A,C,s,t
12	P2-L,A,C,s,t
13	P2-L,A,C,s,t
14	P2-L,A,C,s,t
15	P2-L,A,C,s,t
16	P2-L,A,C,s,t
17	P2-L,A,C,s,t
18	P2-L,A,C,s,t
19	P2-L,A,C,s,t
20	P2-L,A,C,s,t
21	P2-M,A,s,t
22	P2-M,A,C,s,t
23	P2-M,A,C,s,t
24	P2-M,A,s,t
25	P2-M,s,t,H
26	P2-M,s,t

Mode 2 - At each assembly position the insulation resistance shall be as specified between each of the following terminals and the others listed.

P2 - A,C,D,F,H and P.
Remove F from list at steps 21 and 24.
Remove F from list at steps 25 and 26.
Remove A from list at steps 21,24,25 and 26.

4.3.2.1 Classification of characteristics.

4.3.2.1.1 Programmer assembly (open circuit) (6A1A11) (see dwg 8833316).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connector with respect to alignment pins.....	Gage
102.	Defective solder joints.....	Visual/Manual
103.	Fastenings loose or missing.....	Visual/Manual
104.	Component missing.....	Visual
105.	Cracked or broken connectors or damaged pins.....	Visual
106.	Switch assembly mounting screws loose.....	Manual

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107. Objective evidence relative to material improper or missing (see 3.1).....Visual
108. Serialization improper or missing (see 3.4).....Visual

Minor:
201. Evidence of poor workmanship (see 3.8).....Visual
202. Illegible or improper marking.....Visual
203. Improper finish or color.....Visual

4.3.1.1.2 Preparation for delivery (see dwg 8883505).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Packing material and packaging not proper or missing	Visual
102.	Marking not in accordance with dwg	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual

4.4 Test methods.
4.4.1 Calibration capability. Not applicable.
4.4.2 Operational limits.
4.4.2.1 Stepping test. With the assembly in its home position apply a pulsing voltage as specified in 3.5.2.1 and the assembly shall advance through one complete cycle and return to its home position after 26 pulses are applied. Pulses may be counted using a Hewlett-Packard Model 522B counter or other suitable means. Perform the above test two times in succession.
4.4.2.2 Homing test. With the assembly stepped one position off of "home" position apply 28 ± 2 vdc as specified in 3.5.2.2. The time interval between application of the 28 vdc and the closing of the circuit between terminal P2-N and P2-P ("home"position) shall be measured using a Hewlett-Packard Model 522B counter or its equivalent. The Interval measured shall meet the requirements of 3.5.2.2. Perform the above test two times in succession.
4.4.2.3 Continuity and resistance. Momentarily apply 28 ± 2 volts dc terminals P2-m and P2-n as necessary to advance the assembly to the positions indicated in 3.5.2.3, Table I. Resistance measurements, indicating continuity between the terminals listed in Table I shall be measured using a Keithley Milliohmmeter, Model 502, or equivalent. Resistance shall be as specified in 3.5.2.3.

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4.4.2.4 Insulation resistance. Momentarily apply 28 ± 2 volts dc terminal P2-m and P2-n as necessary to advance the assembly to all positions as specified in 3.5.2.4. Insulation resistance at each position shall be measured between the terminals listed in 3.5.2.4 using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma for this test.

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APPENDIX M
PROGRAMMER ASSEMBLY (TDD)
(6A1A12)

The following sections apply specifically to the programmer assembly (TDD.) In addition to the requirements found in this appendix, this assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.
3.5.1 Calibration capability. Not applicable.
3.5.2 Operational limits.
3.5.2.1 Stepping operation. The assembly shall advance one position each time a voltage pulse as specified below is applied between terminals P1-48 and P1-49. Testing shall be performed as specified in 4.4.2.1.

Step pulse "on" time: 45 ± 5 ms
Step pulse "off" time: 225 ± 25 ms
Pulse amplitude: 19 to 23 volts

3.5.2.2 Homing operation. The assembly shall return to position 1 ("home" position) within 2 second when 28 ± 2 vdc is applied between terminals P1-49 and P2J. Testing shall be performed as specified in 4.4.2.4.
3.5.2.3 Continuity and resistance. At each assembly position indicated in Table I, the resistance between the corresponding "Primary" and "Secondary" terminals shall be 0.1 ohm max. when tested as specified in 4.4.2.3 unless otherwise noted.

Table I

Assembly Position	Primary Terminal	Secondary Terminal	Assembly Position	Primary Terminal	Secondary Terminal
1	P1-29	P1-36	5	P2-b	P2-R
1	P2-b	P2-P	6	P2-$\overline{b}$	P2-R
2	P2-$\overline{b}$	P2-R	7	P2-$\overline{b}$	P2-R
3	P2-$\overline{b}$	P2-R	8	P2-$\overline{b}$	P2-R
4	P2-$\overline{b}$	P2-R	9	P2-$\overline{b}$	P2-R
10	P2-$\overline{b}$	P2-R	1	P2-$\overline{R}$	P2-BB
11	P2-$\overline{b}$	P2-S	1	P2-S	P2-V
12	P2-$\overline{b}$	P2-S	1	P2-a	P2-T
13	P2-$\overline{b}$	P2-S	2	P2-$\overline{A}$	P1-1
14	P2-$\overline{b}$	P2-S	2	P1-29	P1-12
15	P2-$\overline{b}$	P2-S	2	P2-W	P1-54

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19	P2-b	P2-S	2	P2-r	P2-c
20	P2-b	P2-S	3	P2-A	P1-2
21	P2-b	P2-a	3	P1-29	P1-12
22	P2-b	P2-a	3	P1-37	P1-39
23	P2-b	P2-a	3	P2-W	P1-54
24	P2-b	P2-a	3	P2-B	P2-C
25	P2-b	P2-a	3	P2-b	P2-d
26	P2-b	P2-a	3	P2-d	P2-s
28	P2-b	P2-L	4	P2-A	P1-8
29	P2-b	P2-L	4	P1-29	P1-13
30	P2-b	P2-L	4	P1-37	P1-38
31	P2-b	P2-L	4	P2-W	P2-Y
32	P2-b	P2-L	4	P2-B	P2-D
33	P2-b	P2-L	4	P2-b	P2-e
34	P2-b	P2-L	4	P2-e	P2-t
35	P2-b	P2-L	5	P2-A	P1-3
36	P2-b	P2-L	5	P1-29	P1-31
37	P2-b	P2-M	5	P1-37	P1-44
38	P2-b	P2-M	5	P2-W	P1-54
39	P2-b	P2-M	5	P2-B	P2-C
40	P2-b	P2-M	5	P2-b	P2-f
41	P2-b	P2-M	5	P2-u	P2-f
42	P2-b	P2-M	6	P2-A	P1-4
43	P2-b	P2-M	6	P1-29	P1-32
44	P2-b	P2-M	6	P1-37	P1-38
45	P2-b	P2-M	6	P2-W	P1-52
46	P2-b	P2-M	6	P2-C	P1-52
47	P2-b	P2-N	6	P2-B	P2-E
48	P2-b	P2-N	6	P2-b	P2-h
49	P2-b	P2-N	6	P2-v	P2-h
50	P2-b	P2-N	7	P2-A	P1-7
51	P2-b	P2-N	7	P1-29	P1-33
52	P2-b	P2-N	7	P1-37	P1-40
20	P2-B	P2-C	29	P1-37	P1-39
20	P2-b	P2-n	30	P2-A	P1-27
21	P2-A	P1-7	30	P1-29	P1-26
21	P1-29	P1-33	31	P1-29	P1-31
21	P2-W	P1-54	31	P1-29	P1-31
21	P2-B	P2-C	32	P2-A	P1-22
21	P2-b	P2-p	32	P1-29	P1-32
22	P2-A	P1-1	33	P2-A	P1-24
22	P1-29	P1-12	33	P1-29	P1-33
22	P2-W	P1-54	34	P2-A	P1-22
22	P2-B	P2-C	34	P1-29	P1-32
22	P2-b	P2-c	35	P2-A	P1-24
23	P2-A	P1-11	35	P1-29	P1-33
23	P2-W	P1-53	36	P2-A	P1-5
23	P2-B	P2-K	36	P1-29	P1-30
23	P2-b	P2-d	37	P2-A	P1-24
24	P2-A	P1-11	37	P1-29	P1-33
24	P2-W	P1-53	38	P2-A	P1-25

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24	P2-B	P2-K	38	P1-29	P1-23
24	P2-b	P2-c	39	P2-A	P1-20
25	P2-$\overline{A}$	P1-$\overline{11}$	39	P1-29	P1-30
25	P2-W	P1-53	40	P2-A	P1-27
25	P2-B	P2-K	40	P1-29	P1-15
25	P2-b	P2-f	41	P2-A	P1-21
26	P2-$\overline{A}$	P1-$\overline{8}$	41	P1-29	P1-31
26	P1-29	P1-35	42	P2-A	P1-22
26	P1-37	P1-44	42	P1-29	P1-16
26	P2-W	P1-53	43	P2-A	P1-24
26	P2-B	P2-K	43	P1-29	P1-33
26	P2-b	P2-h	44	P2-A	P1-22
27	P1-$\overline{29}$	P1-$\overline{17}$	44	P1-29	P1-32
27	P2-b	P2-U	45	P2-A	P1-24
28	P2-$\overline{DD}$	P2-U	45	P1-29	P1-33
28	P2-A	P1-1	46	P2-A	P1-5
28	P1-29	P1-30	46	P1-29	P1-30
28	P2-W	P1-54	47	P2-A	P1-24
28	P2-B	P2-C	48	P1-29	P1-33
28	P2-b	P2-c	48	P2-A	P1-1
29	P2-$\overline{A}$	P1-$\overline{20}$	48	P1-29	P1-30
29	P1-29	P1-30	49	P2-A	P1-28
			50	P2-A	P1-28
7	P2-W	P1-53	13	P2-b	P2-d
7	P2-Z	P1-53	14	P2-$\overline{A}$	P1-$\overline{8}$
7	P2-B	P2-E	14	P1-29	P1-15
7	P2-b	P2-j	12	P1-37	P1-55
7	P2-$\overline{w}$	P2-$\overline{j}$	14	P2-W	P2-D
8	P2-$\overline{A}$	P1-$\overline{4}$	14	P2-B	P2-F
8	P1-29	P1-32	14	P2-b	P2-c
8	P1-37	P1-41	15	P2-$\overline{A}$	P1-$\overline{3}$
8	P2-W	P1-54	15	P1-29	P1-31
8	P2-B	P2-C	15	P1-37	P1-43
8	P2-b	P2-k	15	P2-W	P1-54
8	P2-$\overline{x}$	P2-$\overline{k}$	15	P2-B	P2-C
9	P2-$\overline{A}$	P1-$\overline{7}$	15	P1-51	P2-H
9	P1-29	P1-33	15	P2-b	P2-f
9	P1-37	P1-40	16	P2-$\overline{A}$	P1-$\overline{4}$
9	P2-W	P1-53	16	P1-29	P1-16
9	P2-B	P2-F	16	P1-37	P1-38
9	P2-b	P2-m	16	P2-W	P1-52
9	P2-$\overline{y}$	P2-$\overline{m}$	16	P2-B	P2-E
10	P2-$\overline{A}$	P1-$\overline{5}$	16	P2-b	P2-h
10	P1-29	P1-12	17	P2-$\overline{A}$	P1-$\overline{7}$
10	P1-37	P1-42	17	P1-29	P1-33
10	P2-W	P1-54	17	P1-37	P1-40
10	P2-B	P2-C	17	P2-W	P1-53
10	P2-b	P2-n	17	P2-B	P2-E
10	P2-$\overline{z}$	P2-$\overline{n}$	17	P2-b	P2-j
11	P2-$\overline{A}$	P1-$\overline{7}$	18	P2-$\overline{A}$	P1-$\overline{4}$

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11	P1-29	P1-33	18	P1-29	P1-32
11	P2-W	P1-54	18	P1-37	P1-41
11	P2-B	P2-C	18	P2-W	P1-54
11	P2-b	P2-p	18	P2-B	P2-C
11	P2-$\overline{AA}$	P2-p	18	P2-b	P2-k
12	P2-A	P1-10	19	P2-$\overline{A}$	P1-7
12	P1-29	P1-14	19	P1-29	P1-33
12	P2-W	P1-54	19	P1-37	P1-40
12	P2-B	P2-C	19	P2-W	P1-53
12	P2-b	P2-c	19	P2-B	P2-E
13	P2-$\overline{A}$	P1-2	19	P2-b	P2-M
13	P1-29	P1-12	20	P2-$\overline{A}$	P1-5
13	P1-37	P1-39	20	P1-29	P1-12
13	P2-W	P1-54	20	P1-37	P1-42
13	P2-B	P2-C	20	P2-W	P1-54
51	P2-A	P1-28	1	P2-EE	P1-38
51	P1-29	P1-34	1	P2-W	P1-58
52	P2-A	P1-27	*1	P1-48	P1-50
52	P1-29	P1-35	*2	P1-48	P2-J
1	P2-F	P1-56	*2	P1-46	P2-J
1	P2-B	P1-57	**	P1-29	P1-45

*1.6 ohms max

**Test with 28 volts applied between pins P1-48 and 49. 1.6 ohms max. Do not apply 28v for longer than 1/2 minute.

3.5.2.4 Insulation resistance. Insulation resistance shall be 20 megohms minimum in Mode 1 and Mode 2 below when tested as specified in 4.4.2.4.

Mode 1 - At each assembly position the insulation resistance between each terminal listed below and all other terminals listed and case shall be as specified except for the terminals removed at the indicated assembly positions. Unused terminals may be connected together and tested as one terminal.

P1 - 1,2,3,4,5,7,8,10,11,12,13,14,15,16,17,20,21,22,23,24, 25,26,27,28,30,31,32,33,34,35,36,37,38,39,40,41,42,43, 44,45,46,48,50, and 55.

P2 - C,D,E,F,H,J,K,L,M,N,P,T,V,X,Y,Z,BB,DD,c,d,e,f,h,j,k, m,n, and p.

Unused terminals:

P1 - 18,47,59,60,62,63,64,65,66,67,70,71,72,73,74,75,76,77, 78,79,80 and 82.

P2 - CC,FF and HH.

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Assembly Position	Remove from List
1	P1-50
2 thru 10	P1-46, P2-BB, & J
11 thru 20	P1-46, P2-V, & J
21 thru 26	P1-46, P2-T, & J
27	P1-46, & P2-J

4.3.2.1 Classification of characteristics.

4.3.2.1.1 Programmer assembly (TDD) (6A1A12) (see dwg 8833317).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connector with respect to alignment pins	Gage
102.	Fastenings loose or missing	Visual/Manual
103.	Cracked or broken connectors or damaged pins	Visual
104.	Switch assembly mounting screws loose	Manual
105.	Defective solder joints	Visual/Manual
106.	Component missing	Visual
107.	Objective evidence relative to material improper or missing (see 3.1)	Visual
108.	Serialization improper or missing (see 3.4)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8)	Visual
202.	Illegible or improper marking	Visual
203.	Improper finish or color	Visual

4.3.2.1.2 Preparation for delivery (see dwg 8883505).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Packing material and packaging not proper or missing	Visual
102.	Marking not in accordance with dwg	Visual

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Minor:

201. Evidence of poor workmanship (see 3.8).....Visual

4.4 Test methods.

4.4.1 Calibration capability. Not applicable.

4.4.2 Operational limits.

4.4.2.1 Stepping test. With the assembly in its home position apply a pulsing voltage as specified in 3.5.2.1 and the assembly shall advance through one complete cycle and return to its home position after 52 pulses are applied. Pulses may be counted using a Hewlett-Packard Model 522B counter or other suitable means.

4.4.2.2 Homing test. With the assembly stepped one position off of "home" position apply 28 ± 2 vdc as specified in 3.5.2.2. The time interval between application of the 28 vdc and the closing of the circuit between terminals P1-29 and P1-36 ("home" position) shall be measured using a Hewlett-Packard Model 522B counter or its equivalent. The interval measured shall meet the requirements of 3.5.2.2.

4.4.2.3 Continuity and resistance. Momentarily apply 28 ± 2 volts dc to terminals P1-48 and P1-49 as necessary to advance the assembly to the positions indicated in 3.5.2.3, Table I. Resistance measurements, indicating continuity between the terminals listed in Table I shall be measured using a Keithley Milliohmmeter Model 502, or equivalent. Resistance shall be as specified in 3.5.2.3.

4.4.2.4 Insulation resistance. Momentarily apply 28 ± 2 volts dc to terminals P1-48 and P1-49 as necessary to advance the assembly to all positions as specified in 3.5.2.4. Insulation resistance at each assembly position shall be measured between the terminals listed in 3.5.2.4 using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma for this test.

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APPENDIX N
PANEL ASSEMBLY
(6A1A13)

The following sections apply specifically to the panel assembly. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.
3.5.1 Calibration capability. Not applicable.
3.5.2 Operational limits.
3.5.2.2 Functional tests.

(1) Test numbers. With 350 ± 17.5 volts dc applied through a 100K ± 10% 2 watt resistor to terminals P2d,e,f, the listed test numbers shall appear on the panel when the corresponding listed terminal is connected to the 350 vdc return. The symbol "-" indicates no digit will light. Testing shall be performed in accordance with 4.4.2.1.

Test Number	350VDC Return	Test Number	350 VDC Return
0 - -	P1w	- 3 -	P1b
1 - -	P1x	- 4 -	P1c
2 - -	P1m	- 5 -	P1d
3 - -	P1n	- - 0	P1X
4 - -	P1p	- - 1	P1M
5 - -	P1r	- - 2	P1N
6 - -	P1s	- - 3	P1P
7 - -	P1t	- - 4	P1R
8 - -	P1u	- - 5	P1S
9 - -	P1v	- - 6	P1T
- 0 -	P1j	- - 7	P1U
- 1 -	P1f	- - 8	P1V
- 2 -	P1h	- - 9	P1W

(2) Indicator lights. Each of the indicator lights listed below shall light when 29 ± 3 volts dc are applied to the corresponding listed terminal with the return applied to P2H. With 29 ± 3 vdc applied to P2A and the return connected to P2H, each listed light will light when the lens is depressed with the exception of the "Start" light.

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Indicator Light	Apply 29 ± 3 VDC To Terminal
Testing	P2B
Tester failure	P2C
AK Failure	P2D
AK Batteries	P2E
Go	P2F
Start	P2K

(3) Power light. The power light shall light when 115V, 60 Hz is applied between terminals P2j and P2J through a 56K ± 5%, 1/2 watt resistor.

3.5.2.2 Continuity and resistance. Resistance between the corresponding "primary" and "secondary" terminals listed below shall be specified when tested as specified in 4.4.2.2.

Primary Terminal	Secondary Terminal	Max Resistance (ohms)	Notes
1-P2V	P2U	0.5	
2-P2S	P2m	0.1	
3-P2P	P2N	0.5	
4-P2Z	P2a	0.1	
5-P2a	P2c	0.1	Power switch "On"
6-P2b	P2a	0.1	Power switch "On"
7-P2T	P2U	0.5	See Note 1.
8-P2S	P2N	0.5	See Note 1.
9-P2R	P2N	0.5	See Note 1.
10-P2M	P2N	0.5	See Note 1.
11-P2k	P2W	0.5	Depress and hold "Override"
12-P2Y	P2W	0.5	Depress and hold "Override"
13-P2H	P2X	0.5	Depress and hold "Override"
14-P1j	P1e	0.1	
15-P1f	P1Z	0.1	
16-P1h	P1a	0.1	
17-P1X	P1L	0.1	
18-P1M	P1A	0.1	
19-P1N	P1B	0.1	
20-P1P	P1C	0.1	
21-P1R	P1D	0.1	
22-P1S	P1E	0.1	
23-P1T	P1F	0.1	
24-P1U	P1H	0.1	
25-P1V	P1J	0.1	
26-P1W	P1K	0.1	

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Note 1 For reading apply 29 + 3 vdc between terminals P2L and P2H and depress the "Start" button momentarily. The button shall remain depressed until the 29 vdc is removed.

3.5.2.3 Insulation resistance. The insulation resistance between each of the following terminals and all others listed and case shall be greater than 20 megohms when tested as specified in 4.4.2.3.

Mode 1 "Power" switch "Off".
Terminals: P2H,V,T,S,R,P,M,W,X,Y,k,d,e,f,z,b, and c.

Mode 2 "Power" switch "Off" and "Start" buttong depressed with 28 + 2 volts dc applied between terminals P2L and P2H.

Terminals: P2H,V,U,S,P,W,X,Y,k,d,e,f,Z,b and c.

4.3.2.1 Classification of characteristics.
4.3.2.1.1 Panel assembly (6A1A13) (see dwg 8833345).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect location and orientation of connector with respect to alignment pins	Gage
	102. Fastenings loose or missing	Visual/Manual
	103. Cracked or broken connectors or damaged pins	Visual
	104. Defective solder joints	Visual/Manual
	105. Mounting screws loose	Manual
	106. Component loose or missing	Visual/Manual
	107. Objective evidence relative to material improper or missing (see 3.1)	Visual
	108. Serialization improper or missing (see 3.4)	Visual

4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Packing material and packing not in accordance with MIL-STD-1169 (MU)	Visual

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102. Marking not in accordance with MIL-STD-1169 (MU).....Visual

Minor:

201. Evidence of poorworkmanship (see 3.8).....Visual

4.4 Test methods.

4.4.1 Calibration capability. Not applicable.

4.4.2 Operational limits.

4.4.2.1 Functional tests. Apply the voltage specified and perform the functional tests as specified in 3.5.2.1.

4.4.2.2 Continuity and resistance. Resistance measurements, indicating continuity between the terminals listed in 3.5.2.2 shall be measured using a Keithley Milliohmmeter, Model 502, or equivalent.

4.4.2.3 Insulation resistance. The insulation resistance between the terminals listed in each mode of 3.5.2.3 shall be measured using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma for this test.

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APPENDIX O
RACK ASSEMBLY
(6A1A14)

The following sections apply specifically to the rack assembly. In addition to the requirements of this appendix, this assembly shall meet the requirements found in the main body of MIL-T-48804(PA).

3.5 Operational characteristics.

3.5.1 Calibration capability. Not applicable.

3.5.2 Operational limits.

3.5.2.1 Continuity and resistance. There shall be continuity not to exceed 0.5 ohms between each "Primary" terminal and the corresponding "Secondary" terminal in Table I when tested as specified in 4.4.2.1.

Table I

Primary Terminal	Secondary Terminal	Primary Terminal	Secondary Terminal	Primary Terminal	Secondary Terminal
6J4-A	J13-KK	6J4-R	J13-rr	6J4-f	J13-WW
6J4-A	J13-mm	6J4-R	J13-NN	6J4-g	J13-RR
6J4-B	J13-U	6J4-S	J15-66	6J4-g	J13-tt
6J4-B	J13-v	6J4-T	J13-p	6J4-h	J13-k
6J4-C	J15-67	6J4-T	J13-N	6J4-h	J13-K
6J4-D	J13-T	6J4-U	J15-65	6J4-i	J15-64
6J4-D	J13-u	6J4-V	J13-P	6J4-j	J15-48
6J4-E	J15-23	6J4-V	J13-r	6J4-j	J15-22
6J4-E	J15-49	6J4-W	J13-ww	6J4-k	J13-zz
6J4-F	J13-ff	6J4-W	J13-UU	6J4-k	J13-XX
6J4-F	J13-EE	6J4-X	J13-c	6J4-m	J13-Z
6J4-G	J15-51	6J4-X	J13-C	6J4-m	J13-aa
6J4-G	J15-25	6J4-Y	J13-b	6J4-n	J15-63
6J4-H	J15-50	6J4-Y	J13-B	6J4-p	J13-f
6J4-H	J15-24	6J4-Z	J15-8	6J4-p	J13-F
6J4-J	J13-V	6J4-Z	J15-38	6J4-q	J15-62
6J4-J	J13-w	6J4-A	J15-10	6J4-r	J15-47
6J4-K	J15-32	6J4-A	J15-37	6J4-r	J15-21
6J4-K	J15-2	6J4-b	J13-AA	6J4-t	J13-Pb
6J4-L	J13-e	6J4-b	J13-bb	6J4-t	J13-ZZ
6J4-L	J13-E	6J4-c	J15-7	6J4-s	J15-60
6J4-M	J15-33	6J4-c	J15-36	6J4-u	J15-59
6J4-M	J15-3	6J4-d	J13-j	6J4-w	J13-Aa
6J4-N	J13-S	6J4-d	J13-J	6J4-fw	J13-YY
6J4-N	J13-t	6J4-e	J13-d	6J4-x	J15-35
6J4-P	J13-ss	6J4-e	J13-D	6J4-x	J15-5
6J4-P	J13-PP	6J4-f	J13-yy	6J4-y	J13-h

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6J4-y
6J4-z
6J4-z
6J4-AA
6J4-AA
6J4-BB
6J4-BB
6J4-CC
6J4-CC
6J4-DD
6J4-DD
6J4-EE
6J4-EE
6J4-FF
6J4-FF
6J4-GG
6J4-GG
6J4-HH
J13-s
J13-y
J13-MM
J13-SS
J13-TT
J13-R
J13-X
J13-A
J13-L
J13-M
J13-BB
J13-CC
J13-HH
J13-JJ
J13-LL
J13-a
J13-m
J13-n
J13-pp
J13-uu
J13-vv
J13-cc
J13-dd
J13-jj
J13-kk
J13-nn
J9-Ee
J9-Ff
J9-Hh
J9-Jj
J9-Mm
J9-Nn
J6-B

J13-H
J15-34
J15-4
J13-W
J13-x
J13-DD
J13-ee
J15-II
J15-39
J13-FF
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J15-53
J8-m
J8-u
J8-Vv
J80-V
6J5-C
6J5-M
J8-F

J9-A
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J9-C
J9-D
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J9-F
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J9-K
J9-L
J9-M
J9-N
J9-P
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J9-S
J9-T
J9-U
J9-V
J9-W
J9-X
J9-Y
J9-Z
J9-AA
J9-BB
J9-CC
J9-DD
J9-EE
J9-FF
J9-HH
J9-JJ
J9-KK
J9-LL
J9-MM
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J9-PP
J9-RR
J9-SS
J9-TT
J9-UU
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J10-KK
J10-LL
J10-MM
J10-NN
J10-PP
J10-RR

J6-P
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J8-y
J8-z
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6J3-f
6J3-f
J80-M
6J5-D
6J5-G

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J11-1
J8-E
6J3-d
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J80-N
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J10-c
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J8-V
J8-W
J8-X
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J8-ZZ
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J8-s
J8-t
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J8-jj
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J8-nn
J8-rr
J8-ss
J8-tt
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J8-yy
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J8-Mm
J8-Nn
J8-Pp
J8-Rr
J8-Ss
J8-Tt
J8-Uu
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J7-hh
J7-jj
J7-kk
J7-mm
J7-nn
J7-pp
J7-rr
J7-ss
J7-tt
J7-uu
J7-vv
J7-ww
J7-xx
J7-yy
J7-zz
J7-Aa
J7-Bb
J7-Cc

6J2-F
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6J2-p
6J2-AA
6J3-S
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J17-25
J17-27
J7-A
J7-B
J7-C
J7-D
J7-E
J7-F
J7-H
J7-J
J7-K
J7-M
J7-N
J7-P
J7-R
J7-S
J7-T
J7-U
J7-V
J7-W
J7-X
J7-Y
J7-Z
J7-AA
J7-BB
J7-CC
J7-DD
J12-W
J12-X
J12-Y
J12-Z
J12-a
J12-b
J13-c
J12-d
J12-e
J12-f
J12-h
J12-j
J12-k
J12-m
J12-n
J12-p
J12-r
J12-s

J8-KK
J8-LL
6J5-e
6J5-x
J8-MM
6J5-K
J8-NN
6J5-i
J2-A
J2-N
J2-V
J3-A
J3-D
J3-E
J3-K
J3-N
J3-X
J4-A
J4-D
J4-F
J4-L
J4-K
J4-N
J5-A
J5-C
J5-F
J5-H
J5-J
J5-K
J5-L
J5-N
J5-P
J5-R
J18-E
J18-K
J18-Y
J19-r
J19-s
J19-t
J19-u
J19-v
J19-w
J19-x
J19-m
J19-n
J19-p
J19-e
J19-Z
J19-a
J19-A
J19-B

J7-UU
J7-VV
J7-WW
J7-XX
J7-YY
J7-ZZ
J7-a
J7-b
J7-c
J7-d
J7-e
J7-f
J7-h
J7-j
J7-k
J7-m
J7-n
J7-p
J7-r
J7-s
J7-t
J7-u
J7-v
J7-w
J7-x
J7-y
J7-z
J7-aa
J7-bb
J7-cc
J7-dd
J7-ee
J7-ff
J14-k
J14-m
J14-n
J14-p
J14-r
J14-s
J14-t
J14-u
J14-v
J14-w
J14-CC
J16-L
J16-M
J16-R
J16-S
J16-T
J16-U
J16-V

J14-N
J14-T
J14-X
J14-x
J14-y
J14-z
J14-AA
J14-DD
J16-A
J20-k
J16-C
J16-D
J20-m
J16-F
J16-H
J16-J
J16-K
J16-N
J16-P
J16-b
J16-c
J16-d
J16-e
J16-f
J16-h
J16-j
J16-k
J16-m
J16-n
J16-p
J16-s
J16-u
J18-A
J19-M
J19-N
J19-P
J19-R
J19-S
J19-T
J19-U
J19-V
J19-W
J19-X
J16-t
J18-V
J18-a
J18-c
J18-d
J18-e
J18-f
J18-h

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J7-Dd	J20-U	J12-t	J19-C	J16-W	J18-j
J7-Ee	J20-V	J12-u	J19-D	J16-X	J18-k
J7-Ff	J20-Y	J12-v	J19-E	J16-Y	J18-m
J7-Hh	J20-X	J12-w	J19-F	J16-Z	J18-n
J7-Jj	J20-W	J12-x	J19-H	J16-a	J18-p
J7-Kk	J20-b	J12-y	J19-J	J16-r	J18-J
J7-Mm	J20-c	J12-z	J19-K	J16-v	J18-BB
J7-Nn	6J1-C	J12-AA	J19-L	J18-D	J2-U
J7-Pp	6J1-C	J12-BB	J14-BB	J18-F	J2-T
J7-Rr	J18-b	J12-CC	J16-E	J18-L	J19-b
J7-L	J20-j	J12-DD	J18-DD	J18-M	J19-c
J7-Tt	J16-w	J12-EE	J18-EE	J18-N	J19-d
J7-Ss	Ground lug	J14-P	J19-j	J18-U	J4-P
J12-B	J18-W	J14-R	J19-f	J18-Z	J2-K
J12-C	J2-L	J14-S	J19-h	J4-C	J3-P
J12-D	J2-C	J14-U	J18-R	J4-H	J3-CC
J12-E	J2-M	J14-V	J18-S	J20-B	J5-B
J12-F	J2-E	J14-W	J18-T	J20-S	• J5-D
J12-J	J3-U	J14-Y	J18-r	J20-Z	6J1-A
J12-K	J18-X	J14-Z	J18-s	J20-a	6J1-A
J12-L	J18-H	J14-a	J18-t	6J1-B	Ground lug
J12-N	J3-Y	J14-b	J18-u	Case	Ground lug
J12-P	J2-J	J14-c	J18-v	J8-A	J1-CC
J12-R	J2-P	J14-d	J18-w	J8-SS	J17-44
J12-S	J2-R	J14-e	J18-x	J8-PP	J17-34
J12-T	J2-S	J14-f	J18-y	J80-R	J17-58
J12-U	J18-B	J14-h	J18-z	J8-WW	J17-50
J12-V	J18-C	J14-j	J18-AA	J8-UU	J17-46
J8-TT	J17-45				
J8-VV	J17-48				
J8-Ee	J17-49				
J80-S	J17-57				
J80-H	J17-56				
J9-Kk	J17-43				

3.5.2.2 Insulation resistance. The insulation resistance between each of the following terminals and all others listed and case shall be greater than 20 megohms when tested as specified in 4.4.2.2. Unused terminals may be connected together and tested as one terminal.

Terminals:

6J2 - U and LL
Unused terminals: b,c,d,t,X,S,R,P,N,m,n,CC,B,D,E and J.

6J3 - Unused terminals: A,G,T and g.

6J4 - A,B,C,D,E,F,G,H,J,K,L,M,N,P,R,S,T,U,V,W,X,Y,Z, a,b,c,d,e,f,g,h,i,j,k,m,n,p,q,r,s,t,u,w,x,y,z, AA,BB,CC,DD,EE,FF,GG, and HH.
Unused terminal: v

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6J5 - Unused terminals: H,J,T,U,X,Y,d,g,h,n,q,w, CC,EE,and FF.

J1-Y - Unused terminals: A,B,C,D,E,F,H,J,K,L, and V.

J2 - Unused terminals: B,D,F, and H.

J3 - Unused terminals: B,C,F,H,J,L,M,R,S,T,V,W,Z,AA,BB and DD.

J4 - C and H.
Unused terminals: B,E,J,M, and R.

J5 - B and D.
Unused terminals: E and M.

J6 - Unused terminals: D,L,M, and N.

J7 - A,B,C,D,E,F,H,J,K,L,M,N,P,R,S,T,U,V,W,X,Y,Z, AA,BB,CC,DD,EE,FF,HH,JJ,KK,LL,MM,NN,PP,RR,SS,TT, UU,VV,WW,XX,YY,ZZ,a,b,c,d,e,f,h,j,k,m,n,p,r,s,t, u,v,w,x,y,z,aa,bb,cc,dd,ee,ff,hh,jj,kk,mm,nn,pp, rr,ss,tt,uu,vv,ww,xx,yy,zz,Aa,Bb,Cc,Dd,Ee,Ff,Hh, Jj,Kk,Mm,Nn,Rr,Ss and Tt.
Unused terminals: Uu,Vv and Ww.

J8 - A,B,C,E,F,H,J,K,L,M,N,P,R,S,T,U,V,W,X,Y,Z,AA,BB,CC, DD,EE,FF,HH,JJ,KK,LL,MM,NN,PP,SS,TT,UU,VV,WW,XX,YY, ZZ,a,b,c,d,e,f,k,n,p,s,t,hh,jj,kk,mm,nn,rr,ss,uu,vv, yy,Ee,Kk,Mm,Nn,Pp,Rr,Ss,Tt and Uu.
Unused terminals: D,pp,zz and Ww.

J9 - A,B,C,D,E,F,H,J,K,L,M,N,P,R,S,T,U,V,W,X,U,Z, AA,BB,CC,DD,EE,FF,HH,JJ,KK,LL,MM,NN,PP,RR,SS,TT, UU,VV,WW,XX,YY,ZZ,a,c,d,e,f,h,j,k,m,n,p,r,s,t,u, v,w,x,y,z,aa,cc,dd,ee,ff,hh,jj,kk,mm,nn,pp,rr,ss, uu,vv,ww,xx,yy,Aa,Bb,Cc,Dd,Ee,Ff,Hh,Jj,Mm, and Nn.
Unused terminals: Pp,Rr,Ss,Tt,Uu,Vv,Ww,tt and zz.

J10 - A,B,C,D,E,F,H,J,K,L,M,N,P,R,S,T,U,V,W,X,Y,Z, AA,BB,CC,DD,EE,FF,HH,JJ,KK,LL,MM,NN,PP,RR,SS,TT,UU, VV,WW,XX,YY,ZZ,a,b,c,d,e,f,h,j,k,m,n,p,r,s,t,u,v,w, x,y,z,aa,bb,cc,dd,ee,ff,hh,jj,kk,mm,nn,pp,ss,tt,uu, vv,ww,xx,yy,zz,Aa,Bb,Cc,Dd,Ee,Ff,Hh,Jj,Kk,Mm,Nn,Pp, Rr,Ss,Tt,Uu,Vv, and Ww.

J11 - 1, 24, 54 and 76.
Unused terminals: 78, 79, 80, 82 and 60.

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J12 - B,C,D,E,F,J,K,L,N,P,R,S,T,U,V,W,X,Y,Z,a,b,c,d,e,f, h,j,k,m,n,p,r,s,t,u,v,w,x,y,z,AA,BB,CC,DD, and EE.
Unused terminals: H,M,FF, and HH.

J13 - A,L,M,R,X,a,m,n,s,y,BB,CC,HH,JJ,LL,MM,SS,TT,cc,dd, jj,kk,nn,pp,uu, and vv.
Unused terminals: Cc,Dd,Ee,Ff,Hh,Jj,Kk,Mm,Nn,Pp, Rr,Ss,Tt,Uu,Vv, and Ww.

J14 - P,R,S,U,V,W,Y,Z,a,b,c,d,e,f,h,j,k,m,n,p,r,s,t,u,v, w, and CC,
Unused terminals: EE,FF, and HH.

J15 - Unused terminals: 70,71,72,73,74,75,76,77,78,79, 80, and 82.

J16 - L,M,R,S,T,U,V,W,X,Y,Z,a,r,and v.
Unused terminals: x,y,z,AA,BB,CC,DD,EE,FF, and HH.

J17 - 43,56,57 and 58.
Unused terminals: 18,47,59,60,62,63,64,65,66,67,70, 71,72,73,74,75,76,77,78,79,80,82, and 55.

J18 - D,F,L,M,N,U and Z.
Unused terminals: CC, FF and HH.

J19 - Unused terminals: Y and k.

J20 - Z; - Unused terminals: h,n,p,r,s,t,u,v,w, and x.

J80 - Unused terminals: a,b,c,W,X,Y and Z.

4.3.2.1 Classification of characteristics.
4.3.2.1. Rack assembly (6A1A14) (see dwg 8850424).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location and orientation of connectors	Visual
102.	Component missing	Visual
103.	Fastenings loose or missing	Visual/Manual
104.	Cracked or broken connectors or damaged pins	Visual
105.	Objective evidence relative to material improper or missing (see 3.1)	Visual
106.	Serialization improper or missing (see 3.4)	Visual

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Minor:
201. Evidence of poor workmanship (see 3.8).....Visual
202. Illegible or improper marking.....Visual
203. Improper finish or color.....Visual

4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Packing material and packing not in accordance with MIL-STD-1169 (MU).....	Visual
102.	Marking not in accordance with MIL-STD-1169(MU).....	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.8).....	Visual

4.4 Test methods.
4.4.1 Calibration capability. Not applicable.
4.4.2 Operational limits.
4.4.2.1 Continuity and resistance. Resistance measurements, indicating continuity between the terminals listed in 3.5.2.1, shall be measured using a Keithley Milliohmeter Model 502 or equivalent.
4.4.2.2 Insulation resistance. Insulation resistance shall be measured between the terminals listed in 3.5.2.2 using a test voltage of 500 ± 50 volts dc. Current shall be limited to 5 ma for this test.

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